chapter 10 and chapter 11

10

Signal and System Analysis using the z-Transform

10.1 Introduction

In chapter 7 we have studied the analysis of linear time-invariant continuous-time systems using Laplace transform. In this chapter we will study the counterpart of the Laplace transform, called the z-transform for LTI discrete-time systems. The Laplace transform converts differential equations into algebraic equations. In the same way the z-transform converts difference equations into algebraic equations, there by simplifying the analysis of discrete-time systems. More over, in the z-domain the convolution of two time domain signals is equivalent to multiplication of their corresponding z-transforms. This property simplifies the analysis of the response of an LTI system to various input signals. Therefore, the z-transform is an important tool in the study of discrete-time systems.

10.2 The z-Transform

The z-transform of a discrete-time sequence $x(n)$ is defined as

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} \tag{10.1}$$

where z is a complex variable. In polar form z can be expressed as

$$z = re^{j\omega} \tag{10.2}$$

where r is radius of a circle.

If the sequence $x(n)$ exists for n in the range $-\infty$ to ∞, then Eq. (10.1) represents two sided or the bilateral z-transform. On the otherhand, if the sequence exists only for $n \geq 0$, then the Eq. (10.1) changes to

$$X_+(z) = \sum_{n=0}^{\infty} x(n)z^{-n} \tag{10.3}$$

which is called as one sided z-transform.

By substituting Eq. (10.2) into Eq (10.1) yields

$$X(re^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)(re^{j\omega})^{-n}$$

$$= \sum_{n=-\infty}^{\infty} [x(n)r^{-n}]e^{-j\omega n} \qquad (10.4a)$$

The above Eq. (10.4a) represents the Fourier transform of a signal $x(n)r^{-n}$. Hence the inverse DTFT of $X(re^{j\omega})$ must be $x(n)r^{-n}$. Therefore, we can write

$$x(n)r^{-n} = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(re^{j\omega})e^{j\omega n}d\omega \qquad (10.4b)$$

On multiplying Eq. (10.4b) by r^n, we get

$$x(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(re^{j\omega})(re^{j\omega})^n d\omega \qquad (10.5)$$

$$z = re^{j\omega}$$
$$dz = rje^{j\omega}d\omega$$
$$\Rightarrow d\omega = \frac{dz}{rje^{j\omega}} = \frac{dz}{jz}$$

On substituting $z = re^{j\omega}$ and $d\omega = \frac{dz}{jz}$ in Eq. (10.5), we have

$$x(n) = \frac{1}{2\pi j}\oint_c X(z)z^{n-1}dz \qquad (10.6)$$

where the symbol of $\oint_c$ denotes integration around the circle of radius $|z| = r$ in a counter-clockwise direction.

Definition: The z-transform of an arbitrary signal $x(n)$ is

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} \quad \ldots \qquad (10.7)$$

and the inverse z-transform of $X(z)$ is

$$x(n) = \frac{1}{2\pi j}\oint_c X(z)z^{n-1}dz \quad \ldots \qquad (10.8)$$

The z-transform $X(z)$ is a ratio of two polynomials on z^{-1}, given by

$$X(z) = \frac{b_0 + b_1z^{-1} + b_2z^{-2} + \cdots + b_Mz^{-M}}{1 + a_1z^{-1} + a_2z^{-2} + \cdots + a_Nz^{-N}}. \qquad (10.9)$$

The roots of the numerator polynomial are those values of z for which $X(z) = 0$ and are referred to as the zeros of $X(z)$. The roots of denominator polynomial are known as poles for which $X(z)$ is infinity. Zero locations are denoted by 'o' symbol and the pole locations with '×' symbol in the z-plane.

The z-transform exists when the infinite sum in Eq.(10.7) converges. The sum may not converge for all values of z. The values of z for which the sum in Eq. (10.7) converges is called the Region Of Convergence (ROC). This concept can be explained with the following examples.

Review Questions

1. Define z-transform. What are its applications?
2. What is the relationship between DTFT and z-transform?

Solved Problem 10.1 Find the z-transform and ROC for the signal $x(n) = a^n u(n)$.

Solution:

The given sequence is a causal and infinite duration sequence.

From the definition of z-transform, we can write

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} a^n u(n)z^{-n} \qquad u(n) = 0 \text{ for } n < 0,\ = 1 \text{ for } n \geq 0$$

$$= \sum_{n=0}^{\infty} a^n z^{-n} \qquad (10.10)$$

$$= \sum_{n=0}^{\infty} (az^{-1})^n \qquad (10.11)$$

This is a geometric series of infinite length.

We have

$$a + ar + ar^2 + \cdots \infty = \frac{a}{1-r} \text{ if } |r| < 1 \qquad (10.12)$$

That is, the summation in Eq. (10.11) converges if $|az^{-1}| < 1$ or $|z| > |a|$. Using Eq. (10.12) we can write

$$X(z) = \frac{1}{1-az^{-1}};\ \text{ROC } |z| > |a|, \qquad (10.13)$$

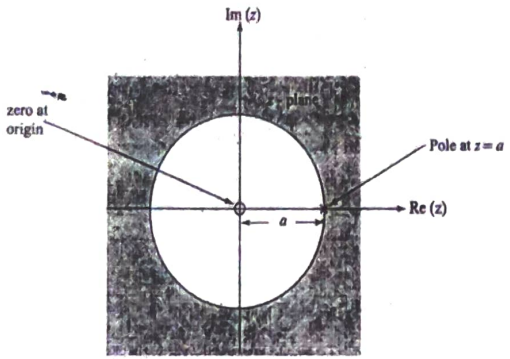

Fig. 10.1 ROC of Example 10.1.

which implies that the ROC is exterior to the circle of radius a as shown in Fig. 10.1.

Solved Problem 10.2 Determine the z-transform of the signal

$x(n) = -b^n u(-n-1)$. Find ROC.

Solution:

The given signal is a non-causal and infinite duration sequence.

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$= -\sum_{n=-\infty}^{\infty} b^n u(-n-1)z^{-n}$$

$u(-n-1) = 0$ for $n \geq 0$
$= 1$ for $n \leq -1$

$$= -\sum_{n=-\infty}^{-1} b^n z^{-n}$$

$$= -\sum_{n=1}^{\infty} b^{-n} z^n = -\sum_{n=1}^{\infty} (b^{-1}z)^n$$

The above series is an infinite series and converges if $|b^{-1}z| < 1$, or $|z| < |b|$. Hence

$$X(z) = -\frac{b^{-1}z}{1-b^{-1}z} = \frac{z}{z-b} = \frac{1}{1-bz^{-1}}; \quad |z| < |b| \qquad (10.14)$$

That is the ROC is interior to the circle of radius b as shown in Fig. 10.2.

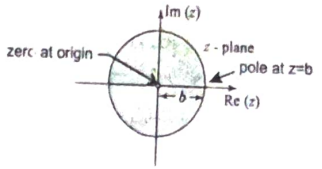

Fig. 10.2 ROC of Example 10.2.

From the examples we find that the z-transform of both sequences are identical even though the sequences are different. Only ROC differentiates them, not the z-transform. Therefore, it is important to know the ROC to determine the correct inverse z-transform.

Solved Problem 10.3 Determine the z-transform of
$x(n) = a^n u(n) - b^n u(-n-1)$ and find ROC.

Solution:

The given sequence is a two-sided infinite duration sequence having values of n; from $-\infty$ to ∞.

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$= \sum_{n=0}^{\infty} a^n z^{-n} + \sum_{n=-1}^{-\infty} b^n z^{-n}$$

$$= \sum_{n=0}^{\infty} (az^{-1})^n + \sum_{n=1}^{\infty} (b^{-1}z)^n \qquad (10.15)$$

The first power series converges if $|az^{-1}| < 1$ or $|z| > |a|$ and the second power series converges if $|b^{-1}z| < 1$ or $|z| < |b|$. If $|b| < |a|$, the two ROCs do not overlap as shown in Fig. 10.3 and $X(z)$ does not exist. If $|b| > |a|$ the two ROCs overlap as shown in Fig. 10.4 and $X(z)$ exists. Therefore the ROC of $X(z)$ is $|a| < |z| < |b|$. That is for an infinite duration two-sided signal the ROC is a ring in the z-plane.

$$X(z) = \frac{z}{z-a} + \frac{z}{z-b} \quad \text{ROC}: |a| < |z| < |b|. \qquad (10.16)$$

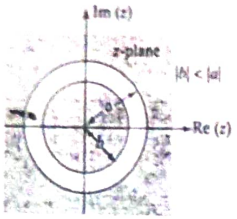

Fig.10.3 ROC of a two sided sequence for $|b| < |a|$.

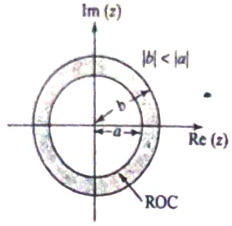

Fig.10.4 ROC of a two sided sequence for $|b| > |a|$.

10.3 z-Transform and ROC of Finite Duration Sequences

10.3.1 Right hand sequence

A right hand sequence is one for which $x(n) = 0$ for all $n < n_0$, where n_0 is positive or negative but finite. If n_0 is greater than or equal to zero, the resulting sequence is causal or a positive time sequence. For such type of sequence the ROC is entire z-plane except at $z = 0$.

Solved Problem 10.4 Find z-transform and ROC of the causal sequence

$$x(n) = \{\underset{\uparrow}{2}, -1, 3, 2, 0, 1\}$$

Solution:

On expanding Eq. (10.7) we get

$$X(z) = \ldots x(-1)z + x(0) + x(1)z^{-1} + x(2)z^{-2} + x(3)z^{-3} + x(4)z^{-4} + x(5)z^{-5} + \cdots \tag{10.17}$$

Given sequence values

$$x(0) = 2;\ x(1) = -1;\ x(2) = 3;\ x(3) = 2;\ x(4) = 0;\ x(5) = 1;$$

Substituting the sequences values in Eq. (10.17) we have

$$X(z) = 2 - z^{-1} + 3z^{-2} + 2z^{-3} + z^{-5} \tag{10.18}$$

The $X(z)$ converges for all values of z except at $z = 0$.

10.3.2 Left hand sequence

A left hand sequence is one for which $x(n) = 0$ for all $n \geq n_0$, where n_0 is positive or negative but finite. If $n_0 \leq 0$, the resulting sequence is anticausal sequence. For such type of sequence the ROC is entire z-plane except at $z = \infty$.

Solved Problem 10.5 Find the z-transform and ROC of the anticausal sequence

$$x(n) = \{3, 2, -1, -4, \underset{\uparrow}{1}\}$$

Solution:

We know

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$
$$= \cdots x(-4)z^4 + x(-3)z^3 + x(-2)z^2 + x(-1)z + x(0) + x(1)z^{-1} + x(2)z^{-2} \ldots \tag{10.19}$$

The given sequence values are

$$x(-4) = 3;\ x(-3) = 2;\ x(-2) = -1;\ x(-1) = -4;\ x(0) = 1$$

On substituting these values in Eq. (10.19) we get

$$X(z) = 3z^4 + 2z^3 - z^2 - 4z + 1 \tag{10.20}$$

The $X(z)$ converges for all values of z except at $z = \infty$.

10.3.3 Two sided sequence

A signal that has finite duration on both the left and right sides is known as two-sided sequence. For such type of sequence, the ROC is entire z-plane except at $z = 0$ and $z = \infty$.

Solved Problem 10.6 Find the z-transform of the sequence

$$x(n) = \{1, 2, 0, -4, \underset{\uparrow}{3}, 2, 1, 6, 5\}$$

Solution:

The sequence values are

$$x(-4) = 1;\ x(-3) = 2;\ x(-2) = 0;\ x(-1) = -4;\ x(0) = 3;\ x(1) = 2$$
$$x(2) = 1;\ x(3) = 6;\ x(4) = 5.$$

On substituting these values

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}, \text{ we get}$$

$$X(z) = z^4 + 2z^3 - 4z + 3 + 2z^{-1} + z^{-2} + 6z^{-3} + 5z^{-4} \qquad (10.21)$$

The $X(z)$ converges for all values except at $z = 0$ and $z = \infty$.

Solved Problem 10.7 Find z-transform of the following sequences

i) $x(n) = u(n) - u(n-3)$

ii) $x(n) = \{1, 2, 3, 2\}$

iii) $x(n) = \{1, 2, \underset{\uparrow}{-1}, 2, 3\}$

Solution:

i) Given $x(n) = u(n) - u(n-3)$

The plot of $x(n)$ is shown in Fig.10.5 from which

$$x(n) = 1 \text{ for } 0 \le n \le 2$$
$$= 0 \text{ otherwise}$$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

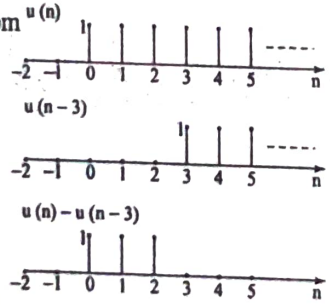

Fig. 10.5

On substituting the sequence values we get

$$X(z) = 1 + z^{-1} + z^{-2} \qquad (10.22)$$

ROC: Entire z-plane except at $z = 0$.

ii) $x(n) = \{1, 2, 3, 2\}$

Given $x(0) = 1; \quad x(1) = 2; \quad x(2) = 3; \quad x(3) = 2$

On substituting the sequence values

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}, \quad \text{we get}$$

$$X(z) = 1 + 2z^{-1} + 3z^{-2} + 2z^{-3}$$

ROC: Entire z-plane except at $z = 0$.

iii) $x(n) = \{1, 2, \underset{\uparrow}{-1}, 2, 3\}$

$$X(z) = z^2 + 2z - 1 + 2z^{-1} + 3z^{-2} \qquad (10.23)$$

ROC: Entire z-plane except at $z = 0$ and $z = \infty$.

Practice Problem 10.1

Find the z-transform of the following

(i) $x_1(n) = \{\underset{\uparrow}{1}, 2, -1, 3, 2, 1\}$

(ii) $x_2(n) = \{1, 2, -1, 0, 1, \underset{\uparrow}{2}\}$

(iii) $x_3(n) = \{3, 2, 1, \underset{\uparrow}{0}, 3, 1, -2\}$

(iv) $x_4(n) = (e)^n u(n)$

(v) $x_5(n) = (-1)^n u(n)$

(vi) $x_6(n) = (0.5)^n$ for $-\infty < n < 0$
$= (0.2)^n$ for $n \ge 0$

Ans:

(i) $X_1(z) = 1 + 2z^{-1} - z^{-2} + 3z^{-3} + 2z^{-4} + z^{-5}$
ROC: Entire z-plane except at $z = 0$.

(ii) $X_2(z) = z^5 + 2z^4 - z^3 + z + 2$
ROC: Entire z-plane except at $z = \infty$.

(iii) $X_3(z) = 3z^3 + 2z^2 + z + 3z^{-1} + z^{-2} - 2z^{-3}$
ROC: Entire z-plane except at $z = 0$ and $z = \infty$.

(iv) $\dfrac{1}{1 - ez^{-1}}$ ROC: $|z| > e$

(v) $\dfrac{1}{1 + z^{-1}}$ ROC: $|z| > 1$

(vi) $\dfrac{1}{1 - 0.2z^{-1}} - \dfrac{1}{1 - 0.5z^{-1}}$
ROC: $0.2 < |z| < 0.5$

10.4 Properties of Region of Convergence

1. The ROC is a ring or disk in the z-plane centered at the origin.
2. The ROC cannot contain any poles.
3. If $x(n)$ is a finite duration, causal sequence then the ROC is the entire z-plane except at $z = 0$.
4. If $x(n)$ is a finite duration, anti-causal sequence then the ROC is the entire z-plane except at $z = \infty$.
5. If $x(n)$ is a finite duration, two-sided sequence the ROC is entire z-plane except at $z = 0$ and $z = \infty$.

6. If $x(n)$ is an infinite duration, two-sided sequence the ROC will consist of a ring in the z-plane, bounded on the interior and exterior by a pole, not containing any poles.
7. The ROC of an LTI stable system contains the unit circle (to be discussed in section (10.9).
8. The ROC must be a connected region.

Review Questions

1. Prove that for causal sequences ROC is in exterior of circle of some radius r. JNTU March 2006
2. Explain the properties of ROC of $X(z)$. JNTU Nov 2005 AU, 2006
3. What is the significance of ROC?

Solved Problem 10.8 Find the z-transform of the signal $x(n) = [\sin \omega_o n]u(n)$ and find ROC.

Solution:

Given $x(n) = (\sin \omega_o n)u(n)$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} (\sin \omega_o n)u(n)z^{-n} = \sum_{n=0}^{\infty} (\sin \omega_o n)z^{-n}$$

$$= \sum_{n=0}^{\infty} \left[\frac{e^{j\omega_o n} - e^{-j\omega_o n}}{2j}\right] z^{-n} \qquad \boxed{\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}}$$

$$= \frac{1}{2j} \sum_{n=0}^{\infty} \left(e^{j\omega_o n} - e^{-j\omega_o n}\right) z^{-n}$$

$$= \frac{1}{2j} \left[\sum_{n=0}^{\infty} e^{j\omega_o n} z^{-n} - \sum_{n=0}^{\infty} e^{-j\omega_o n} z^{-n}\right]$$

$$= \frac{1}{2j} \left[\sum_{n=0}^{\infty} \left(e^{j\omega_o} z^{-1}\right)^n - \sum_{n=0}^{\infty} \left(e^{-j\omega_o} z^{-1}\right)^n\right]$$

If $|e^{j\omega_o} z^{-1}| < 1$ and $|e^{-j\omega_o} z^{-1}| < 1$, the above series converges.

That is for $|z| > 1$, the series converges and we get

$$X(z) = \frac{1}{2j}\left[\frac{1}{1 - e^{j\omega_o} z^{-1}} - \frac{1}{1 - e^{-j\omega_o} z^{-1}}\right]$$

$$\boxed{\sum_{n=0}^{\infty} = 1 + r + r^2 + \cdots + \infty = \frac{1}{1-r} \text{ if } |r| < 1}$$

$$= \frac{1}{2j}\left[\frac{1 - e^{-j\omega_o} z^{-1} - 1 + e^{j\omega_o} z^{-1}}{(1 - e^{j\omega_o} z^{-1})(1 - e^{-j\omega_o} z^{-1})}\right]$$

$$= \frac{1}{2j}\left[\frac{e^{j\omega_o} z^{-1} - e^{-j\omega_o} z^{-1}}{1 - e^{j\omega_o} z^{-1} - e^{-j\omega_o} z^{-1} + z^{-2}}\right]$$

$$= \frac{(\sin \omega_o) z^{-1}}{1 - 2(\cos \omega_o) z^{-1} + z^{-2}} \qquad \text{ROC}: |z| > 1 \qquad (10.24)$$

Solved Problem 10.9 Determine the z-transform, ROC and pole-zero locations of $X(z)$ for

$$x(n) = \left(\frac{2}{3}\right)^n u(n) + \left(-\frac{1}{2}\right)^n u(n)$$

Solution:

Given $x(n) = \left(\frac{2}{3}\right)^n u(n) + \left(-\frac{1}{2}\right)^n u(n)$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} \left[\left(\frac{2}{3}\right)^n u(n) + \left(-\frac{1}{2}\right)^n u(n)\right] z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n z^{-n} + \sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n z^{-n} = \sum_{n=0}^{\infty} \left(\frac{2}{3} z^{-1}\right)^n + \sum_{n=0}^{\infty} \left(-\frac{1}{2} z^{-1}\right)^n$$

The first summation converges for $|\frac{2}{3} z^{-1}| < 1$ or $|z| > \frac{2}{3}$ and the second summation converges for $|-\frac{1}{2} z^{-1}| < 1$ or $|z| > \frac{1}{2}$. Therefore $X(z)$ converges for $|z| > \frac{2}{3}$.

Now

$$X(z) = \frac{1}{1 - \frac{2}{3} z^{-1}} + \frac{1}{1 + \frac{1}{2} z^{-1}} \qquad \left(\because\ 1 + r + r^2 + \cdots + \infty = \frac{1}{1-r}\right)$$

$$= \frac{z}{z - \frac{2}{3}} + \frac{z}{z + \frac{1}{2}}$$

$$= \frac{2z^2 - \frac{1}{6}z}{\left(z-\frac{2}{3}\right)\left(z+\frac{1}{2}\right)}$$

$$= \frac{z\left(2z-\frac{1}{6}\right)}{\left(z-\frac{2}{3}\right)\left(z+\frac{1}{2}\right)} \quad \text{ROC}: |z| > \frac{2}{3} \qquad (10.25)$$

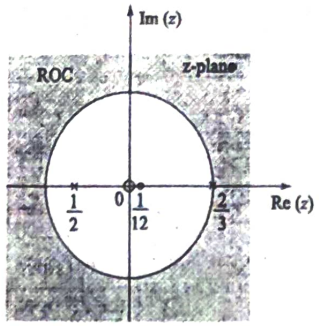

Fig.10.6

The poles of $X(z)$ are at $z = \frac{2}{3}$ and at $z = -\frac{1}{2}$ and the zeros of $X(z)$ are at $z = 0$ and at $z = \frac{1}{12}$.

Solved Problem 10.10 Find z-transform and ROC of $x(n) = a^{|n|}; |a| < 1$

Solution:

$$X(z) = \sum_{n=-\infty}^{\infty} a^{|n|} z^{-n}$$

$$= \sum_{n=-\infty}^{-1} a^{|n|} z^{-n} + \sum_{n=0}^{\infty} a^{|n|} z^{-n}$$

$$= \sum_{n=1}^{\infty} (az)^n + \sum_{n=0}^{\infty} (az^{-1})^n$$

$$= \frac{az}{1-az} + \frac{1}{1-az^{-1}}$$

$$= -\frac{z}{z-\frac{1}{a}} + \frac{z}{z-a} \quad \text{ROC}: |a| < |z| < \frac{1}{|a|} \qquad (10.26)$$

$\sum_{n=1}^{\infty} (az)^n$ converges for $|az| < 1$ or $|z| < \frac{1}{|a|}$.
$\sum_{n=0}^{\infty} (az^{-1})^n$ converges for $|az^{-1}| < 1$ or $|z| > |a|$.

Solved Problem 10.11 Find the z-transform of the following signals

(i) $x(n) = u(n)$; (ii) $x(n) = \delta(n)$; (iii) $x(n) = \left(\frac{1}{2}\right)^n u(-n)$;

(iv) $x(n) = \left(\frac{1}{2}\right)^n u(n) + \left(\frac{1}{3}\right)^n u(n)$ AU, Nov 2008

Solution:

i) Given

$$x(n) = u(n)$$

$u(n) = 1$ for $n \geq 0$
$= 0$ for $n < 0$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = \sum_{n=-\infty}^{\infty} u(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} z^{-n} = \sum_{n=0}^{\infty} (z^{-1})^n$$

$\sum_{n=0}^{\infty} (z^{-1})^n$ converges if $|z^{-1}| < 1$ or $|z| > 1$

$$= \frac{1}{1-z^{-1}} \quad \text{ROC}: |z| > 1 \qquad (10.27)$$

ii) $x(n) = \delta(n)$

$$X(z) = \sum_{n=-\infty}^{\infty} \delta(n) z^{-n}$$

$\delta(n) = 1$ for $n = 0$
$= 0$ for $n \neq 0$

$$= 1 \quad \text{ROC: Entire } z\text{-plane} \qquad (10.28)$$

iii) $x(n) = \left(\frac{1}{2}\right)^n u(-n)$

$u(-n) = 0$ for $n > 0$
$= 1$ for $n \leq 0$

$$X(z) = \sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^n u(-n) z^{-n}$$

$$= \sum_{n=-\infty}^{0} \left(\frac{1}{2}\right)^n z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^{-n} z^n$$

$$= \sum_{n=0}^{\infty} (2z)^n$$

The above infinite series converges if $|2z| < 1$, i.e., $|z| < 1/2$

$$X(z) = \frac{1}{1-2z} = \frac{-0.5}{z-0.5} \quad \text{ROC}: |z| < \frac{1}{2} \qquad (10.29)$$

iv)

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n} + \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n z^{-n}$$

$$= \underbrace{\sum_{n=0}^{\infty}\left(\frac{1}{2}z^{-1}\right)^n}_{\text{converges for } |z|>\frac{1}{2}} + \underbrace{\sum_{n=0}^{\infty}\left(\frac{1}{3}z^{-1}\right)^n}_{\text{converges for } |z|>\frac{1}{3}}$$

$$X(z) = \frac{1}{1-\frac{1}{2}z^{-1}} + \frac{1}{1-\frac{1}{3}z^{-1}} \qquad \text{ROC}: |z| > \frac{1}{2} \tag{10.30}$$

Practice Problem 10.2

Find z-transform and ROC of the following signals

(i) $x(n) = [3(3)^n - 4(2)^n]u(n)$

(ii) $x(n) = [\cos \omega_0 n]u(n)$

Ans:

(i) $\dfrac{3}{1-3z^{-1}} - \dfrac{4}{1-2z^{-1}}$ ROC: $|z| > 3$

(ii) $\dfrac{1-(\cos\ \omega_0)z^{-1}}{1-(2\cos\ \omega_0)z^{-1}+z^{-2}}$ ROC: $|z| > 1$

Table 10.1 z-transform pairs.

	Sequence $x(n)$	z-Transform $X(z)$	ROC
1.	$\delta(n)$	1	All z
2.	$u(n)$	$\dfrac{1}{1-z^{-1}}$	$\|z\| > 1$
3.	$-u(-n-1)$	$\dfrac{1}{1-z^{-1}}$	$\|z\| < 1$
4.	$\delta(n-m)$	z^{-m}	All z except at $z=0$ (if $m>0$) All z except at $z=\infty$ (if $m<0$)
5.	$a^n u(n)$	$\dfrac{1}{1-az^{-1}}$	$\|z\| > \|a\|$
6.	$-a^n u(-n-1)$	$\dfrac{1}{1-az^{-1}}$	$\|z\| < \|a\|$
7.	$na^n u(n)$	$\dfrac{az^{-1}}{(1-az^{-1})^2}$	$\|z\| > \|a\|$
8.	$-na^n u(-n-1)$	$\dfrac{az^{-1}}{(1-az^{-1})^2}$	$\|z\| < \|a\|$
9.	$[\cos \omega_0 n]u(n)$	$\dfrac{1-(\cos\omega_0)z^{-1}}{1-(2\cos\omega_0)z^{-1}+z^{-2}}$	$\|z\| > 1$
10.	$[\sin \omega_0 n]u(n)$	$\dfrac{(\sin\omega_0)z^{-1}}{1-(2\cos\omega_0)z^{-1}+z^{-2}}$	$\|z\| > 1$
11.	$r^n \cos \omega_0 n u(n)$	$\dfrac{1-(r\cos\omega_0)z^{-1}}{1-2r(\cos\ \omega_0)z^{-1}+r^2z^{-2}}$	$\|z\| > \|r\|$
12.	$r^n \sin \omega_0 n u(n)$	$\dfrac{r\sin\omega_0 z^{-1}}{1-2r\cos\omega_0 z^{-1}+r^2z^{-2}}$	$\|z\| > \|r\|$
13.	$\dfrac{1}{n}, n>0$	$-\ln(1-z^{-1})$	$1 < \|z\|$
14.	$n^k a^n\ k<0$	$-\left(-z\dfrac{d}{dz}\right)^n \dfrac{1}{1-az^{-1}}$	$\|z\| < \|a\|$
15.	$a^{\|n\|}$ all n	$\dfrac{1-a^2}{(1-az)(1-az^{-1})}$	$\|a\| < \|z\| < \left\|\dfrac{1}{a}\right\|$

10.5 Properties of z-Transform

The properties of z-transform can be used to find closed-form expression for the z-transform of a given sequence. In this section, we will study such properties that make the z-transform a powerful tool for the analysis and design of discrete-time LTI systems. Many of these properties are analogous to those of the DTFT. In general, both two-sided and one-sided z-transforms have almost same properties. The properties which are applicable to one sided z-transform are specified by $X_+(z) = Z\{x(n)\}$. Let the ROC of $X(z)$ be $R_1 < |z| < R_2$.

10.5.1 Linearity

The linearity property states that the z-transform of a weighted sum of two signals is equal to the weighted sum of the individual z-transforms.

If

$$X_1(z) = Z\{x_1(n)\} \text{ and } X_2(z) = Z\{x_2(n)\},$$

then

$$Z\{ax_1(n) + bx_2(n)\} = aX_1(z) + bX_2(z) \tag{10.31}$$

Proof:

$$Z\{ax_1(n) + bx_2(n)\} = \sum_{n=-\infty}^{\infty} [ax_1(n) + bx_2(n)]z^{-n}$$

$$= a\sum_{n=-\infty}^{\infty} x_1(n)z^{-n} + b\sum_{n=-\infty}^{\infty} x_2(n)z^{-n}$$

$$= aX_1(z) + bX_2(z)$$

The ROC for the transform of a sum of sequences is the intersection of the ROCs of the individual transforms.

10.5.2 Time shifting

a) If $X(z) = Z\{x(n)\}$ and the initial conditions for $x(n)$ are zero, then

$$Z\{x(n-m)\} = z^{-m}X(z) \tag{10.32}$$

where m is a positive or a negative integer.

Proof:

$$Z\{x(n-m)\} = \sum_{n=-\infty}^{\infty} x(n-m)z^{-n}$$

Let $n-m = p$ then $n = p+m$. Now

$$Z\{x(n-m)\} = \sum_{p=-\infty}^{\infty} x(p)z^{-(m+p)}$$

$$= z^{-m}\sum_{p=-\infty}^{\infty} x(p)z^{-p} = z^{-m}X(z)$$

$$\boxed{Z\{x(n-m)\} = z^{-m}X(z)}$$

If $m > 0$ the ROC of $z^{-m}X(z)$ is the same as that of $X(z)$ except for $n = 0$.

b) If $X_+(z) = Z\{x(n)\}$. Then

$$\text{(i)} \quad Z\{x(n-m)\} = z^{-m}\left\{X_+(z) + \sum_{k=1}^{m} x(-k)z^{k}\right\} \tag{10.33}$$

$$\text{(ii)} \quad Z\{x(n+m)\} = z^{m}\left\{X_+(z) - \sum_{k=0}^{m-1} x(k)z^{-k}\right\} \tag{10.34}$$

where m is a positive integer in both cases.

(i) *Proof:*

$$Z\{x(n-m)\} = \sum_{n=0}^{\infty} x(n-m)z^{-n}$$

$$= z^{-m}\sum_{n=0}^{\infty} x(n-m)z^{-(n-m)}$$

let $n-m = l$, then

$$Z\{x(n-m)\} = z^{-m}\sum_{l=-m}^{\infty} x(l)z^{-l}$$

$$= z^{-m}\left[\sum_{l=0}^{\infty} x(l)z^{-l} + \sum_{l=-m}^{-1} x(l)z^{-l}\right]$$

Substituting $l = -k$, we obtain

$$Z\{x(n-m)\} = z^{-m}\left[X_+(z) + \sum_{k=1}^{m} x(-k)z^{k}\right]$$

(ii)

$$Z\{x(n+m)\} = \sum_{n=0}^{\infty} x(n+m)z^{-n}$$

$$= z^{m}\sum_{n=0}^{\infty} x(n+m)z^{-(n+m)}$$

let $l = n+m$, then

$$Z\{x(n+m)\} = z^{m}\sum_{l=m}^{\infty} x(l)z^{-l}$$

$$= z^{m}\left[\sum_{l=0}^{\infty} x(l)z^{-l} - \sum_{l=0}^{m-1} x(l)z^{-l}\right]$$

$$= z^{-m}\left[X_+(z) - \sum_{k=0}^{m-1} x(k)z^{-k}\right]$$

A typical application of this property is in finding the output $y(n)$ of a system described in difference equation for an input $x(n)$.

Solved Problem 10.12 Find the z-transform of the sequence

$$x(n) = a^{n-1}u(n-1)$$

Solution:

The z-transform of the sequence

$x(n) = a^n u(n)$ is given by

$$X(z) = \frac{1}{1-az^{-1}} \quad \text{ROC: } |z| > |a|$$

By using time shifting property of z-transform we have

$$Z\{x(n-k)\} = z^{-k}X(z)$$

In the same way, we have

$$Z\{a^{n-1}u(n-1)\} = \frac{z^{-1}}{1-az^{-1}} = \frac{1}{z-a} \quad \text{ROC: } |z| > |a| \qquad (10.35)$$

10.5.3 Multiplication by an exponential sequence

If $X(z) = Z\{x(n)\}$ then

$$Z[a^n x(n)] = X(a^{-1}z) \qquad (10.36)$$

Proof:

$$Z\{a^n x(n)\} = \sum_{n=-\infty}^{\infty} a^n x(n) z^{-n} = \sum_{n=-\infty}^{\infty} x(n)(a^{-1}z)^{-n}$$

$$= X(a^{-1}z)$$

The ROC is $R_1 < |a^{-1}z| < R_2$

that is $|a|R_1 < |z| < |a|R_2$

Solved Problem 10.13 Find z - transform of the sequence $x(n) = a^n \cos\frac{n\pi}{2}$

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Solution:

From table (10.1) we know

$$Z[\cos\omega_0 n\, u(n)] = \frac{1-(\cos\omega_0)z^{-1}}{1-(2\cos\omega_0)z^{-1}+z^{-2}}$$

for $\omega_0 = \frac{\pi}{2}$

$$Z\left[\cos\frac{\pi}{2}nu(n)\right] = \frac{1}{1+z^{-2}}$$

Using multiplication by exponential sequence property

$$Z[a^n x(n)] = X(a^{-1}z)$$

$$Z\left[a^n \cos\frac{n\pi}{2}u(n)\right] = \frac{1}{1+(a^{-1}z)^{-2}} = \frac{1}{1+\frac{a^2}{z^2}} = \frac{z^2}{z^2+a^2} \qquad (10.37)$$

10.5.4 Time reversal

If $X(z) = Z\{x(n)\}$, then

$$Z\{x(-n)\} = X(z^{-1}) \qquad (10.38)$$

Proof:

$$Z\{x(-n)\} = \sum_{n=-\infty}^{\infty} x(-n)z^{-n}$$

let $l = -n$, then

$$Z\{x(-n)\} = \sum_{n=-\infty}^{\infty} x(l)(z^{-1})^{-l}$$

$$= X(z^{-1})$$

The ROC is $|R_1| < |z^{-1}| < R_2$, that is $\frac{1}{R_2} < |z| < \frac{1}{R_1}$

10.5.5 Multiplication by n

If $Z\{x(n)\} = X(z)$, then

$$Z\{nx(n)\} = -z\frac{d}{dz}X(z) \qquad (10.39)$$

Proof:

We know

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$Z\{nx(n)\} = \sum_{n=-\infty}^{\infty} nx(n)z^{-n}$$

$$= z\sum_{n=-\infty}^{\infty} nx(n)z^{-n-1}$$

$$= z\sum_{n=-\infty}^{\infty} x(n)\left[nz^{-(n+1)}\right]$$

$$= z\sum_{n=-\infty}^{\infty} x(n)\left\{-\frac{d}{dz}(z^{-n})\right\}$$

$$= -z\sum_{n=-\infty}^{\infty} x(n)\left\{\frac{d}{dz}(z^{-n})\right\}$$

$$= -z\frac{d}{dz}\left\{\sum_{n=-\infty}^{\infty} x(n)z^{-n}\right\}$$

$$= -z\frac{d}{dz}X(z)$$

In the same way

$$Z\{n^k x(n)\} = \left(-z\frac{d}{dz}\right)^k X(z) \tag{10.40}$$

Solved Problem 10.14 Determine the z-transform of $x(n) = nu(n)$

Solution:

The z-transform of unit step sequence is given by

$$Z[u(n)] = \frac{z}{z-1}$$

$$Z[nu(n)] = -z\frac{d}{dz}\left(\frac{z}{z-1}\right)$$

$$= \frac{z}{(z-1)^2} \quad \text{ROC: } |z| > 1 \tag{10.41}$$

10.5.6 Convolution

One of the most important properties of z-transforms used in analysis of discrete-time system is convolution property. According to this property, the z-transform of convolution of two signals is equal to the multiplication of their z-transform.

$$Z\{x(n) * h(n)\} = X(z)H(z) \tag{10.42}$$

where $x(n) * h(n)$ denotes the linear convolution of sequences.

Proof:

Let $y(n) = x(n) * h(n)$, then we have

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

Taking z-transform on both sides we obtain

$$Y(z) = \sum_{n=-\infty}^{\infty}\left[\sum_{k=-\infty}^{\infty} x(k)h(n-k)\right]z^{-n}$$

Interchanging the order of summation results

$$Y(z) = \sum_{k=-\infty}^{\infty} x(k)z^{-k}\sum_{n=-\infty}^{\infty} h(n-k)z^{-(n-k)}$$

Replacing $(n-k)$ by l in the second summation yields

$$Y(z) = \sum_{k=-\infty}^{\infty} x(k)z^{-k}\sum_{l=-\infty}^{\infty} h(l)z^{-l}$$

$$= X(z)H(z)$$

Solved Problem 10.15 If $x(n) = x_1(n) * x_2(n)$ where $x_1(n) = \left(\frac{1}{3}\right)^n u(n)$ and $x_2(n) = \left(\frac{1}{5}\right)^n u(n)$, find $X(z)$ by using convolution property for z-transform.

Solution:

Given $x_1(n) = \left(\frac{1}{3}\right)^n u(n)$ and $x_2(n) = \left(\frac{1}{5}\right)^n u(n)$

$$X_1(z) = \frac{1}{1-\frac{1}{3}z^{-1}} \quad \text{ROC: } |z| > \frac{1}{3}$$

$$X_2(z) = \frac{1}{1-\frac{1}{5}z^{-1}} \quad \text{ROC: } |z| > \frac{1}{5}$$

$$\boxed{Z[a^n u(n)] = \frac{1}{1-az^{-1}}}$$

$$x(n) = x_1(n) * x_2(n)$$

$$X(z) = Z[x(n)] = Z[x_1(n) * x_2(n)] = X_1(z)X_2(z)$$

$$= \frac{1}{1-\frac{1}{3}z^{-1}} \frac{1}{1-\frac{1}{5}z^{-1}}$$

$$X(z) = \frac{1}{1-\frac{1}{3}z^{-1}} \cdot \frac{1}{1-\frac{1}{5}z^{-1}} \quad \text{ROC:} |z| > \frac{1}{3}$$

Solved Problem 10.16 Using z-transform, find the convolution of two sequences

$$x_1(n) = \{\ ,2,-1,0,3\}; \quad x_2(n) = \{1,2,-1\}$$

Solution:

From convolution property we have

$$Z\{x_1(n) * x_2(n)\} = X_1(z)X_2(z) \quad \text{which implies that}$$

$$x_1(n) * x_2(n) = Z^{-1}[X_1(z)X_2(z)]$$

$$X_1(z) = 1 + 2z^{-1} - z^{-2} + 3z^{-4}$$

$$X_2(z) = 1 + 2z^{-1} - z^{-2}$$

$$X_1(z)X_2(z) = (1 + 2z^{-1} - z^{-2} + 3z^{-4})(1 + 2z^{-1} - z^{-2})$$

$$= 1 + 4z^{-1} + 2z^{-2} - 4z^{-3} + 4z^{-4} + 6z^{-5} - 3z^{-6}$$

$$x(n) = \{1,4,2,-4,4,6,-3\}$$

10.5.7 Time expansion

The signal

$$x_k(n) = \begin{cases} x(n/k) & \text{if } n \text{ is a multiple of } k \\ 0 & \text{if } n \text{ is not a multiple of } k \end{cases}$$

has $k-1$ zeros inserted between successive values of the original signal.

$$\text{If} \quad Z\{x(n)\} = X(z)$$

$$\text{then} \quad Z\{x_k(n)\} = X(z^k) \tag{10.43}$$

Proof:

$$Z\{x_k(n)\} = \sum_{n=-\infty}^{\infty} x(n/k)z^{-n}$$

Let $n/k = l$, then

$$Z\{x_k(n)\} = \sum_{l=-\infty}^{\infty} x(l)z^{-kl}$$

$$= \sum_{l=-\infty}^{\infty} x(l)(z^k)^{-l} = X(z^k)$$

10.5.8 Conjugation

$$\text{If} \quad Z\{x(n)\} = X(z)$$

$$\text{then} \quad Z\{x^*(n)\} = X^*(z^*) \tag{10.44}$$

Proof:

$$Z\{x^*(n)\} = \sum_{n=-\infty}^{\infty} x^*(n)z^{-n}$$

$$= \left[\sum_{n=-\infty}^{\infty} x(n)(z^*)^{-n}\right]^*$$

$$= [X(z^*)]^* = X^*(z^*)$$

10.5.9 Complex convolution theorem

The z-transform of the product of two sequences can be related to the z-transform of the individual sequences through complex convolution theorem. This theorem states that if

$$x_3(n) = x_1(n)x_2(n)$$

then

$$X_3(z) = \frac{1}{2\pi j} \oint_c X_1(v)\, X_2\left(\frac{z}{v}\right) v^{-1} dv \tag{10.45}$$

The convergence region for $X_3(z)$ consists of all z such that if v is in the region of convergence for $X_1(z)$, then $\frac{z}{v}$ is in the region of convergence for $X_2(z)$. The contour of integration c is a closed contour inside the intersection of the convergence regions for $X_1(v)$ and $X_2\left(\frac{z}{v}\right)$.

Proof:

$$X_3(z) = \sum_{n=-\infty}^{\infty} x_3(n) z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x_1(n)\, x_2(n) z^{-n} \tag{10.46}$$

From Eq. (10.8) we have

$$x_1(n) = \frac{1}{2\pi j} \oint_c X_1(v)\, v^{n-1} dv \tag{10.47}$$

on substituting Eq. (10.47) in Eq. (10.46) yields

$$X_3(z) = \sum_{n=-\infty}^{\infty} x_2(n)\, z^{-n} \frac{1}{2\pi j} \oint_c X_1(v) v^{n-1} dv \tag{10.48}$$

Interchanging the order of summation and integration results

$$X_3(z) = \frac{1}{2\pi j} \oint_c X_1(v) \sum_{n=-\infty}^{\infty} x_2(n) \left(\frac{z}{v}\right)^{-n} v^{-1} dv$$

$$= \frac{1}{2\pi j} \oint_c X_1(v) X_2\left(\frac{z}{v}\right) v^{-1} dv \tag{10.49}$$

If c is unit circle, we can write Eq. (10.49) so that it looks more like a convolution.

Let $v = e^{j\theta}$ and $z = e^{j\omega}$,

then $dv = je^{j\theta} d\theta$

Substituting all the above relations in Eq. (10.49) we get

$$X_3(z) = \frac{1}{2\pi j} \oint_c X_1(e^{j\theta}) X_2(e^{j(\omega-\theta)}) e^{-j\theta} \cdot je^{j\theta} d\theta$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(e^{j\theta}) X_2(e^{j(\omega-\theta)}) d\theta \tag{10.50}$$

10.5.10 Parseval's relation

Let us consider two complex valued sequences $x_1(n)$ and $x_2(n)$. Parseval's relation states that

$x_2^*(n)$ is complex conjugate of $x_2(n)$

$$\sum_{n=-\infty}^{\infty} x_1(n) x_2^*(n) = \frac{1}{2\pi j} \oint_c X_1(v) X_2^*\left(\frac{1}{v^*}\right) v^{-1} dv \tag{10.51}$$

where the contour integration must be in the overlap of the region of convergence of $X_1(v)$ and $X_2^*\left(\frac{1}{v^*}\right)$.

Proof:

Let $y(n) = x_1(n) x_2^*(n)$

$$Y(z) = \sum_{n=-\infty}^{\infty} x_1(n) x_2^*(n) z^{-n} \tag{10.52}$$

Using complex convolution theorem we can write above equation as follows

$$Y(z) = \frac{1}{2\pi j} \oint_c X_1(v) X_2^*\left(\frac{z^*}{v^*}\right) v^{-1} dv \tag{10.53}$$

Evaluating Eq. (10.52) for $z = 1$ gives

$$\sum_{n=-\infty}^{\infty} x_1(n) x_2^*(n) = \frac{1}{2\pi j} \oint_c X_1(v) X_2^*\left(\frac{1}{v^*}\right) v^{-1} dv \tag{10.54}$$

If $X_1(z)$ and $X_2(z)$ both converge on the unit circle, we can choose $v = e^{j\omega}$ and Eq. (10.54) becomes

$$\sum_{n=-\infty}^{\infty} x_1(n) x_2^*(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(e^{j\omega}) X_2^*(e^{j\omega}) d\omega \tag{10.55}$$

which is parseval's relation in terms of Fourier transform. If $x_1(n) = x_2(n) = x(n)$, where $x(n)$ is a real sequence Eq. (10.54) becomes

$$\sum_{n=-\infty}^{\infty} x^2(n) = \frac{1}{2\pi j} \oint_c X(v) X(v^{-1}) v^{-1} dv \tag{10.56}$$

where c is a closed contour in the ROC of $X(z)$.

10.5.11 Correlation

If $X_1(z) = Z\{x_1(n)\}$ and $X_2(z) = Z\{x_2(n)\}$

then $$Z[r_{x_1x_2}(l)] = Z\left[\sum_{n=-\infty}^{\infty} x_1(n)x_2(n-l)\right]$$
$$= R_{x_1x_2}(z) = X_1(z)X_2(z^{-1}) \tag{10.57}$$

10.5.12 Initial value theorem

$$\text{If} \quad X_+(z) = Z\{x(n)\}, \quad \text{then}$$
$$x(0) = \lim_{z\to\infty} X_+(z) \tag{10.58}$$

Proof:

$$X_+(z) = \sum_{n=0}^{\infty} x(n)z^{-n} = x(0) + x(1)z^{-1} + x(2)z^{-2} + \cdots$$

As $z \to \infty$, all the terms vanish except $x(0)$. Therefore

$$\lim_{z\to\infty} X_+(z) = \lim_{z\to\infty} \sum_{n=0}^{\infty} x(n)z^{-n} = x(0)$$

That is, $$x(0) = \lim_{z\to\infty} X_+(z)$$

10.5.13 Final value theorem

If $X_+(z) = Z\{x(n)\}$, where the ROC for $X_+(z)$ includes, but is not necessarily confined to $|z| > 1$ and $(z-1)X_+(z)$ has no poles on or outside the unit circle, then

$$x(\infty) = \lim_{z\to 1}(z-1)X_+(z) \tag{10.59}$$

Proof:

$$Z\{x(n+1)\} - Z\{x(n)\} = \lim_{k\to\infty} \sum_{n=0}^{k} [x(n+1) - x(n)]z^{-n}$$

$$zX_+(z) - z\,x(0) - X_+(z) = \lim_{k\to\infty} \sum_{n=0}^{k} [x(n+1) - x(n)]z^{-n}$$

$$(z-1)X_+(z) - z\,x(0) = \lim_{k\to\infty} \sum_{n=0}^{k} [x(n+1) - x(n)]z^{-n}$$

Now let $z \to 1$

$$\lim_{z\to 1}(z-1)X_+(z) - x(0) = \lim_{k\to\infty}\{[x(1) - x(0)] + [x(2) - x(1)] + [x(3) - x(2)]$$
$$+ \ldots\ldots\ldots + [x(k+1) - x(k)]\}$$
$$= x(\infty) - x(0)$$

Therefore

$$x(\infty) = \lim_{z\to 1}(z-1)X_+(z)$$

or

$$x(\infty) = \lim_{z\to 1}(1 - z^{-1})X(z)$$

This theorem enables us to find the steady state value of $x(n)$ without solving for the entire sequence.

Review Questions

1. State and prove convolution theorem of z-transform. JNTU March 2006
2. State and prove scaling and time shifting properties of z-transform. JNTU May 2005
3. Derive an expression for $H(z)$ from convolution sum. JNTU Nov. 2004
4. Prove the differentiation property of z-transform. JNTU May 2004
5. State and prove initial and final value theorem of z-transform. JNTU May 2004, AU, May 2004
6. State and prove the time shifting property of z-transform Nov. 2003

Solved Problem 10.17 Determine the z-transform of the following

$$x(n) = r^n(\sin\omega_o n)u(n)$$

Solution:

From solved problem (10.8)

$$Z\{\sin\omega_o n u(n)\} = \frac{(\sin\omega_o)z^{-1}}{1 - 2(\cos\omega_o)z^{-1} + z^{-2}} \quad \text{ROC}: |z| > 1$$

From multiplication by exponential sequence property of z-transform

$$Z\{a^n x(n)\} = X(a^{-1}z)$$

Similarly

$$Z[r^n \sin(\omega_o n)u(n)] = \frac{(\sin\omega_o)(r^{-1}z)^{-1}}{1-2(\cos\omega_o)(r^{-1}z)^{-1}+(r^{-1}z)^{-2}}$$

$$= \frac{r(\sin\omega_o)z^{-1}}{1-2r(\cos\omega_o)z^{-1}+r^2z^{-2}} \quad \text{ROC: } |z|>|r| \qquad (10.60)$$

Practice Problem 10.3

Find z-transform of the following

(i) $x(n) = na^n u(n)$ (iii) $x(n) = (-1)^n \cos\left(\frac{\pi}{3}n\right)u(n)$

(ii) $x(n) = 2^n \sin\left(\frac{\pi}{4}n\right)u(n)$ (iv) $x(n) = n^2 u(n)$

Ans:

(i) $X(z) = \frac{az}{(z-a)^2}$ ROC: $|z|>a$ (iii) $X(z) = \frac{1.414z^{-1}}{1-2.828z^{-1}+4z^{-2}}$ ROC: $|z|>2$

(ii) $X(z) = \frac{z(z+0.5)}{z^2+z+1}$ ROC: $|z|>1$ (iv) $X(z) = \frac{z(z+1)}{(z-1)^3}$ ROC: $|z|>1$

Solved Problem 10.18 Determine the signal $x(n)$ whose z-transform is given by

$$X(z) = \log(1-az^{-1}); \quad |z|>|a|$$

Solution:

Given

$$X(z) = \log(1-az^{-1})$$

Differentiating on both sides, we get

$$\frac{d}{dz}X(z) = \frac{1}{1-az^{-1}}(az^{-2}) = \frac{az^{-2}}{1-az^{-1}}$$

Multiplying both sides with $-z$

$$-z\frac{d}{dz}X(z) = \frac{-az^{-1}}{1-az^{-1}}$$

$$= -az^{-1}\left[\frac{1}{1-az^{-1}}\right]$$

$$= -az^{-1}[Z\{a^n u(n)\}]$$

$$= -aZ[a^{n-1}u(n-1)] \qquad (10.61)$$

From differentiation property, we know

$$Z[nx(n)] = -z\frac{d}{dz}[X(z)] \qquad (10.62)$$

Comparing Eq. (10.61) with Eq. (10.62) we obtain

$$nx(n) = -a[a^{n-1}u(n-1)]$$

$$x(n) = \frac{-a[a^{n-1}u(n-1)]}{n} = \frac{-a^n u(n-1)}{n}$$

Practice Problem 10.4

Determine the z-transform and ROC of the following

(i) $\delta(n-1)$; (ii) $\delta(n+1)$; (iii) $\left(\frac{1}{2}\right)^n [u(n)-u(n-10)]$

Ans:

(i) z^{-1} (ii) z (iii) $\frac{1}{1-\frac{1}{2}z^{-1}} - \frac{\left(\frac{1}{2}z^{-1}\right)^{10}}{1-\frac{1}{2}z^{-1}}$

Practice Problem 10.5

Determine a sequence $x(n)$ whose z-transform is

(i) $X(z) = e^z + e^{1/z}; z \neq 0$; (ii) $X(z) = \ell n(1+z^{-1}) : |z|>0$

Ans:

(i) $\left\{\cdots \frac{1}{3!}, \frac{1}{2!}, 1, 2, 1, \frac{1}{2!}, \frac{1}{3!}, \cdots\right\}$ (ii) $\frac{-(-1)^n u(n-1)}{n}$

Practice Problem 10.6

Using convolution property find the z-transform of the following

(i) $x(n) = x_1(n) * x_2(n); x_1(n) = \{\underset{\uparrow}{1}, 2, 3, 4\}; x_2(n) = \{2, \underset{\uparrow}{-1}, 0, 3\}$.

(ii) $x_1(n) = (0.5)^n u(n)$; $x_2(n) = (0.2)^n u(n)$

Ans:

(i) $2z+3+4z^{-1}+8z^{-2}+2z^{-3}+9z^{-4}+12z^{-5}$

(ii) $\frac{5}{3}(0.5)^n u(n) - \frac{2}{3}(0.2)^n u(n)$

Solved Problem 10.19 Find $x(\infty)$ if $X(z)$ is given by

(i) $\dfrac{z+2}{(z-0.8)^2}$ (ii) $\dfrac{z+1}{3(z-1)(z+0.9)}$ (iii) $\dfrac{3z+4}{(z-1)(z+1)}$

Solution:

(a) Given

$$X(z)=\frac{z+2}{(z-0.8)^2}$$

By inspection, we can find that the ROC of $X(z)$ is $|z|>0.8$ and $(z-1)X(z)$ has no poles on or outside the unit circle. Therefore

$$x(\infty)=\lim_{z\to 1}(z-1)\frac{(z+2)}{(z-0.8)^2}=0$$

(b) Given

$$X(z)=\frac{z+1}{3(z-1)(z+0.9)}$$

$$(z-1)X(z)=\frac{z+1}{3(z+0.9)}$$

$(z-1)X(z)$ has no poles on or outside the unit circle.

$$x(\infty)=\lim_{z\to 1}(z-1)X(z)$$

$$=\lim_{z\to 1}\frac{(z+1)}{3(z+0.9)}$$

$$=\frac{2}{3(1.9)}=0.3508$$

(c) Given

$$X(z)=\frac{3z+4}{(z-1)(z+1)}$$

$$(z-1)X(z)=\frac{3z+4}{z+1}$$

$$x(\infty)=\lim_{z\to 1}(z-1)X(z)$$

$$=\lim_{z\to 1}\frac{3z+4}{z+1}$$

In this case $(z-1)X(z)$ has a pole on the unit circle, so $x(\infty)$ is not defined.

Practice Problem 10.7

(i) If $X(z)=\dfrac{0.5(1-0.5z^{-1})}{(1-0.25z^{-1})(1-0.75z^{-1})(1-z^{-1})}$

find the steady state value of $x(n)$ if it exists.

(ii) Find $x(\infty)$ if $X(z)$ is given by

$$X(z)=\frac{(2z+1)(z-10)}{z(z+2)}$$

(iii) Find the initial and final values of the function $X(z)=\dfrac{1+z^{-1}}{1-0.25z^{-2}}$

AU, Nov 2008

Ans:

(i) 1.33 (ii) $\pm\infty$

Solved Problem 10.20 Determine the z-transform of the following signal

$$x(n)=\frac{1}{2}(n^2+n)\left(\frac{1}{3}\right)^{n-1}u(n-1)$$

Solution:

Given

$$x(n)=\frac{1}{2}(n^2+n)\left(\frac{1}{3}\right)^{n-1}u(n-1)$$

$$=\frac{1}{2}n^2\left(\frac{1}{3}\right)^{n-1}u(n-1)+\frac{1}{2}n\left(\frac{1}{3}\right)^{n-1}u(n-1)$$

$$Z\left[\left(\frac{1}{3}\right)^n u(n)\right]=\frac{z}{z-\frac{1}{3}}$$

By using time-shifting property

$$Z\left[\left(\frac{1}{3}\right)^{n-1}u(n-1)\right]=\frac{1}{z-\frac{1}{3}}$$

By using multiplication property

$$Z\left[n\left(\frac{1}{3}\right)^{n-1}u(n-1)\right]=-z\frac{d}{dz}\left(\frac{1}{z-\frac{1}{3}}\right)=-z\left[\frac{-1}{\left(z-\frac{1}{3}\right)^2}\right]=\frac{z}{\left(z-\frac{1}{3}\right)^2}$$

$$Z\left[n^2\left(\frac{1}{3}\right)^{n-1}u(n-1)\right] = -z\frac{d}{dz}\left[\frac{z}{\left(z-\frac{1}{3}\right)^2}\right] = -z\left[\frac{\left(z-\frac{1}{3}\right)^2 - 2z\left(z-\frac{1}{3}\right)}{\left(z-\frac{1}{3}\right)^4}\right]$$

$$= \frac{z\left(z+\frac{1}{3}\right)}{\left(z-\frac{1}{3}\right)^3}$$

Now

$$X(z) = \frac{1}{2}\left[\frac{z\left(z+\frac{1}{3}\right)}{\left(z-\frac{1}{3}\right)^3} + \frac{z}{\left(z-\frac{1}{3}\right)^2}\right]$$

$$= \frac{z^2}{\left(z-\frac{1}{3}\right)^3} \quad \text{ROC: } |z| > \frac{1}{3}$$

Solved Problem 10.21 Express the z-transform of

$$y(n) = \sum_{k=-\infty}^{n} x(k)$$

in terms of $X(z)$.

Solution:

Given

$$y(n) = \sum_{k=-\infty}^{n} x(k)$$

$$y(n-1) = \sum_{k=-\infty}^{n-1} x(k)$$

$$y(n) - y(n-1) = \sum_{k=-\infty}^{n} x(k) - \sum_{k=-\infty}^{n-1} x(k) = x(n)$$

Taking z-transform on both sides

$$Y(z) - z^{-1}Y(z) = X(z)$$

$$\Rightarrow Y(z) = \frac{X(z)}{1-z^{-1}}$$

10.6 The Inverse z-Transform

The process of finding $x(n)$ from its z-transform $X(z)$ is called the inverse z-transform and is denoted as follows

$$x(n) = Z^{-1}[X(z)]$$

There are four methods that are often used to find the inverse z-transform.

1. Long division method.
2. Partial fraction method.
3. Residue method.
4. Convolution method.

10.6.1 Long division method

The z-transform of a sequence $x(n)$ is

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} \tag{10.63}$$

From Eq. (10.63) we find that $X(z)$ have both positive powers of z as well as negative powers of z. If the sequence $x(n)$ is causal, then

$$X(z) = \sum_{n=0}^{\infty} x(n)z^{-n} \tag{10.64}$$

have only negative powers of z, with ROC: $|z| > r$. If the sequence is non-causal, $X(z)$ will have positive powers of z with ROC: $|z| < r$.

If $X(z)$ is ratio of two polynomials

$$X(z) = \frac{N(z)}{D(z)} = \frac{b_0 + b_1z^{-1} + \cdots + b_Mz^{-M}}{1 + a_1z^{-1} + \cdots + a_Nz^{-N}} \tag{10.65}$$

we can generate a series in z by dividing the numerator of Eq. (10.65) by the denominator. For example,

If $X(z)$ converges for $|z| > r$, we obtain the series as follows

$$X(z) = x(0) + x(1)z^{-1} + x(2)x^{-2} + \cdots \tag{10.66}$$

which converges for the same range of z.[1] Eq. (10.66) is obtained by beginning the division with constant (Highest power of z). Here the $x(n)$ is in negative powers of z. If $X(z)$ converges for $|z| < r$, then $X(z)$ is to be expressed as positive powers of z. In this case, we should begin our division with lowest order[2] (highest negative) of z to obtain

[1] If $X(z)$ in Eq. (10.65) is expressed in positive powers of z, then Eq. (10.66) is obtained by beginning the division with highest power of z.

[2] If $X(z)$ in Eq. (10.65) is expressed in positive powers of z, then Eq. (10.67) is obtained by beginning the division with lowest order of z.

$$X(z) = x(0) + x(-1)z + x(-2)z^2 + \cdots \tag{10.67}$$

From Eq. (10.67) we can identify that the sequence is non-causal.

Solved Problem 10.22 Using long division, determine the inverse z-transform of

$$X(z) = \frac{1+2z^{-1}}{1-2z^{-1}+z^{-2}}$$

if (a) $x(n)$ is causal and (b) is anticausal.

Solution:

(a) Given $X(z) = \dfrac{1+2z^{-1}}{1-2z^{-1}+z^{-2}}$

$$1+4z^{-1}+7z^{-2}+10z^{-3}+13z^{-4}+16z^{-5}$$

$$\begin{array}{l|l}
1-2z^{-1}+z^{-2} & 1+2z^{-1} \\
 & 1-2z^{-1}+z^{-2} \\
\hline
 & 4z^{-1}-z^{-2} \\
 & 4z^{-1}-8z^{-2}+4z^{-3} \\
\hline
 & 7z^{-2}-4z^{-3} \\
 & 7z^{-2}-14z^{-3}+7z^{-4} \\
\hline
 & 10z^{-3}-7z^{-4} \\
 & 10z^{-3}-20z^{-4}+10z^{-5} \\
\hline
 & 13z^{-4}-10z^{-5} \\
 & 13z^{-4}-26z^{-5}+13z^{-6} \\
\hline
 & 16z^{-5}-13z^{-6} \\
 & 16z^{-5}-32z^{-6}+16z^{-7} \\
\hline
 & 19z^{-6}-16z^{-7}
\end{array}$$

The sequence $x(n)$ is causal and $X(z)$ is expressed in negative powers of z. Therefore we begin the division with a constant.

$$X(z) = 1+4z^{-1}+7z^{-2}+10z^{-3}+13z^{-4}+16z^{-5}+\cdots$$
$$x(n) = \{\underset{\uparrow}{1},4,7,10,13,16,19,\ldots\}$$

(b)

$$2z+5z^2+8z^3+11z^4+14z^5\ldots$$

$$\begin{array}{l|l}
z^{-2}-2z^{-1}+1 & 2z^{-1}+1 \\
 & 2z^{-1}-4+2z \\
\hline
 & 5-2z \\
 & 5-10z+5z^2 \\
\hline
 & 8z-5z^2 \\
 & 8z-16z^2+8z^3 \\
\hline
 & 11z^2-8z^3 \\
 & 11z^2-22z^3+11z^4 \\
\hline
 & 14z^3-11z^4 \\
 & 14z^3-28z^4+14z^5 \\
\hline
 & 17z^4-14z^5
\end{array}$$

Since $x(n)$ is non-causal we begin the division with highest negative of z.

$$X(z) = 2z+5z^2+8z^3+11z^4+14z^5+\ldots$$
$$x(n) = \{\ldots,14,11,8,5,2,\underset{\uparrow}{0}\}$$

Solved Problem 10.23 Using long division, determine the inverse z-transform of

$$X(z) = \frac{z+1}{z^2-3z+2}$$

if (a) $x(n)$ is causal and (b) is anticausal

Solution:

(a)

$$z^{-1}+4z^{-2}+10z^{-3}+22z^{-4}$$

$$\begin{array}{l|l}
z^2-3z+2 & z+1 \\
 & z-3+2z^{-1} \\
\hline
 & 4-2z^{-1} \\
 & 4-12z^{-1}+8z^{-2} \\
\hline
 & 10z^{-1}-8z^{-2} \\
 & 10z^{-1}-30z^{-2}+20z^{-3} \\
\hline
 & 22z^{-2}-20z^{-3} \\
 & 22z^{-2}-66z^{-3}+44z^{-4} \\
\hline
 & 46z^{-3}-44z^{-4}
\end{array}$$

$$X(z) = z^{-1} + 4z^{-2} + 10z^{-3} + 22z^{-4} + \ldots$$
$$x(n) = \{0, 1, 4, 10, 22, \ldots\}$$
$$\uparrow$$

(b)

$$2 - 3z + z^2 \;\Big|\; 0.5 + 1.25z + 1.625z^2 + 1.8125z^3 \ldots$$

$$1 + z$$
$$1 - 1.5z + 0.5z^2$$
$$2.5z - 0.5z^2$$
$$2.5z - 3.75z^2 + 1.25z^3$$
$$3.25z^2 - 1.25z^3$$
$$3.25z^2 - 4.875z^3 + 1.625z^2$$
$$3.625z^3 - 1.625z^2$$
$$\vdots$$

$$X(z) = 0.5 + 1.25z + 1.625z^2 + 1.8125z^3 \ldots$$
$$x(n) = \{\ldots, 1.8125, 1.625, 1.25, 0.5\}$$
$$\uparrow$$

Solved Problem 10.24 Find the first four terms of causal signal whose z-transform is $X(z) = \dfrac{4z^{-1}}{2 - 2z^{-1} + z^{-2}}$

Solution:

$$2 - 2z^{-1} + z^{-2} \;\Big|\; 2z^{-1} + 2z^{-2} + z^{-3} - \tfrac{1}{2}z^{-5} - \tfrac{1}{2}z^{-6}$$

$$4z^{-1}$$
$$4z^{-1} - 4z^{-2} + 2z^{-3}$$
$$4z^{-2} - 2z^{-3}$$
$$4z^{-2} - 4z^{-3} + 2z^{-4}$$
$$2z^{-3} - 2z^{-4}$$
$$2z^{-3} - 2z^{-4} + z^{-5}$$
$$-z^{-5}$$
$$-z^{-5} + z^{-6} - \tfrac{1}{2}z^{-7}$$
$$-z^{-6} + \tfrac{1}{2}z^{-7}$$
$$-z^{-6} + z^{-7} - z^{-8}$$
$$-\tfrac{1}{2}z^{-7} + z^{-8}$$

$$x(n) = \{0, 2, 2, 1, 0, -\frac{1}{2}, -\frac{1}{2} \ldots\}$$

Practice Problem 10.8

Find the inverse z-transform of $X(z)$ using power series method, given $X(z) = \dfrac{1}{1 - az^{-1}}; |z| > a.$

JNTU March 2004

Hint:

$$H(z) = \frac{1}{1 - az^{-1}} = (1 - az^{-1})^{-1} = 1 + az^{-1} + a^2z^{-2} + \cdots$$

$$(\because\ (1-x)^{-1} = 1 + x + x^2 + \cdots)$$

$$x(n) = \{1, a, a^2 \cdots\} = a^n u(n) \quad \text{since ROC:} |z| > a$$

Practice Problem 10.9

Given $X(z) = \dfrac{1}{1 - az^{-1}}$ $|z| < a$ find $x(n)$ using long division method

JNTU May 2005, March 2006

Hint:

$$H(z) = \frac{1}{1 - az^{-1}} = (1 - az^{-1})^{-1} = 1 + az^{-1} + a^2z^{-2} + \cdots$$

$$x(n) = \{1, a, a^2 \cdots\}$$
$$= -a^n u(-n-1) \quad \text{since ROC:} |z| < a$$

Practice Problem 10.10

Find the inverse z-transform of the following using long division method.

(i) $X(z) = \dfrac{1}{1 + 3z^{-1} + 2z^{-2}}$ ROC : $|z| > 2$

(ii) $X(z) = \dfrac{z+1}{z^2 + 5z + 4}$ ROC : $|z| < 1$

(iii) $X(z) = \dfrac{1}{1 - 4z^{-1} + 6z^{-2}}$

JNTU Nov 2005

Ans:

(iii) $\{1, 4, 10, 16, 4, -80, \cdots\}$

10.6.2 Partial fraction expansion method

Consider a rational function $X(z)$ given by

$$X(z) = \frac{N(z)}{D(z)} = \frac{b_0 + b_1 z^{-1} + \cdots + b_M z^{-M}}{1 + a_1 z^{-1} + \cdots + a_N z^{-N}} \tag{10.68}$$

If $M < N$ and $a_N \neq 0$, the rational function $X(z)$ is said to be proper. On the other hand if $M \geq N$ it is called improper. An improper rational function $X(z)$ can always be written as the sum of a polynomial and a proper rational function. That is

$$X(z) = \frac{N(z)}{D(z)} = \underbrace{C_0 + C_1 z^{-1} + \cdots + C_{M-N} z^{-(M-N)}}_{\text{Polynomial}} + \underbrace{\frac{N_1(z)}{D(z)}}_{\substack{\text{proper} \\ \text{rational} \\ \text{function}}} \tag{10.69}$$

The inverse z-transform of polynomial $X(z)$ as shown in Eq.(10.69) can be easily found. Therefore, let us focus our attention on the inversion of proper rational function.

Let us consider a rational function of the form of Eq. (10.68). Multiply numerator and denominator by z^N to eliminate negative powers of z. This results in

$$X(z) = \frac{b_0 z^N + b_1 z^{N-1} + \cdots + b_M z^{N-M}}{z^N + a_1 z^{N-1} + \cdots + a_N} \tag{10.70}$$

Divide Eq. (10.70) by z to get

$$\frac{X(z)}{z} = \frac{b_0 z^{N-1} + b_1 z^{N-2} + \cdots + b_M z^{N-M-1}}{z^N + a_1 z^{N-1} + \cdots + a_N} \tag{10.71}$$

$$= \frac{b_0 z^{N-1} + b_1 z^{N-2} + \cdots + b_M z^{N-M-1}}{(z-p_1)(z-p_2)\cdots(z-p_N)} \tag{10.72}$$

For distinct poles the Eq. (10.72) can be expanded in the form

$$\frac{X(z)}{z} = \frac{c_1}{z-p_1} + \frac{c_2}{z-p_2} + \cdots\cdots + \frac{c_N}{z-p_N} \tag{10.73}$$

The coefficients $c_1, c_2, \ldots, c_N$ can be determined using the formula

$$c_k = (z-p_k)\frac{X(z)}{z}\bigg|_{z=p_k} \qquad k = 1,2,\ldots,N \tag{10.74}$$

If $X(z)$ has a pole of multiplicity l, then denominator contains the factor $(z-p_k)^l$, and the expansion of the form of Eq. (10.74) is no longer valid. Let us assume $l = 2$ and

$$\frac{X(z)}{z} = \frac{1}{(z-p_1)(z-p_2)^2} \tag{10.75}$$

The Eq. (10.75) can be expressed in partial fraction form as

$$\frac{X(z)}{z} = \frac{c_1}{z-p_1} + \frac{c_2}{z-p_2} + \frac{c_3}{(z-p_2)^2} \tag{10.76}$$

where

$$c_1 = (z-p_1)\frac{X(z)}{z}\bigg|_{z=p_1}$$

$$c_2 = \frac{d}{dz}\left\{(z-p_2)^2\frac{X(z)}{z}\right\}\bigg|_{z=p_2} \quad \text{and}$$

$$c_3 = (z-p_2)^2\,\frac{X(z)}{z}\bigg|_{z=p_2} \tag{10.77}$$

To consider a general case of l-repeated roots, let

$$\frac{X(z)}{z} = \frac{R(z)}{(z-p_k)^l}$$

$$= \frac{c_{1k}}{z-p_k} + \frac{c_{2k}}{(z-p_k)^2} + \cdots + \frac{c_{ik}}{(z-p_k)^i} + \cdots + \frac{c_{lk}}{(z-p_k)^l}\cdots \tag{10.78}$$

where i is any term in the partial fraction expansion and $R(z)$ is defined as

$$R(z) = \frac{X(z)}{z}(z-p_k)^l \tag{10.79}$$

Multiplying Eq. (10.78) by $(z-p_k)^l$ yields

$$R(z) = c_{1k}(z-p_k)^{l-1} + c_{2k}(z-p_k)^{l-2} + \cdots + c_{l-1k}(z-p_k) + c_{lk} \tag{10.80}$$

For this equation 10.80 we can find a method to evaluate the coefficients. If $z = p_k$ all the terms in Eq. (10.80) disappear except c_{lk} which can be evaluated. Next, differentiate the equation once with respect to z. The term c_{lk} will vanish but c_{l-1k} will remain without a multiplying function z. Again c_{l-1k} can be evaluated by letting $z = p_k$. To find general term c_{ik}, differentiate Eq. (10.80) $(l-i)$ times and

let $z = p_k$; then

$$c_{ik} = \frac{1}{(l-i)!}\frac{d^{l-i}}{dz^{l-i}}R(z)\bigg|_{z=p_k}$$

or

$$c_{ik} = \frac{1}{(l-i)!}\frac{d^{l-i}}{dz^{l-i}}\left[(z-p_k)^l\frac{X(z)}{z}\right]\bigg|_{z=p_k} \tag{10.81}$$

If $X(z)$ has a complex pole then the partial fraction can be expressed as follows

$$\frac{X(z)}{z} = \frac{c_1}{z-p_1} + \frac{c_1^*}{z-p_1^*}$$

where c_1^* is a complex conjugate of c_1 and p_1^* is a complex conjugate of p_1.

In otherwords, complex conjugate poles result in complex conjugate coefficients in the partial-fraction expansion.

Table 10.2 Inverse z-transforms used for partial fraction expansion method.

S.No.	$X(z)$	$x(n) = Z^{-1}[X(z)]$ for ROC: $\lvert z\rvert > \lvert a\rvert$
1.	$\frac{z}{z-a}$	$a^n u(n)$
2.	$\frac{z}{(z-a)^2}$	$na^{n-1}u(n)$
3.	$\frac{z}{(z-a)^3}$	$\frac{n(n-1)a^{n-2}u(n)}{2!}$
4.	$\frac{z}{(z-a)^4}$	$\frac{n(n-1)(n-2)}{3!}a^{n-3}u(n)$
⋮	⋮	⋮
k	$\frac{z}{(z-a)^k}$	$\frac{n(n-1)\ldots(n-(k-2))a^{n-k+1}}{(k-1)!}u(n)$

Solved Problem 10.25 Find the inverse of z-transform of the following

i) $X(z) = \dfrac{\frac{1}{4}z^{-1}}{\left(1-\frac{1}{2}z^{-1}\right)\left(1-\frac{1}{4}z^{-1}\right)}$, ROC: $|z| > \frac{1}{2}$.

ii) $X(z) = \dfrac{z^2}{(1-az)(z-a)}$ ROC: $a < |z| < \frac{1}{a}$.

iii) $X(z) = \dfrac{1}{1+z^{-1}+z^{-2}}$ ROC: $|z| > 1$.

iv) $X(z) = \dfrac{z(z-1)}{(z+2)^3(z+1)}$ ROC: $|z| > 2$.

v) $X(z) = \dfrac{8z-9}{z^2-5z+6}$ ROC: $|z| > 3$.

Solution:

i) Given

$$X(z) = \frac{\frac{1}{4}z^{-1}}{\left(1-\frac{1}{2}z^{-1}\right)\left(1-\frac{1}{4}z^{-1}\right)};\ |z| > \frac{1}{2}.$$

On multiplying numerator and denominator with z^2, we get

$$X(z) = \frac{\frac{1}{4}z}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{4}\right)}$$

The above equation can be expressed in partial fraction as follows

$$\frac{X(z)}{z} = \frac{c_1}{z-\frac{1}{2}} + \frac{c_2}{z-\frac{1}{4}}$$

where c_1 and c_2 can be evaluated using Eq. (10.74).

That is

$$c_1 = \left(z-\frac{1}{2}\right)\frac{\frac{1}{4}}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{4}\right)}\bigg|_{z=\frac{1}{2}} = 1$$

$$c_2 = \left(z-\frac{1}{4}\right)\frac{\frac{1}{4}}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{4}\right)}\bigg|_{z=\frac{1}{4}} = -1$$

$$\frac{X(z)}{z} = \frac{1}{z-\frac{1}{2}} - \frac{1}{z-\frac{1}{4}}$$

$$X(z) = \frac{z}{z-\frac{1}{2}} - \frac{z}{z-\frac{1}{4}} \quad \text{ROC: } |z| > \frac{1}{2}$$

Since ROC is $|z| > \frac{1}{2}$, the sequence is causal and using table (10.2) we can write

$$x(n) = \left(\frac{1}{2}\right)^n u(n) - \left(\frac{1}{4}\right)^n u(n) \qquad (10.82)$$

ii) Given

$$X(z) = \frac{z^2}{(1-az)(z-a)}$$

Dividing $X(z)$ by z, we get

$$\frac{X(z)}{z} = \frac{z}{(1-az)(z-a)} = \frac{-z}{a\left(z-\frac{1}{a}\right)(z-a)}$$
$$= \frac{-z/a}{\left(z-\frac{1}{a}\right)(z-a)}$$
$$= \frac{c_1}{z-\frac{1}{a}} + \frac{c_2}{z-a}$$

c_1 and c_2 can be evaluated using Eq. (10.74)

$$c_1 = \left(z-\frac{1}{a}\right)\frac{-z/a}{\left(z-\frac{1}{a}\right)(z-a)}\bigg|_{z=\frac{1}{a}} = \frac{-1/a^2}{\frac{1}{a}-a} = \frac{-\frac{1}{a^2}}{\frac{1-a^2}{a}} = -\frac{1}{a(1-a^2)}$$

$$c_2 = (z-a)\frac{-z/a}{(z-a)\left(z-\frac{1}{a}\right)}\bigg|_{z=a} = \frac{-1}{a-\frac{1}{a}} = \frac{a}{1-a^2}$$

$$\frac{X(z)}{z} = \frac{-1}{a(1-a^2)}\cdot\frac{1}{z-\frac{1}{a}} + \frac{a}{1-a^2}\cdot\frac{1}{z-a}$$

$$X(z) = \frac{-1}{a(1-a^2)}\frac{z}{z-\frac{1}{a}} + \frac{a}{1-a^2}\frac{z}{z-a}$$

The ROC is $a < |z| < \frac{1}{a}$, the pole $z = \frac{1}{a}$ provides the noncausal term and the pole $z = a$ provides the causal term.

Therefore

$$x(n) = \frac{1}{a(1-a^2)}\left(\frac{1}{a}\right)^n u(-n-1) + \frac{a}{1-a^2}(a)^n u(n) \qquad (10.83)$$

iii) Given

$$X(z) = \frac{1}{1+z^{-1}+z^{-2}}$$

Multiply numerator and denominator by z^2.

$$X(z) = \frac{z^2}{z^2+z+1}$$

$$\frac{X(z)}{z} = \frac{z}{z^2+z+1}$$

The poles of $X(z)$ are complex conjugate given by -0.5 ± 0.866

$$= \frac{z}{(z+0.5+j0.866)(z+0.5-j0.866)}$$

If the poles are complex conjugate, then $\frac{X(z)}{z}$ can be split in to partial fraction as

$$\frac{X(z)}{z} = \frac{c}{z-(-0.5+j0.866)} + \frac{c^*}{z-(-0.5-j0.866)}$$

$$c = \frac{z}{[z-(-0.5-j0.866)]}\bigg|_{z=-0.5+j0.866}$$
$$= \frac{-0.5+j0.866}{j1.732} = 0.5+j0.288$$

Similarly

$$c^* = 0.5-j0.288$$

$$\frac{X(z)}{z} = \frac{0.5+j0.288}{z+0.5-j0.866} + \frac{0.5-j0.288}{z+0.5+j0.866}$$

$$X(z) = \frac{(0.5+j0.288)z}{z+0.5-j0.866} + \frac{(0.5-j0.288)z}{z+0.5+j0.866}$$

On taking inverse z-transform on both sides, we have

$$x(n) = (0.5+j0.288)(-0.5+j0.866)^n u(n) + (0.5-j0.288)(-0.5-j0.866)^n u(n)$$
$$= (0.5+j0.288)\left[\cos\frac{2\pi}{3} + j\sin\frac{2\pi}{3}\right]^n u(n) + (0.5-j0.288)\left[\cos\frac{2\pi}{3} - j\sin\frac{2\pi}{3}\right]^n u(n)$$
$$= (0.5+j0.288)\left(\cos\frac{2n\pi}{3} + j\sin\frac{2n\pi}{3}\right)u(n) + (0.5-j0.288)\left(\cos\frac{2n\pi}{3} - j\sin\frac{2n\pi}{3}\right)u(n)$$
$$= \left[\cos\frac{2n\pi}{3} - 0.576\sin\frac{2n\pi}{3}\right]u(n)$$

$$= 1.15\left[0.866\cos\frac{2n\pi}{3} - 0.5\sin\frac{2n\pi}{3}\right]$$

$$= 1.15\left[\cos\left(\frac{2n\pi}{3}+\frac{\pi}{6}\right)\right]u(n) \qquad (10.84)$$

iv) $X(z) = \dfrac{z(z-1)}{(z+2)^3(z+1)}$

On dividing $X(z)$ by z, we obtain

$$\frac{X(z)}{z} = \frac{z-1}{(z+2)^3(z+1)}$$

$$= \frac{c_1}{z+2} + \frac{c_2}{(z+2)^2} + \frac{c_3}{(z+2)^3} + \frac{c_4}{z+1}$$

c_1, c_2 and c_3 can be obtained using Eq. (10.81)

$$c_1 = \frac{1}{2!}\frac{d^2}{dz^2}\left[\cancel{(z+2)^3}\,\frac{(z-1)}{\cancel{(z+2)^3}\,(z+1)}\right]\Bigg|_{z=-2}$$

$$= \frac{1}{2}\left[\frac{-4}{(z+1)^3}\right]\Bigg|_{z=-2} = 2$$

$$c_2 = \frac{1}{1!}\frac{d}{dz}\left[\cancel{(z+2)^3}\,\frac{(z-1)}{\cancel{(z+2)^3}\,(z+1)}\right]\Bigg|_{z=-2}$$

$$= \frac{2}{(z+1)^2}\Bigg|_{z=-2} = 2$$

$$c_3 = \cancel{(z+2)^3}\,\frac{z-1}{\cancel{(z+2)^3}\,(z+1)}\Bigg|_{z=-2} = 3$$

$$c_4 = \cancel{(z+1)}\,\frac{(z-1)}{(z+2)^3\,\cancel{(z+1)}}\Bigg|_{z=-1} = -2$$

$$X(z) = \frac{2z}{z+2} + \frac{2z}{(z+2)^2} + \frac{3z}{(z+2)^3} - \frac{2z}{z+1}$$

By using table 10.2, we can write the inverse z- transform as

$$x(n) = 2(-2)^n u(n) + 2n(-2)^{n-1}u(n) + 3\frac{n(n-1)(-2)^{n-2}u(n)}{2!} - 2(-1)^n u(n)$$

$$= 2(-2)^n u(n) - n(-2)^n u(n) + \frac{3}{8}n(n-1)(-2)^n u(n) - 2(-1)^n u(n)$$

$$= \left[2 - n + \frac{3}{8}n(n-1)\right](-2)^n u(n) - 2(-1)^n u(n)$$

v) $X(z) = \dfrac{8z-19}{z^2-5z+6}$

$$\frac{X(z)}{z} = \frac{8z-19}{z(z^2-5z+6)}$$

$$= \frac{c_1}{z} + \frac{c_2}{z-2} + \frac{c_3}{z-3}$$

$$= \frac{-19}{6z} + \frac{3}{2(z-2)} + \frac{5}{3(z-3)}$$

$$X(z) = \frac{-19}{6} + \frac{3z}{2(z-2)} + \frac{5z}{3(z-3)}$$

$$c_1 = \cancel{z}\,\frac{8z-19}{\cancel{z}(z^2-5z+6)}\Bigg|_{z=0} = \frac{-19}{6}$$

$$c_2 = \cancel{(z-2)}\,\frac{8z-19}{z\,\cancel{(z-2)}\,(z-3)}\Bigg|_{z=2} = \frac{3}{2}$$

$$c_3 = \cancel{(z-3)}\,\frac{8z-19}{z(z-2)\,\cancel{(z-3)}}\Bigg|_{z=3} = \frac{5}{3}$$

Taking inverse z-transform

$$x(n) = \frac{-19}{6}\,\delta(n) + \frac{3}{2}(2)^n u(n) + \frac{5}{3}(3)^n u(n) \qquad (10.85)$$

Practice Problem 10.11

Find the inverse z-transform of the function

$$X(z) = \frac{1}{1+2z^{-1}+2z^{-2}}$$

Ans:

$$2^{(n+1)/2}\left[\cos(3n+1)\frac{\pi}{4}\right]$$

Practice Problem 10.12

Determine the causal signal $x(n)$ having z-transform

$$X(z) = \frac{1}{(1-2z^{-1})(1-z^{-1})^2}$$

Ans: $x(n) = 4(2)^n u(n) - 3u(n) - nu(n)$

Practice Problem 10.13

Find the inverse z-transform of the function

$$X(z) = \frac{z-4}{z^2-5z+6}$$

Ans:

$$\frac{-2}{3}\,\delta(n) + \left[2^n - \frac{1}{3}(3)^n\right]u(n)$$

Table 10.2 z-transform properties.

Property	Sequence	z-transform	ROC
	$x(n)$	$X(z)$	$R: R_1 < \|z\| < R_2$
	$x_1(n)$	$X_1(z)$	R_1
	$x_2(n)$	$X_2(z)$	R_2
Linearity	$a_1x_1(n)+a_2x_2(n)$	$a_1X_1(z)+a_2X_2(z)$	Intersection of R_1 and R_2
Time shifting	$x(n-m)$	$z^{-m}X(z)$	Same as $X(z)$ except $z=0$ if $m>0$ and $z=\infty$ if $m<0$
Multiplication by exponential sequence	$a^n x(n)$	$X(a^{-1}z)$	$\|a\|R_1 < \|z\| < \|a\|R_2$
Time reversal	$x(-n)$	$X(z^{-1})$	$\frac{1}{R_2} < \|z\| < \frac{1}{R_1}$
Differentiation in the z-domain	$nx(n)$	$-z\frac{d}{dz}X(z)$	$R_1 < \|z\| < R_2$
Conjugation	$x^*(n)$	$X^*(z^*)$	$R_1 < \|z\| < R_2$
Convolution	$x_1(n) * x_2(n)$	$X_1(z)X_2(z)$	At least the intersection of R_1 and R_2
Correlation	$x_1(l) * x_2(-l)$	$X_1(z)X_2(z^{-1})$	At least the intersection of ROC of $X_1(z)$ and $X_2(z^{-1})$
Multiplication	$x_1(n)x_2(n)$	$\frac{1}{2\pi j}\oint_c X_1(v)X_2\left(\frac{z}{v}\right)v^{-1}dv$	At least $R_{1l}R_{2l} < \|z\| < R_{1u}R_{2u}$
Initial value theorem	$x(0) = \lim_{z\to\infty} X(z)$		
Final value theorem	$x(\infty) = \lim_{z\to 1}(1-z^{-1})X(z)$, if $(1-z^{-1})X(z)$ has no pole on or outside the unit circle.		
Parseval's relation	$\sum_{n=-\infty}^{\infty} x_1(n)x_2^*(n) = \frac{1}{2\pi j}\oint_c X_1(v)X_2^*\left(\frac{1}{v^*}\right)v^{-1}dv$		

Solved Problem 10.26 Determine all possible signals $x(n)$ associated with z-transform

$$X(z) = \frac{5z^{-1}}{(1-2z^{-1})(1-3z^{-1})}$$

Solution:

Given

$$X(z) = \frac{5z^{-1}}{(1-2z^{-1})(1-3z^{-1})}$$

Multiplying numerator and denominator with z^2, we obtain

$$X(z) = \frac{5z}{(z-2)(z-3)}$$

$$\frac{X(z)}{z} = \frac{5}{(z-2)(z-3)} = \frac{A}{z-2} + \frac{B}{z-3}$$

$$A = \cancel{(z-2)}\left.\frac{5}{\cancel{(z-2)}(z-3)}\right|_{z=2} = -5$$

$$B = \cancel{(z-3)}\left.\frac{5}{(z-2)\cancel{(z-3)}}\right|_{z=3} = 5$$

$$X(z) = \frac{-5z}{z-2} + \frac{5z}{z-3} \tag{10.86}$$

The pole-zero plot of $X(z)$ is shown in Fig. 10.7.

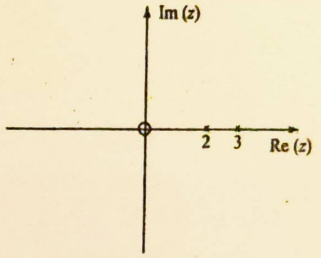


Fig. 10.7

For the ROC: $|z| > 3$ shown in Fig. 10.8, the signal $x(n)$ is causal and all terms in Eq. (10.86) are causal terms.

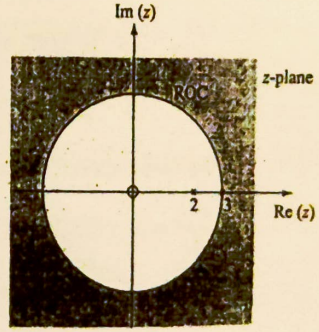


Fig. 10.8

$$x(n) = -5(2)^n u(n) + 5(3)^n u(n)$$

For ROC: $|z| < 2$, shown in Fig. 10.9, the signal $x(n)$ is anticausal and all terms in Eq (10.86) are anticausal terms.

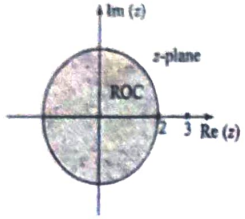


Fig. 10.9

Therefore

$$x(n) = 5(2)^n u(-n-1) - 5(3)^n u(-n-1)$$

For ROC $2 < |z| < 3$, shown in Fig. 10.10 the signal $x(n)$ is two sided. The pole $z = 2$ provides causal term and the pole $z = 3$ provides the noncausal term. Therefore

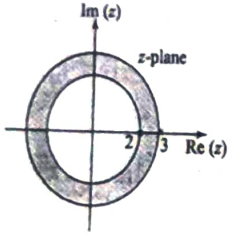


Fig. 10.10

$$x(n) = -5(2)^n u(n) - 5(3)^n u(-n-1)$$

Practice Problem 10.14

Find the inverse z-transform of $X(z) = \dfrac{z}{(z+2)(z-3)}$ when

ROC is (i) $|z| < 2$ (ii) ROC: $2 < |z| < 3$ JNTU Nov. 2 '04

Ans:

(i) $\frac{1}{5}[(-3)^n - (-2)^n]u(-n-1)$ (ii) $\frac{1}{5}[(-3)^n u(-n-1) - (-2)^n u(n)]$

Practice Problem 10.15

Determine all possible signals $x(n)$ associated with z-transform

(i) $X(z) = \dfrac{1}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}$

(ii) $X(z) = \dfrac{z(z^2 - 4z + 5)}{(z-1)(z-2)(z-3)}$

Ans:

(i) $1.15\left(\frac{1}{2}\right)^n \cos\left(\frac{n\pi}{3} - 29.54^\circ\right)u(n)$ ROC: $|z| > 0.5$

$-1.15\left(\frac{1}{2}\right)^n \cos\left(\frac{n\pi}{3} - 29.54^\circ\right)u(-n-1)$ ROC: $|z| < 0.5$

(ii) $-(2)^n u(n) + u(n) - (3)^n u(-n-1)$ ROC: $2 < |z| < 3$

$u(n) - (2)^n u(n) + (3)^n u(n)$ ROC: $|z| > 3$

$[-1 + 2^n - (3)^n]u(-n-1)$ ROC: $|z| < 1$

10.6.3 Residue method

The inverse z-transform of $X(z)$ can be obtained using Eq. (10.8) that is

$$x(n) = \frac{1}{2\pi j}\oint_c X(z)z^{n-1}dz \tag{10.87}$$

where c is a circle in the z-plane in the ROC of $X(z)$.

The Eq. (10.87) can be evaluated by finding sum of all residues of the poles that are inside the circle c. Therefore Eq. (10.87) can be rewritten as

$$x(n) = \sum (\text{residues of } X(z)z^{n-1} \text{ at the poles inside } c)$$

$$= \sum_i (z - z_i)X(z)z^{n-1}\Big|_{z=z_i} \tag{10.88}$$

If $X(z)z^{n-1}$ has no poles inside the contour c for one or more values of n, then $x(n) = 0$ for these values. The residues of the poles can be found by using cauchy residue theorem.

Cauchy residue theorem

Let $f(z)$ be a function of the complex variable z and c be a closed path in the z-plane. If the derivative $\frac{d}{dz}f(z)$ exists on and inside the contour c and if $X(z)$ has no poles at $z = z_0$, then

$$\frac{1}{2\pi j}\oint_c \frac{f(z)}{z-z_0}dz = f(z_0) \text{ if } z_0 \text{ is inside } c$$

$$= 0 \text{ if } z_0 \text{ is outside } c \qquad (10.89)$$

If the $(k+1)^{\text{th}}$ order derivative of $f(z)$ exists and $f(z)$ has no poles at $z = z_0$, then

$$\frac{1}{2\pi j}\oint_c \frac{f(z)}{(z-z_0)^k}dz = \frac{1}{(k-1)!}\frac{d^{k-1}}{dz^{k-1}}f(z)\bigg|_{z=z_0} \text{ if } z_0 \text{ is inside } c \qquad (10.90)$$

$$= 0 \text{ if } z_0 \text{ is outside } c$$

The values on the right hand side of Eq. (10.89) and Eq. (10.90) are called residues of the pole at $z = z_0$. If there are more than one pole inside the contour the integral can then be evaluated by adding up the results from each pole. For example, if there are n unique poles $z_1, z_2, \ldots, z_n$ and $f(z)$ containing no poles inside c, then

$$\frac{1}{2\pi j}\oint_c \frac{f(z)dz}{(z-z_1)(z-z_2)\ldots(z-z_n)} = \sum_{i=1}^{n}\lim_{z\to z_i}[(z-z_i)\,\psi(z)] \qquad (10.91)$$

where

$$\psi(z) = \frac{f(z)}{(z-z_1)(z-z_2)\ldots(z-z_n)} \qquad (10.92)$$

If any of poles are of multiplicity greater than one, the formula given in Eq. (10.90) is used to determine their contribution to the integral, while Eq. (10.89) is used for simple poles.

Solved Problem 10.27 Using residue method find the inverse z-transform of

$$X(z) = \frac{1+3z^{-1}}{1+3z^{-1}+2z^{-2}} \quad |z| > 2$$

Solution:

Given

$$X(z) = \frac{1+3z^{-1}}{1+3z^{-1}+2z^{-2}} = \frac{z(z+3)}{z^2+3z+2} = \frac{z(z+3)}{(z+1)(z+2)}$$

From Eq. (10.87), we know that

$$x(n) = \frac{1}{2\pi j}\oint_c X(z)z^{n-1}dz$$

$$= \sum \text{residues of } X(z)z^{n-1} \text{ at poles of } X(z)z^{n-1} \text{ with in } c$$

$$= \sum \text{residues of } \frac{z(z+3)}{(z+1)(z+2)}z^{n-1} \text{ at poles of same with in } c$$

The closed contour c, being in the ROC $|z| > 2$ encloses the poles at $z = -1$ and $z = -2$

$$x(n) = \sum \text{ residues of } \frac{(z+3)z^n}{(z+1)(z+2)} \text{ at poles } z = -1 \text{ and } z = -2$$

$$= \text{residue of } \frac{(z+3)z^n}{(z+1)(z+2)} \text{ at } z = -1 + \text{residue of } \frac{(z+3)z^n}{(z+1)(z+2)} \text{ at } z = -2$$

$$= \cancel{(z+1)}\,\frac{(z+3)z^n}{\cancel{(z+1)}(z+2)}\bigg|_{z=-1} + \cancel{(z+2)}\,\frac{(z+3)z^n}{(z+1)\cancel{(z+2)}}\bigg|_{z=-2}$$

$$= 2(-1)^n - (-2)^n$$

For $n < 0$ we can prove $x(n) = 0$. For $n = -1$

$$X(z)z^{n-1} = \frac{z+3}{z(z+1)(z+2)}$$

$$x(-1) = \sum \text{residues of } \frac{z+3}{z(z+1)(z+2)} \text{ at } z = 0, -1 \text{ and } -2$$

$$= \text{residue of } \frac{z+3}{z(z+1)(z+2)} \text{ at } z=0 + \text{residue of } \frac{z+3}{z(z+1)(z+2)} \text{ at } z=-1$$

$$+ \text{residue of } \frac{z+3}{z(z+1)(z+2)} \text{ at } z = -2$$

$$= \cancel{z}\,\frac{z+3}{\cancel{z}(z+1)(z+2)}\bigg|_{z=0} + \cancel{(z+1)}\,\frac{z+3}{z\,\cancel{(z+1)}(z+2)}\bigg|_{z=-1}$$

$$+ \cancel{(z+2)}\,\frac{z+3}{z(z+1)\,\cancel{(z+2)}}\bigg|_{z=-2}$$

$$= \frac{3}{2} - 2 + \frac{1}{2} = 0.$$

Similarly we can prove that $x(n) = 0$ for other negative values of n.

Therefore for $n<0$ $\quad x(n)=0$

$$x(n)=2(-1)^n u(n)-(-2)^n u(n)$$

Solved Problem 10.28 Determine the inverse z-transform using the complex integral

(a) $X(z)=\dfrac{2z^{-1}}{\left(1-\frac{1}{4}z^{-1}\right)^2}$ ROC: $|z|>\dfrac{1}{4}$

(b) $X(z)=\dfrac{z^{-1}}{(1-2z^{-1})(1-3z^{-1})}$ ROC: $2<|z|<3$

(c) $X(z)=\dfrac{z}{(z-1)^3}$ JNTU May 2005, Nov 2005

(d) $X(z)=\dfrac{z^2}{(z-a)^2}$ JNTU Nov 2005, AU 2002

Solution:

(a) Given $X(z)=\dfrac{2z^{-1}}{\left(1-\frac{1}{4}z^{-1}\right)^2}$

$$=\frac{2z}{\left(z-\frac{1}{4}\right)^2}$$

Using complex integral $x(n)=\dfrac{1}{2\pi j}\oint_c X(z)z^{n-1}dz$

For $n\geq 0$

$$x(n)=\frac{1}{2\pi j}\oint_c \frac{2z}{\left(z-\frac{1}{4}\right)^2}\cdot z^{n-1}dz$$

$$=\frac{1}{2\pi j}\oint_c \frac{2z^n}{\left(z-\frac{1}{4}\right)^2}dz$$

$$=\text{residue of }\frac{2z^n}{\left(z-\frac{1}{4}\right)^2}\text{ at the pole }z=\frac{1}{4}\text{ of multiplicity 2}$$

Therefore using Eq. (10.90) we have

$$x(n)=\frac{1}{1!}\frac{d}{dz}\left[\left(z-\frac{1}{4}\right)^2\frac{2z^n}{\left(z-\frac{1}{4}\right)^2}\right]\Bigg|_{z=\frac{1}{4}}$$

$$=2n(z)^{n-1}\Big|_{z=\frac{1}{4}}$$

$$=2n\left(\frac{1}{4}\right)^{n-1}$$

For $n=-1$

$$x(-1)=\frac{1}{2\pi j}\oint_c \frac{1}{z\left(z-\frac{1}{4}\right)^2}dz$$

$$=\cancel{1}\cdot\frac{1}{\cancel{z}\left(z-\frac{1}{4}\right)^2}\Bigg|_{z=0}+\frac{1}{1!}\frac{d}{dz}\left[\cancel{\left(z-\frac{1}{4}\right)^2}\cdot\frac{1}{z\cancel{\left(z-\frac{1}{4}\right)^2}}\right]\Bigg|_{z=\frac{1}{4}}$$

$$=16-16\quad=0$$

i.e., for $n<0$ $\quad x(n)=0$

Therefore

$$x(n)=2n\left(\frac{1}{4}\right)^{n-1}u(n)$$

(b) Given

$$X(z)=\frac{z^{-1}}{(1-2z^{-1})(1-3z^{-1})}\quad \text{ROC}: 2<|z|<3$$

$$=\frac{z}{(z-2)(z-3)}$$

The contour of integration c lies in the annular region of ROC, and the inverse z- transform is

$$x(n)=-\sum\text{residue of }X(z)z^{n-1}\text{ at pole }z=3\text{ for }n<0$$

$$=\sum\text{residue of }X(z)z^{n-1}\text{ at pole }z=2\text{ for }n\geq 0$$

For $n<0$

$$x(n)=-(z-3)\frac{z^n}{(z-2)(z-3)}\Bigg|_{z=3}$$

$$=-(3)^n$$

For $n \geq 0$

$$x(n) = \left.\frac{(z-2)}{(z-2)(z-3)}\frac{z^n}{}\right|_{z=2}$$
$$= -(2)^n$$

Therefore

$$x(n) = -(2)^n u(n) - (3)^n u(-n-1)$$

(c) $$x(n) = \frac{1}{2\pi j}\oint_c X(z)z^{n-1}dz$$

For $n \geq 0$ $x(n) = \frac{1}{2\pi j}\oint_c \frac{z}{(z-1)^3}z^{n-1}dz$

= residue of $\frac{z^n}{(z-1)^3}$ at the pole $z = 1$ of multiplicity 3

$$x(n) = \frac{1}{2!}\frac{d^2}{dz^2}(z-1)^3\left.\frac{z^n}{(z-1)^3}\right|_{z=1}$$
$$= \frac{1}{2}n(n-1)z^{n-2}\Big|_{z=1}$$
$$= \frac{n(n-1)}{2}$$

For $n = -1$

$$x(-1) = \frac{1}{2\pi j}\oint \frac{z}{(z-1)^3}z^{-2}dz$$

= residue of $\frac{1}{z(z-1)^3}$ at the pole $z = 1$ of multiplicity 3 + residu

$\frac{1}{z(z-1)^3}$ at the pole $z = 0$

$$= \frac{1}{2!}\frac{d^2}{dz^2}\left[(z-1)^3\frac{1}{z(z-1)^3}\right]\Big|_{z=1} + z\left(\frac{1}{z(z-1)^3}\right)\Big|_{z=0} = 1-1 =$$

Hence $x(n) = \frac{n(n-1)}{2}u(n)$

(d)

$$x(n) = \frac{1}{2\pi j}\oint_c X(z)z^{n-1}dz$$

For $n \geq 0$ $x(n) = \frac{1}{2\pi j}\oint_c \frac{z^2}{(z-a)^2}z^{n-1}dz$

$$= \frac{1}{1!}\frac{d}{dz}\left[(z-a)^2\frac{z^{n+1}}{(z-a)^2}\right]\Big|_{z=a}$$
$$= (n+1)z^n|_{z=a}$$
$$= (n+1)a^n$$

We can prove

For $n < 0$ $x(n) = 0$

$\Rightarrow$ $x(n) = (n+1)a^n u(n)$

Practice Problem 10.16

Find the inverse z-transform of the following using Residue method

(i) $X_1(z) = \frac{z^2+z}{(z-1)(z-3)}$ ROC: $|z| > 3$

(ii) $X_2(z) = \frac{1-\frac{1}{4}z^{-1}}{1-\frac{1}{9}z^{-1}}$ ROC: $|z| > \frac{1}{3}$

(iii) $X_3(z) = \frac{z^2}{\left(z-\frac{1}{4}\right)^2}$ ROC: $|z| > \frac{1}{4}$

Ans:

(i) $2(3)^n u(n) - u(n)$

(ii) $\frac{1}{8}\left(\frac{1}{3}\right)^n + \frac{7}{8}\left(-\frac{1}{3}\right)^n u(n).$

(iii) $(n+1)\left(\frac{1}{4}\right)^n u(n)$

10.6.4 Convolution method

In this method, the given $X(z)$ is splitted into $X_1(z)$and $X_2(z)$ such that $X(z) = X_1(z)X_2(z)$. Next we find $x_1(n)$ and $x_2(n)$ by taking inverse z- transform of $X_1(z)$ and $X_2(z)$ respectively.

From convolution property of z- transform we know

$$Z[x_1(n) * x_2(n)] = X_1(z)X_2(z) = X(z) \qquad (10.93)$$

From Eq. (10.93) we know that the convolution of $x_1(n)$and $x_2(n)$ is the inverse z- transform of $X(z)$. Thus $x(n)$ can be obtained by convolving $x_1(n)$and $x_2(n)$.

Solved Problem 10.29 Repeat the Solved problem 10.27 using convolution method.

Solution:

Given $X(z) = \frac{1+3z^{-1}}{1+3z^{-1}+2z^{-2}}$; $|z| > 2$

$$= \frac{z(z+3)}{(z+1)(z+2)}$$

let $X(z) = X_1(z)X_2(z)$

where $X_1(z) = \frac{z}{z+1}$

$$X_2(z) = \frac{z+3}{z+2} = \frac{z}{z+2} + z^{-1}\frac{3z}{z+2}$$

Then $x_1(n) = (-1)^n u(n)$ and

$$x_2(n) = (-2)^n u(n) + 3(-2)^{n-1}u(n-1)$$

$$x(n) = x_1(n) * x_2(n)$$

$$= (-1)^n u(n) * [(-2)^n u(n) + 3(-2)^{n-1}u(n-1)]$$

$$= (-1)^n u(n) * (-2)^n u(n) + 3(-1)^n u(n) * (-2)^{n-1}u(n-1)$$

$$= \sum_{k=0}^{n}(-1)^k u(k)(-2)^{n-k}u(n-k) + 3\sum_{k=0}^{n-1}(-1)^k u(k)(-2)^{n-k-1}u(n-k-1)$$

$$= (-2)^n \sum_{k=0}^{n}(-1)^k(-2)^{-k} + 3(-2)^{n-1}\sum_{k=0}^{n-1}(-1)^k(-2)^{-k}$$

$$= (-2)^n \sum_{k=0}^{n}(0.5)^k + 3(-2)^{n-1}\sum_{k=0}^{n-1}(0.5)^k \qquad \boxed{\sum_{k=0}^{n} a^k = \frac{1-a^{n+1}}{1-a}}$$

$$= (-2)^n\left[\frac{1-(0.5)^{n+1}}{1-0.5}\right] + 3(-2)^{n-1}\left[\frac{1-(0.5)^n}{1-0.5}\right]$$

$$= (-2)^n \cdot 2[1-0.5(0.5)^n] + 3(-2)^{n-1}\cdot 2[1-(0.5)^n]$$

$$= (-2)^n 2[1-0.5(0.5)^n] - 3(-2)^n[1-(0.5)^n]$$

$$= 2(-2)^n - (-1)^n - 3(-2)^n + 3(-1)^n$$

$$= 2(-1)^n - (-2)^n \quad \text{for } n \geq 0$$

$$x(n) = [2(-1)^n - (-2)^n]u(n)$$

Solved Problem 10.30 Use convolution theorem, find the inverse z - transform of $Y(z) = \frac{z}{(z-1)^3}$ JNTU Nov. 2005

Solution:

$$Y(z) = \frac{z}{(z-1)^3}$$

$$= Y_1(z)\,Y_2(z)$$

$$= \frac{z}{z-1}\,\frac{1}{(z-1)^2}$$

where $Y_1(z) = \frac{z}{z-1}$ and $Y_2(z) = \frac{1}{(z-1)^2}$

$$y_1(n) = u(n) \text{ and}$$

$$y_2(n) = (n-1)u(n-1)$$

$$y(n) = y_1(n) * y_2(n)$$

$$= \sum_{k=-\infty}^{\infty}(k-1)u(k-1)u(n-k)$$

We know

$$(k-1)u(k-1) = 0 \text{ for } k < 1$$

and $u(n-k) = 1$ for $k \leq n$

$$\Rightarrow \quad y(n) = \sum_{k=1}^{n} = \frac{n(n-1)}{2}u(n)$$

Practice Problem 10.17

Find the z- transform of the following using convolution method

(i) $X_1(z) = \frac{z^2}{(z-2)(z-4)}$

(ii) $X_2(z) = \frac{1+3z^{-1}}{(1-0.5z^{-1})(1-0.4z^{-1})}$

(iii) $X_3(z) = \frac{1}{1-3z^{-1}+2z^{-2}}$

Ans:

(i) $-(2)^n u(n) + 2(4)^n u(n)$

(ii) $35(0.5)^n u(n) - 34(0.4)^n u(n)$

(iii) $2(2)^n u(n) - u(n)$

Review Questions

1. What are the methods by which inverse z-transforms can be found out? JNTU May 2005, March 2006
2. Explain the residual method of obtaining inverse z-transform.
3. Explain the convolution method of obtaining inverse z-transform.

10.7 The System Function

Consider an Nth order linear time invariant discrete-time system specified by the difference equation

$$\sum_{k=0}^{N} a_k y(n-k) = \sum_{k=0}^{M} b_k x(n-k) \tag{10.94}$$

Expanding Eq. (10.94) yields

$$a_0 y(n) + a_1 y(n-1) + a_2 y(n-2) + \ldots + a_N y(n-N)$$
$$= b_0 x(n) + b_1 x(n-1) + \ldots + b_M x(n-M)$$

Taking z - transform on both sides and applying zero initial conditions, we obtain

$$a_0 Y(z) + a_1 z^{-1} Y(z) + a_2 z^{-2} Y(z) + \ldots + a_N z^{-N} Y(z)$$
$$= b_0 X(z) + b_1 z^{-1} X(z) + \ldots + b_M z^{-M} X(z)$$
$$Y(z)\left[a_0 + a_1 z^{-1} + a_2 z^{-2} + \ldots + a_N z^{-N}\right]$$
$$= X(z)\left[b_0 + b_1 z^{-1} + b_2 z^{-2} + \ldots + b_M z^{-M}\right]$$
$$\frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \ldots + b_M z^{-M}}{a_0 + a_1 z^{-1} + a_2 z^{-2} + \ldots + a_N z^{-N}}$$
$$= \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

$$\boxed{H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}} \tag{10.95}$$

Where $H(z)$ is known as transfer function or system function. $Y(z)$ is the z-transform of the output sequence $y(n)$ and $X(z)$ is the z-transform of the input sequence $x(n)$. Therefore system function can be defined as the ratio of z-transform of output to z-transform of input assuming zero initial conditions.

If the input $x(n)$ is an impulse sequence then $X(z)$ is equal to 1. The Eq. (10.95) can be written as $Y(z) = H(z)$. That is, the system function $H(z)$ is the z-transform of the impulse response $h(n)$.

The poles and zeros of the system function offer an insight into the system characteristics. The poles of the system are defined as the values of z for which the system function $H(z)$ is infinity and the zeros of the system are the values of z for which the system function $H(z)$ is zero. The system function of Eq. (10.95) can be expressed in the pole - zero form as shown below.

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_M z^{-M}}{a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_N z^{N}}$$
$$= \frac{b_0\left[1 + b_1' z^{-1} + b_2' z^{-2} + \cdots + b_M' z^{-M}\right]}{a_0\left[1 + a_1' z^{-1} + a_2' z^{-2} + \cdots + a_N' z^{-N}\right]} \tag{10.96}$$

Factorizing the numerator and denominator polynomials, we get

$$H(z) = \frac{K \prod_{k=1}^{N} (1 - c_k z^{-1})}{\prod_{k=1}^{N} (1 - d_k z^{-1})} \tag{10.97}$$

where the c_k and d_k are the poles and zeros of the system and

$$K = \frac{b_0}{a_0} \tag{10.98}$$

Equivalently Eq. (10.97) can also be expressed as follows

$$H(z) = \frac{K z^M \prod_{k=1}^{M} (z - c_k)}{z^N \prod_{k=1}^{N} (z - p_k)}$$
$$= \frac{K \prod_{k=1}^{M} (z - c_k)}{z^{N-M} \prod_{k=1}^{N} (z - p_k)} \tag{10.99}$$

Solved Problem 10.31 Plot the pole-zero pattern and determine which of the following systems are stable.

(i) $y(n) = y(n-1) - 0.5y(n-2) + x(n) + x(n-1)$

(ii) $y(n) = 1.8y(n-1) - 0.72y(n-2) + x(n) + 0.5x(n-1)$

Solution:

Given

$$y(n) = y(n-1) - 0.5y(n-2) + x(n) + x(n-1)$$

Taking z-transform on both sides and assuming zero initial conditions, we get

$$Y(z) = z^{-1}Y(z) - 0.5z^{-2}Y(z) + X(z) + z^{-1}X(z)$$

$$Y(z)\left[1 - z^{-1} + 0.5z^{-2}\right] = X(z)\left(1 + z^{-1}\right)$$

$$\Rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{1+z^{-1}}{1-z^{-1}+0.5z^{-2}}$$

$$= \frac{z(z+1)}{z^2 - z + 0.5}$$

$$= \frac{z(z+1)}{(z-0.5-j0.5)(z-0.5+j0.5)}$$

when initial conditions are zero
$Z\{y(n-1)\} = z^{-1}Y(z)$
$Z\{y(n-2)\} = z^{-2}Y(z)$
$Z\{x(n-1)\} = z^{-1}X(z)$

The zeros of $H(z)$ are $0, -1$.

The poles of $H(z)$ are $0.5 + j0.5, 0.5 - j0.5$.

The plot of pole-zero pattern is shown in Fig. 10.11.

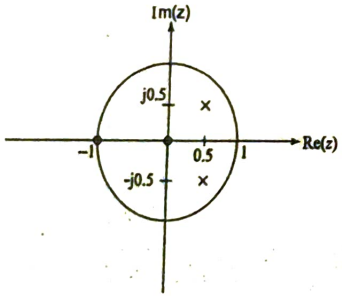


Fig. 10.11

All the poles are inside the unit circle. Therefore the system is stable.

(ii) $y(n) = 1.8y(n-1) - 0.72y(n-2) + x(n) + 0.5x(n-1)$

Applying z-transform on both sides and assuming zero initial condition yields

$$Y(z) = 1.8z^{-1}Y(z) - 0.72z^{-2}Y(z) + X(z) + 0.5z^{-1}X(z)$$

$$Y(z)\left[1 - 1.8z^{-1} + 0.72z^{-2}\right] = X(z)\left[1 + 0.5z^{-1}\right]$$

$$\Rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{1+0.5z^{-1}}{1-1.8z^{-1}+0.72z^{-2}}$$

$$= \frac{z(z+0.5)}{z^2 - 1.8z + 0.72}$$

$$= \frac{z(z+0.5)}{(z-1.2)(z-0.6)}$$

The zeros of $H(z)$ are $0, -0.5$. The poles of $H(z)$ are 1.2 and 0.6. The plot of pole-zero pattern is shown in Fig. 10.12. A pole is outside the unit circle. Therefore system is unstable.

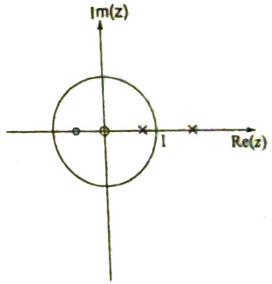


Fig. 10.12

Practice Problem 10.18

Find the system function and plot the pole zero pattern for the following systems

(i) $y(n) - 2y(n-1) + 3y(n-2) = x(n) + x(n-1)$

(ii) $y(n) + 6y(n-1) + 12y(n-2) = x(n-1) + 2x(n-2)$

Ans (i) $H(z) = \dfrac{1+z^{-1}}{1-2z^{-1}+3z^{-2}}$ (ii) $H(z) = \dfrac{z^{-1}+2z^{-2}}{1+6z^{-1}+12z^{-2}}$

(iii) Obtain the poles and zero of the system $y(n) = ay(n-1) + x(n)$

AU Nov 2008

(iv) Draw the pole - zero plot for

$$y(n) - 2y(n-1) + 2y(n-2) = x(n) + \frac{1}{2}x(n-1)$$ AU Nov 2008

10.8 Relationship between z-Transform and DTFT

In section (10.3.1), we studied that the Fourier transform exists for a discrete-time sequence $x(n)$ if and only if the sequence is absolutely summable, that is, the sequence has to satisfy the condition

$$\sum_{n=-\infty}^{\infty} |x(n)| < \infty. \tag{10.100}$$

The DTFT does not exists for the sequences that are growing exponentially (Ex: $a^n u(n), a > 1$), since they are not absolutely summable. Therefore DTFT method of analysing a system can be applied for a limited class of signals. Moreover, this method can be applied only to asymptotically stable systems and it cannot be used for unstable or even marginally stable systems. That is, the DTFT can be used for the systems whose system function $H(z)$ have poles inside the unit circle.

Let us now see the relationship between the transfer function $H(e^{j\omega})$ and the system function $H(z)$ of a system. This relationship helps us in evaluating the Fourier transform in the z-plane.

We know the z-transform of the impulse response $h(n)$ is given by

$$H(z) = \sum_{n=-\infty}^{\infty} h(n) z^{-n}, \tag{10.101}$$

$$\text{where} \quad z = re^{j\omega} \tag{10.102}$$

Substituting Eq. (10.102) in Eq. (10.101) we obtain

$$H(re^{j\omega}) = \sum_{n=-\infty}^{\infty} h(n) r^n e^{-j\omega n} \tag{10.103}$$

The Fourier transform of $h(n)$ is given by

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h(n) e^{-j\omega n} \tag{10.104}$$

The Eq. (10.104) and Eq. (10.103) are identical when $r = 1$.

In z-plane this corresponds to the locus of points on the unit circle $|z| = 1$. Hence $H(e^{j\omega})$ is equal to $H(z)$ when evaluated along the unit circle.

That is

$$H(e^{j\omega}) = H(z)\big|_{z=e^{j\omega}} \tag{10.105}$$

For $H(e^{j\omega})$ to exist, the ROC of $X(z)$ must include the unit circle. Since ROC cannot contain any poles of $H(z)$, all the poles must lie inside the unit circle.

Therefore we can conclude that Fourier transform can be obtained for any sequence $x(n)$, from its z- transform $X(z)$ if the poles of $X(z)$ are inside the unit circle.

Solved Problem 10.32 Find the DTFT for the following sequences

(i) $X(z) = \dfrac{z}{(z-0.5)(z+0.7)}$

(ii) $X(z) = \dfrac{z}{(z+1.2)(z+0.7)}$

Solution:

(i) Given $X(z) = \dfrac{z}{(z-0.5)(z+0.7)}$

The poles of the system function are $z_1 = 0.5$ and $z_2 = -0.7$. Both poles are inside the unit circle. Therefore

$$H(e^{j\omega}) = H(z)\big|_{z=e^{j\omega}}$$

$$= \frac{e^{j\omega}}{(e^{j\omega}-0.5)(e^{j\omega}+0.7)}$$

(ii) Given $X(z) = \dfrac{z}{(z+1.2)(z+0.7)}$

The poles of $X(z)$ are $z = -1.2$ and $z_2 = -0.7$. One poles is out side the unit circle. Therefore $X(e^{j\omega})$ does not exist.

Review Questions

1. Define system function.
2. What is the relation between Fourier transform and z-transform?
3. What is the condition on $h(n)$ to get $H(e^{j\omega})$ from $H(z)$?

Practice Problem 10.19

Find DTFT of the following sequences

1. $X(z) = \dfrac{z}{(z-0.2)(z-0.9)}$
2. $X(z) = \dfrac{z^{-1}}{1-4z^{-1}+3z^{-2}}$

10.9 Stability Criterion

In this section we study the relationship between the stability of a system and the location of poles in z-plane. From section (3.13) we know that the necessary and sufficient condition for a causal linear time-invariant system to be BIBO stable is

$$\sum_{n=0}^{\infty} h(n) < \infty \tag{10.106}$$

The system function of a causal LTI system can be obtained using Eq. (10.101)

$$H(z) = \sum_{n=0}^{\infty} h(n) z^{-n} \tag{10.107}$$

The magnitude of z-transform

$$|H(z)| = \sum_{n=0}^{\infty} |h(n) z^{-n}| \leq \sum_{n=0}^{\infty} |h(n)||z^{-n}| \tag{10.108}$$

The evaluation of $|H(z)|$ on unit circle yields

$$|H(z)| \leq \sum_{n=0}^{\infty} |h(n)| < \infty \quad (\because \quad |z| = 1 \quad \text{for unit circle}) \tag{10.109}$$

Therefore for a stable system the ROC of a system function includes unit circle.

For a causal system the ROC is exterior of the circle of radius r. Consequently, a causal and stable system must have a system function that converges for $|z| > r < 1$. Since ROC cannot contain any pole of $H(z)$; a causal linear time invariant system is BIBO stable if and only if all the poles of $H(z)$ are inside the unit circle. In Fig. 10.13 and Fig. 10.14 the pole locations of the system function of a causal LTI system and corresponding impulse responses are shown. From the figure 10.13 we can find the relationship between the stability of a system and the location of poles.

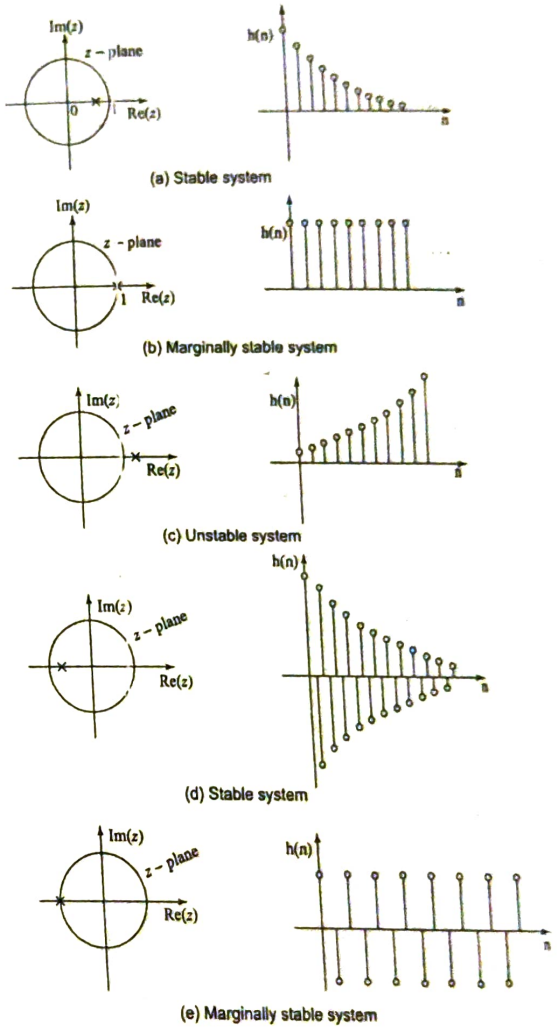


Fig. 10.13

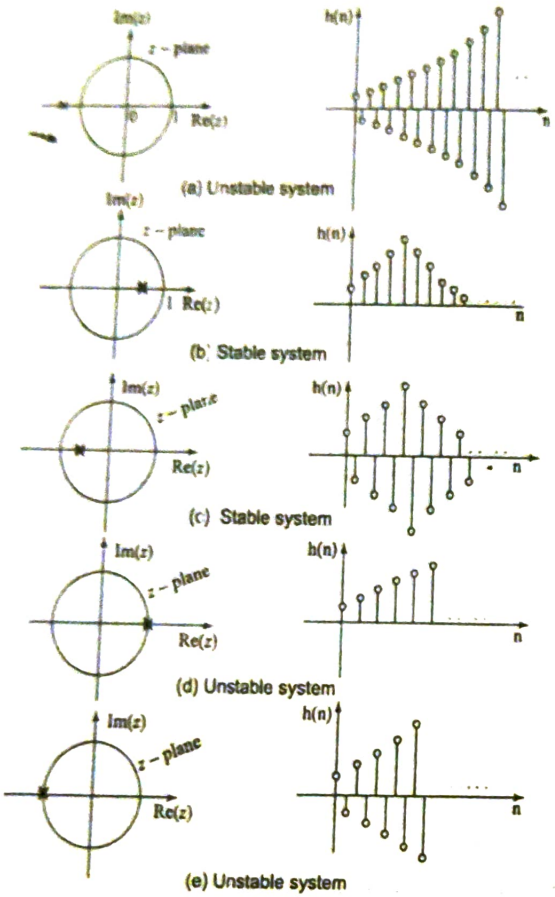


Fig. 10.14

Review Questions

1. Prove that for a stable system ROC of a system function includes unit circle.
2. What is the condition on pole-locations for stability?

10.10 Solution of Difference Equations using z-Transform

In section 4.1, we studied the classical method of solving difference equations. In this section we will study about the solution of linear difference equations using z-transform. As Laplace transform converts the differential equation, into an algebraic equation, the z-transform converts the difference equation into algebraic equation in z-domain. By applying inverse z-transform to z-domain algebraic solution, time domain solution can be obtained.

Solved Problem 10.33 Find the impulse response and step response for the following systems.

a) $y(n)-\frac{3}{4}y(n-1)+\frac{1}{8}y(n-2)=x(n)$

b) $y(n)=x(n)+2x(n-1)-4x(n-2)+x(n-3)$

Solution:

Given

$$y(n)-\frac{3}{4}y(n-1)+\frac{1}{8}y(n-2)=x(n)$$

Taking z-transform on both sides, we get

$$Y(z)-\frac{3}{4}\left[z^{-1}Y(z)+y(-1)\right]+\frac{1}{8}\left[z^{-2}Y(z)+z^{-1}y(-1)+y(-2)\right]=X(z)$$

Substituting initial conditions $y(-1)=y(-2)=0$ yields

$$Y(z)-\frac{3}{4}z^{-1}Y(z)+\frac{1}{8}z^{-2}Y(z)=X(z)$$

$$\Rightarrow \frac{Y(z)}{X(z)}=\frac{1}{1-\frac{3}{4}z^{-1}+\frac{1}{8}z^{-2}}$$

Impulse response for $x(n)=\delta(n)$; $X(z)=1$

$$\Rightarrow Y(z)=\frac{1}{1-\frac{3}{4}z^{-1}+\frac{1}{8}z^{-2}}=\frac{z^2}{z^2-\frac{3}{4}z+\frac{1}{8}}$$

$$\frac{Y(z)}{z}=\frac{z}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{4}\right)}=\frac{A}{z-\frac{1}{2}}+\frac{B}{z-\frac{1}{4}}$$

$$A=\left(z-\frac{1}{2}\right)\frac{z}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{4}\right)}\Bigg|_{z=\frac{1}{2}}=\frac{\left(\frac{1}{2}\right)}{\frac{1}{2}-\frac{1}{4}}=2$$

$$B=\left(z-\frac{1}{4}\right)\frac{z}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{4}\right)}\Bigg|_{z=\frac{1}{4}}=\frac{\frac{1}{4}}{\left(\frac{1}{4}-\frac{1}{2}\right)}=-1$$

$$\Rightarrow \frac{Y(z)}{z}=\frac{2}{z-\frac{1}{2}}-\frac{1}{z-\frac{1}{4}}$$

$$Y(z)=2\frac{z}{z-\frac{1}{2}}-\frac{z}{z-\frac{1}{4}}$$

Taking inverse z-transform we get $y(n)=2\left(\frac{1}{2}\right)^n u(n)-\left(\frac{1}{4}\right)^n u(n)$

Step response For a unit step input $x(n)=u(n);\ X(z)=\frac{z}{z-1}$

$$\frac{Y(z)}{X(z)}=\frac{1}{1-\frac{3}{4}z^{-1}+\frac{1}{8}z^{-2}}$$

$$Y(z)=\frac{z}{z-1}\quad\frac{z^2}{z^2-\frac{3}{4}z+\frac{1}{8}}$$

$$\frac{Y(z)}{z}=\frac{z^2}{(z-1)\left(z-\frac{1}{2}\right)\left(z-\frac{1}{4}\right)}$$

$$=\frac{A}{z-1}+\frac{B}{z-\frac{1}{2}}+\frac{C}{z-\frac{1}{4}}$$

$$A=\cancel{(z-1)}\frac{z^2}{\cancel{(z-1)}\left(z-\frac{1}{2}\right)\left(z-\frac{1}{4}\right)}\Bigg|_{z=1}$$

$$=\frac{(1)^2}{\left(1-\frac{1}{2}\right)\left(1-\frac{1}{4}\right)}=\frac{1}{\left(\frac{1}{2}\right)\left(\frac{3}{4}\right)}=\frac{8}{3}$$

$$B=\left(z-\frac{1}{2}\right)\frac{z^2}{(z-1)\left(z-\frac{1}{2}\right)\left(z-\frac{1}{4}\right)}\Bigg|_{z=\frac{1}{2}}$$

$$=\frac{\left(\frac{1}{2}\right)^2}{\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-\frac{1}{4}\right)}=\frac{\frac{1}{4}}{\left(-\frac{1}{2}\right)\left(\frac{1}{4}\right)}=-2$$

$$C=\left(z-\frac{1}{4}\right)\frac{z^2}{(z-1)\left(z-\frac{1}{2}\right)\left(z-\frac{1}{4}\right)}\Bigg|_{z=\frac{1}{4}}$$

$$=\frac{\left(\frac{1}{4}\right)^2}{\left(\frac{1}{4}-1\right)\left(\frac{1}{4}-\frac{1}{2}\right)}=\frac{\frac{1}{16}}{\left(\frac{-3}{4}\right)\left(\frac{-1}{4}\right)}=\frac{1}{3}$$

$$\frac{Y(z)}{z}=\frac{8}{3(z-1)}-\frac{2}{z-\frac{1}{2}}+\frac{1}{3\left(z-\frac{1}{4}\right)}$$

$$Y(z)=\frac{8}{3}\cdot\frac{z}{z-1}-2\frac{z}{z-\frac{1}{2}}+\frac{1}{3}\quad\frac{z}{z-\frac{1}{4}}$$

Taking inverse $-z$ transform yields

$$y(n)=\frac{8}{3}u(n)-2\left(\frac{1}{2}\right)^n u(n)+\frac{1}{3}\left(\frac{1}{4}\right)^n u(n)$$

ii) $y(n)=x(n)+2x(n-1)-4x(n-2)+x(n-3)$

Taking z-transform on both sides

$$Y(z)=X(z)+2z^{-1}X(z)-4z^{-2}X(z)+z^{-3}X(z)$$

$$\Rightarrow \frac{Y(z)}{X(z)}=1+2z^{-1}-4z^{-2}+z^{-3}$$

For an impulse $\delta(n)$ the z-transform is equal to 1. Therefore

$$Y(z)=1+2z^{-1}-4z^{-2}+z^{-3}$$

$$y(n)=\{1,2,-4,1\}$$
$$\uparrow$$

Step response

For $x(n)=u(n);X(z)=\frac{1}{1-z^{-1}}$

Now

$$Y(z)=\frac{1}{1-z^{-1}}+2\frac{z^{-1}}{1-z^{-1}}-\frac{4z^{-2}}{1-z^{-1}}+\frac{z^{-3}}{1-z^{-1}}$$

Using time shifting property of z-transform, we get

$$y(n)=u(n)+2u(n-1)-4u(n-2)+u(n-3)$$

Solved Problem 10.34 Find the output $y(n)$ of a linear time invariant discrete time system specified by the equation

$$y(n)-\frac{3}{2}y(n-1)+\frac{1}{2}y(n-2)=2x(n)+\frac{3}{2}x(n-1) \qquad (10.110)$$

When the initial condition are $y(-1)=0, y(-2)=1$ and the input $x(n)=\left(\frac{1}{4}\right)^n u(n)$.

Solution:

Given

$$y(n)-\frac{3}{2}y(n-1)+\frac{1}{2}y(n-2)=2x(n)+\frac{3}{2}x(n-1)$$

Taking z- transform on both sides, we get

$$Y(z)-\frac{3}{2}[z^{-1}Y(z)+y(-1)]+\frac{1}{2}[z^{-2}Y(z)+z^{-1}y(-1)+y(-2)]$$
$$=2X(z)+\frac{3}{2}[z^{-1}X(z)+x(-1)] \qquad (10.111)$$

We have

$$y(-1)=0;\quad y(-2)=1 \quad \text{and} \quad x(-1)=0$$

Substituting above values in Eq. (10.111), we get

$$Y(z)-\frac{3}{2}z^{-1}Y(z)+\frac{1}{2}[z^{-2}Y(z)+1]=X(z)\left[2+\frac{3}{2}z^{-1}\right]$$

For input $x(n)=\left(\frac{1}{4}\right)^n u(n);\ X(z)=\frac{1}{1-\frac{1}{4}z^{-1}}$

$$\Rightarrow Y(z)\left[1-\frac{3}{2}z^{-1}+\frac{1}{2}z^{-2}\right]=-\frac{1}{2}+\frac{1}{1-\frac{1}{4}z^{-1}}\left(2+\frac{3}{2}z^{-1}\right)$$

$$Y(z)=\frac{-1}{2\left(1-\frac{3}{2}z^{-1}+\frac{1}{2}z^{-2}\right)}+\frac{\left(2+\frac{3}{2}z^{-1}\right)}{\left(1-\frac{1}{4}z^{-1}\right)\left(1-\frac{3}{2}z^{-1}+\frac{1}{2}z^{-2}\right)}$$

$$=\frac{-z^2}{2\left(z^2-\frac{3}{2}z+\frac{1}{2}\right)}+\frac{z^2\left(2z+\frac{3}{2}\right)}{\left(z-\frac{1}{4}\right)\left(z^2-\frac{3}{2}z+\frac{1}{2}\right)}$$

$$=\frac{-z^2}{2(z-1)\left(z-\frac{1}{2}\right)}+\frac{z^2\left(2z+\frac{3}{2}\right)}{\left(z-\frac{1}{4}\right)(z-1)\left(z-\frac{1}{2}\right)}$$

$$=Y_1(z)\ +\ Y_2(z)$$

$$Y_1(z)=\frac{-z^2}{2(z-1)\left(z-\frac{1}{2}\right)}$$

$$\frac{Y_1(z)}{z}=\frac{-z}{2(z-1)\left(z-\frac{1}{2}\right)}$$

$$=\frac{A}{z-1}+\frac{B}{z-\frac{1}{2}}$$

$$=\frac{-1}{z-1}+\frac{1}{2\left(z-\frac{1}{2}\right)}$$

$$Y_1(z)=\frac{-z}{z-1}+\frac{z}{2\left(z-\frac{1}{2}\right)}$$

$$y_1(n)=-u(n)+0.5\left(\frac{1}{2}\right)^n u(n)$$

$$Y_2(z)=\frac{z^2\left(2z+\frac{3}{2}\right)}{\left(z-\frac{1}{4}\right)(z-1)\left(z-\frac{1}{2}\right)}$$

$$\frac{Y_2(z)}{z}=\frac{z\left(2z+\frac{3}{2}\right)}{\left(z-\frac{1}{4}\right)(z-1)\left(z-\frac{1}{2}\right)}$$

$$=\frac{A_1}{z-\frac{1}{4}}+\frac{B_1}{z-1}+\frac{C_1}{z-\frac{1}{2}}$$

$$=\frac{8}{3\left(z-\frac{1}{4}\right)}+\frac{28}{3(z-1)}-\frac{10}{z-\frac{1}{2}}$$

$$Y_2(z)=\frac{8}{3}\frac{z}{z-\frac{1}{4}}+\frac{28}{3}\frac{z}{z-1}-10\frac{z}{z-\frac{1}{2}}$$

$$y_2(n)=\frac{8}{3}\left(\frac{1}{4}\right)^n u(n)+\frac{28}{3}u(n)-10\left(\frac{1}{2}\right)^n u(n)$$

$$A=\left.\cancel{(z-1)}\,\frac{-z}{2\,\cancel{(z-1)}\left(z-\frac{1}{2}\right)}\right|_{z=1}$$
$$=\frac{-1}{2\left(1-\frac{1}{2}\right)}=-1$$

$$B=\left.\left(\cancel{z-\frac{1}{2}}\right)\frac{-z}{2(z-1)\left(\cancel{z-\frac{1}{2}}\right)}\right|_{z=\frac{1}{2}}$$
$$=\frac{-\frac{1}{2}}{2\left(\frac{1}{2}-1\right)}=\frac{-\frac{1}{2}}{2\left(-\frac{1}{2}\right)}=\frac{1}{2}$$

$$A_1=\left.\left(\cancel{z-\frac{1}{4}}\right)\frac{z\left(2z+\frac{3}{2}\right)}{\left(\cancel{z-\frac{1}{4}}\right)(z-1)\left(z-\frac{1}{2}\right)}\right|_{z=\frac{1}{4}}$$
$$=\frac{\frac{1}{4}\left(\frac{1}{2}+\frac{3}{2}\right)}{\left(\frac{1}{4}-1\right)\left(\frac{1}{4}-\frac{1}{2}\right)}=\frac{\frac{1}{4}(2)}{\left(\frac{-3}{4}\right)\left(-\frac{1}{4}\right)}$$
$$=\frac{1}{2}\cdot\frac{16}{3}=\frac{8}{3}$$

$$B_1=\left.\cancel{(z-1)}\,\frac{z\left(2z+\frac{3}{2}\right)}{\cancel{(z-1)}\left(z-\frac{1}{4}\right)\left(z-\frac{1}{2}\right)}\right|_{z=1}$$
$$=\frac{1\left(2+\frac{3}{2}\right)}{\left(\frac{3}{4}\right)\left(\frac{1}{2}\right)}=\frac{\frac{7}{2}}{\frac{3}{8}}=\frac{7}{2}\cdot\frac{8}{3}=\frac{28}{3}$$

$$C_1=\left.\left(\cancel{z-\frac{1}{2}}\right)\frac{z\left(2z+\frac{3}{2}\right)}{(z-1)\left(z-\frac{1}{4}\right)\left(\cancel{z-\frac{1}{2}}\right)}\right|_{z=1}$$
$$=\frac{\frac{1}{2}\left(1+\frac{3}{2}\right)}{\left(-\frac{1}{2}\right)\left(\frac{1}{4}\right)}=\frac{5}{4}\left(\frac{-8}{1}\right)=-10$$

$$y(n)=\frac{25}{3}u(n)+\frac{8}{3}\left(\frac{1}{4}\right)^n u(n)-\frac{19}{2}\left(\frac{1}{2}\right)^n u(n)$$

Solved Problem 10.35 Using z-transform, determine the response of the linear time - invariant system with difference equation

$$y(n)-2r\cos\theta y(n-1)+r^2y(n-2)=x(n)$$

to an excitation $x(n)=a^nu(n)$

Solution:

Given $\quad y(n)-2r\cos\theta y(n-1)+r^2y(n-2)=x(n)$

Taking z-transform and both sides and applying zero initial conditions yields

$$Y(z)-2r\cos\theta z^{-1}Y(z)+r^2z^{-2}Y(z)=X(z)$$

$$\Rightarrow \quad \frac{Y(z)}{X(z)}=\frac{1}{1-2r\cos\theta z^{-1}+r^2z^{-2}}$$

For $x(n)=a^nu(n)$; $X(z)=\dfrac{1}{1-az^{-1}}$

$$\text{i.e., } Y(z)=\frac{1}{(1-az^{-1})(1-2r\cos\theta z^{-1}+r^2z^{-2})}$$

$$=\frac{1}{(1-az^{-1})(1-re^{j\theta}z^{-1})(1-re^{-j\theta}z^{-1})}$$

$$=\frac{z^3}{(z-a)(z-re^{j\theta})(z-re^{-j\theta})}$$

$$\frac{Y(z)}{z}=\frac{z^2}{(z-a)(z-re^{j\theta})(z-re^{-j\theta})}$$

$$=\frac{A}{z-a}+\frac{B}{z-re^{j\theta}}+\frac{C}{z-re^{-j\theta}}$$

$$Y(z)=\frac{a^2}{a^2-2ar\cos\theta+r^2}\cdot\left(\frac{z}{z-a}\right)+\frac{r^2e^{j2\theta}}{(re^{j\theta}-a)(re^{j\theta}-re^{-j\theta})}\frac{z}{z-re^{j\theta}}$$

$$+\frac{r^2e^{-j2\theta}}{(re^{-j\theta}-a)(re^{-j\theta}-re^{j\theta})}\frac{z}{z-re^{-j\theta}}$$

Taking inverse z-transform on both sides and simplifying, we get

$$y(n)=\frac{a^2}{a^2-2ar\cos\theta+r^2}a^nu(n)+\frac{r^{n+1}}{\sin\theta}\left[\frac{r\sin(n+1)\theta-a\sin(n+2)\theta}{a^2-2ar\cos\theta+r^2}\right]u(n)$$

Solved Problem 10.36 Find the response of

$$y(n)+y(n-1)-2y(n-2)=u(n-1)+2u(n-2)$$

due to $y(-1)=0.5; y(-2)=0.25$

Solution:

Given

$$y(n)+y(n-1)-2y(n-2)=u(n-1)+2u(n-2)$$

Taking z-transform on both sides, we get

$$Y(z)+[z^{-1}Y(z)+y(-1)]-2[z^{-2}Y(z)+z^{-1}y(-1)+y(-2)]$$

$$=\frac{z^{-1}}{1-z^{-1}}+\frac{2z^{-2}}{1-z^{-1}}$$

$$\Rightarrow Y(z)+z^{-1}Y(z)+0.5-2[z^{-2}Y(z)+0.5z^{-1}+0.25]$$

$$=\frac{z^{-1}}{1-z^{-1}}+\frac{2z^{-2}}{1-z^{-1}}$$

$$Y(z)[1+z^{-1}-2z^{-2}]=z^{-1}+\frac{z^{-1}}{1-z^{-1}}+\frac{2z^{-2}}{1-z^{-1}}$$

$$Z[u(n)]=\frac{1}{1-z^{-1}}$$

$$Z(u(n-1))=\frac{z^{-1}}{1-z^{-1}}$$

$$Y(z)=\frac{z^{-1}}{1+z^{-1}-2z^{-2}}+\frac{z^{-1}+2z^{-2}}{(1-z^{-1})(1+z^{-1}-2z^{-2})}$$

$$=\frac{z}{z^2+z-2}+\frac{z(z+2)}{(z-1)(z^2+z-2)}$$

$$Y(z)=Y_1(z)+Y_2(z)$$

Taking inverse z-transform, we get

$$y(n)=y_1(n)+y_2(n)$$

$$Y_1(z)=\frac{z}{(z+2)(z-1)}$$

$$\frac{Y_1(z)}{z}=\frac{A}{z+2}+\frac{B}{z-1}$$

$$=\frac{-1}{3(z+2)}+\frac{1}{3(z-1)}$$

$$Y_1(z)=\frac{-1}{3}\frac{z}{z+2}+\frac{1}{3}\frac{z}{z-1}$$

$$y_1(n)=\frac{-1}{3}(-2)^nu(n)+\frac{1}{3}u(n)$$

$$A=(z+2)\frac{1}{(z+2)(z-1)}\bigg|_{z=-2}=\frac{-1}{3}$$

$$B=(z-1)\frac{1}{(z+2)(z-1)}\bigg|_{z=1}=\frac{1}{3}$$

$$Y_2(z) = \frac{z(z+2)}{(z-1)^2(z+2)} = \frac{z}{(z-1)^2}$$

$$y_2(n) = nu(n)$$

$$\Rightarrow y(n) = \frac{1}{3}u(n) + nu(n) - \frac{1}{3}(-2)^n u(n)$$

Practice Problem 10.20

Determine the system function and impulse response $h(u)$ of the system . AU Nov 2008

$$y(n) - \frac{1}{2}y(n-1) = 2x(n)$$

with zero initial conditions

Ans:

$$H(z) = \frac{2}{1 - \frac{1}{2}z^{-1}}\ ;\ u(n) = 2\left(\tfrac{1}{2}\right)^n u(n)$$

Practice Problem 10.21

Determine the system function for the causal LTI system with difference equation

$$y(n) - \frac{1}{2}y(n-1) + \frac{1}{4}y(n-2) = x(n)$$

Find $y(n)$ if $x(n) = \left(\frac{1}{3}\right)^n u(n)$

10.11 Relationship between s-plane and z-plane

Consider a continuous system $H(s)$. Let the input and output of the system be $x(t)$ and $y(t)$ respectively.

The Laplace transform of the input signal $x(t)$ is defined as

$$X(s) = \int_0^\infty x(t)e^{-st}dt$$

The discrete version of the signal $x(t)$ can be obtained by multiplying $x(t)$ with sequence of impulse spaced T seconds apart. That is

$$x^*(t). = \sum_{n=0}^{\infty} x(nT)\delta(t - nT)$$

The Laplace transform of $x^*(t)$ can be written as

$$L[x^*(t)] = X^*(s) = \left[\sum_{n=0}^{\infty} x(nT)\delta(t-nT)\right]$$

$$= \sum_{n=0}^{\infty} x(nT)L[\delta(t-nT)]$$

$$= \sum_{n=0}^{\infty} x(nT)e^{-nTs} \qquad (10.112)$$

The z-transform of the signal $x(nT)$ is given by

$$Z[x^*(t)] = Z[x(nT)] = X(z) = \sum_{n=0}^{\infty} x(nT)z^{-n} \qquad (10.113)$$

Comparing Eq. (10.112) and Eq. (10.113) we find that the relationship between s-plane and z-plane can be described by the equation

$$z = e^{sT} \qquad (10.114)$$

we know that $z = re^{j\omega}$ (10.115)

and

$$s = \sigma + j\Omega \qquad (10.116)$$

Substituting Eq. (10.115) and Eq. (10.116) into Eq. (10.114) yields

$$z = re^{j\omega} = e^{(\sigma + j\Omega)T}$$

$$= e^{\sigma T}e^{j\Omega T}$$

$$\Rightarrow z = re^{j\omega} = e^{\sigma T}e^{j\Omega T}$$

From the above relation, we find that $|z| = e^{\sigma T}$ and $\omega = \Omega T$. If $\sigma = 0, |z| = 1$, for $\sigma < 0, |z| < 1$ and for $\sigma > 0, |z| > 1$. When $\sigma = 0, |z| = 1$, the $j\Omega$-axis of s-plane maps into the unit circle. The left half of s-plane where $\sigma < 0$ maps into the inside the unit circle ($|z| < 1$). The right half of s-plane where $\sigma > 0$ maps into the outside of the unit circle. The mapping is shown in Fig. 10.15.

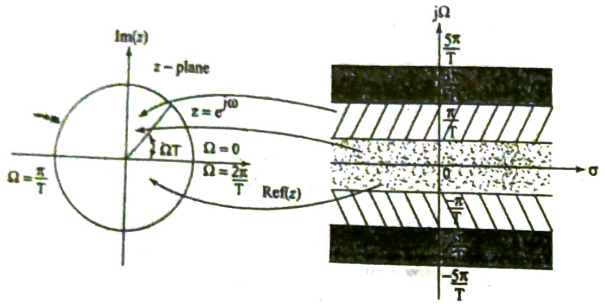


Fig. 10.15

Review Questions

1. Bring out the relationship between s-plane and z-plane

Solved Problem 10.37 Consider the system

$$H(z) = \frac{0.2z}{(z+0.4)(z-0.2)} \quad \text{ROC}: |z| > 0.4$$

(a) Find the impulse response function of the system.
(b) Is DTFT exists for the system? if so, how?
(c) Find the DTFT.

Solution:

Given

$$H(z) = \frac{0.2z}{(z+0.4)(z-0.2)}$$

$$h(n) = Z^{-1}[H(z)]$$

$$\frac{H(z)}{z} = \frac{0.2}{(z+0.4)(z-0.2)}$$

$$= \frac{A}{z+0.4} + \frac{B}{z-0.2}$$

$$= -\frac{1}{3}\cdot\frac{1}{z+0.4} + \frac{1}{3}\frac{1}{z-0.2}$$

$$A = (z+0.4)\frac{0.2}{(z+0.4)(z-0.2)}\Big|_{z=-0.4}$$

$$= \frac{0.2}{-0.6} = \frac{-1}{3}$$

$$B = (z-0.2)\frac{0.2}{(z+0.4)(z-0.2)}\Big|_{z=0.2}$$

$$= \frac{0.2}{0.6} = \frac{1}{3}$$

$$H(z) = -\frac{1}{3}\frac{z}{z+0.4} + \frac{1}{3}\frac{z}{z-0.2}$$

Taking inverse of z-transform on both sides, we get

$$h(n) = -\frac{1}{3}(-0.4)^n u(n) + \frac{1}{3}(0.2)^n u(n)$$

$$Z^{-1}\left[\frac{z}{z-a}\right] = a^n u(n)$$

b) The poles of the system fuction $H(z)$ are $z_1 = -0.4$ and $z_2 = 0.2$

Both poles lies inside the unit circle therefore the DTFT exists for the system.

c) $H(e^{j\omega}) = H(z)|_{z=e^{j\omega}}$

$$= \frac{0.2z}{(z+0.4)(z-0.2)}\Big|_{z=e^{j\omega}}$$

$$= \frac{0.2e^{j\omega}}{(e^{j\omega}+0.4)(e^{j\omega}-0.2)}$$

Solved Problem 10.38 Solve the following difference equations for $y(n)$ using z-transform and the specified initial conditions.

i) $y(n) - y(n-1) + \frac{1}{4}y(n-2) = x(n) \quad n \geq 0$

where $x(n) = 2\left(\frac{1}{8}\right)^n$; $y(-1) = 2$ and $y(-2) = 4$

ii) $y(n) = 0.5y(n-1) + x(n), n \geq 0$

$$x(n) = \cos\frac{\pi}{3}nu(n)$$

$$y(-1) = 0$$

Solution:

Given

$$y(n) - y(n-1) + \frac{1}{4}y(n-2) = x(n)$$

Taking z-transform on both sides

$$Y(z) - [z^{-1}Y(z) + y(-1)] + \frac{1}{4}[z^{-2}Y(z) + z^{-1}y(-1) + y(-2)] = X(z)$$

$$Y(z)\left[1 - z^{-1} + \frac{1}{4}z^{-2}\right] - 2 + \frac{1}{2}z^{-1} + 1 = X(z)$$

$$Y(z)\left[1-z^{-1}+\frac{1}{4}z^{-2}\right]=1-0.5z^{-1}+X(z) \qquad (10.117)$$

Given

$$x(n)=2\left(\frac{1}{8}\right)^n u(n)$$

$$X(z)=2\left(\frac{1}{1-\frac{1}{8}z^{-1}}\right) \qquad (10.118)$$

Substituting Eq.(10.118) in Eq.(10.117) yields

$$Y(z)\left[1-z^{-1}+\frac{1}{4}z^{-2}\right]=1-0.5z^{-1}+\frac{2}{1-\frac{1}{8}z^{-1}}$$

$$Y(z)=\frac{1-0.5z^{-1}}{1-z^{-1}+\frac{1}{4}z^{-2}}+\frac{2}{\left(1-z^{-1}+\frac{1}{4}z^{-2}\right)\left(1-\frac{1}{8}z^{-1}\right)}$$

$$=\frac{z(z-0.5)}{z^2-z+\frac{1}{4}}+\frac{2z^3}{\left(z^2-z+\frac{1}{4}\right)\left(z-\frac{1}{8}\right)}$$

$$=\frac{z(z-0.5)}{\left(z-\frac{1}{2}\right)^2}+\frac{2z^3}{\left(z-\frac{1}{2}\right)^2\left(z-\frac{1}{8}\right)}$$

$$=\frac{z}{z-\frac{1}{2}}+\frac{2z^3}{\left(z-\frac{1}{2}\right)^2\left(z-\frac{1}{8}\right)}$$

$$=Y_1(z)+Y_2(z)$$

$$\Rightarrow \quad y(n)=y_1(n)+y_2(n)$$

$$Y_1(z)=\frac{z}{z-\frac{1}{2}}$$

$$y_1(n)=\left(\frac{1}{2}\right)^n u(n)$$

$$Y_2(z)=\frac{2z^3}{\left(z-\frac{1}{2}\right)^2\left(z-\frac{1}{8}\right)}$$

$$\frac{Y_2(z)}{z}=\frac{2z^2}{\left(z-\frac{1}{2}\right)^2\left(z-\frac{1}{8}\right)}$$

$$=\frac{A}{z-\frac{1}{2}}+\frac{B}{\left(z-\frac{1}{2}\right)^2}+\frac{C}{z-\frac{1}{8}}$$

$$=\frac{16}{9}\frac{1}{z-\frac{1}{2}}+\frac{4}{3\left(z-\frac{1}{2}\right)^2}+\frac{2}{9}\cdot\frac{1}{z-\frac{1}{8}}$$

$$Y_2(z)=\frac{16}{9}\frac{z}{z-\frac{1}{2}}+\frac{4}{3}\cdot\frac{z}{\left(z-\frac{1}{2}\right)^2}+\frac{2}{9}\frac{z}{z-\frac{1}{8}}$$

$$y_2(n)=\frac{16}{9}\left(\frac{1}{2}\right)^n u(n)+\frac{8}{3}n\left(\frac{1}{2}\right)^n u(n)+\frac{2}{9}\left(\frac{1}{8}\right)^n u(n)$$

$$A=\frac{1}{1!}\frac{d}{dz}\left(z-\frac{1}{2}\right)^2\frac{2z^2}{\left(z-\frac{1}{2}\right)^2\left(z-\frac{1}{8}\right)}\Bigg|_{z=\frac{1}{2}}$$

$$=\frac{\left(z-\frac{1}{8}\right)4z-2z^2}{\left(z-\frac{1}{8}\right)^2}\Bigg|_{z=\frac{1}{2}}$$

$$=\frac{\left(\frac{1}{2}-\frac{1}{8}\right)2-\frac{1}{2}}{\left(\frac{1}{2}-\frac{1}{8}\right)^2}$$

$$=\frac{\frac{3}{4}-\frac{1}{2}}{(3/8)^2}=\frac{1}{4}\cdot\frac{8}{3}\cdot\frac{8}{3}$$

$$=\frac{16}{9}$$

$$B=\left(z-\frac{1}{2}\right)^2\frac{2z^2}{\left(z-\frac{1}{2}\right)^2\left(z-\frac{1}{8}\right)}\Bigg|_{z=\frac{1}{2}}$$

$$=\frac{2(1/4)}{\frac{1}{2}-\frac{1}{8}}=(1/2)\cdot\frac{8}{3}=\frac{4}{3}$$

$$C=\left(z-\frac{1}{8}\right)\frac{2z^2}{\left(z-\frac{1}{2}\right)^2\left(z-\frac{1}{8}\right)}\Bigg|_{z=\frac{1}{8}}$$

$$=\frac{2\left(\frac{1}{8}\right)^2}{\left(-\frac{3}{8}\right)^2}=2\frac{1}{64}\cdot\frac{64}{9}$$

$$=\frac{2}{9}$$

$$y(n)=y_1(n)+y_2(n)$$

$$=\frac{25}{9}\left(\frac{1}{2}\right)^n u(n)+\frac{8}{3}n\left(\frac{1}{2}\right)^n u(n)+\frac{2}{9}\left(\frac{1}{8}\right)^n u(n)$$

ii) $y(n)=0.5y(n-1)+x(n) \quad n\geq 0$

$$x(n)=\cos\frac{\pi}{3}nu(n)$$

Applying z-transform on both sides and substituting $y(-1)=0$ yields

$$Y(z)=0.5z^{-1}Y(z)+X(z)$$

$$Y(z)[1-0.5z^{-1}]=X(z)$$

$$\frac{Y(z)}{X(z)}=\frac{1}{1-0.5z^{-1}}$$

For $x(n)=\cos\frac{\pi}{3}n u(n); X(z)=\frac{z^2-0.5z}{z^2-z+1}$

$$Y(z)=\frac{z^2-0.5z}{(z^2-z+1)(1-0.5z^{-1})}=\frac{z^2(z-0.5)}{(z^2-z+1)(z-0.5)}$$

$$=\frac{z^2}{z^2-z+1}$$

$$\frac{Y(z)}{z}=\frac{z}{z^2-z+1}=\frac{0.5-j0.288}{z-(0.5+j0.866)}+\frac{0.5+j0.288}{z-(0.5-j0.866)}$$

$$Y(z)=\frac{(0.5-j0.288)z}{z-(0.5+j0.866)}+\frac{(0.5+j0.288)z}{z-(0.5-j0.866)}$$

$$y(n)=(0.5-j0.288)(0.5+j0.866)^n u(n)$$
$$+(0.5+j0.288)(0.5-j0.866)^n u(n)$$

$$y(n)=(0.5-j0.288)\left(\cos\frac{\pi}{3}n+j\sin\frac{\pi}{3}n\right)+(0.5+j0.288)$$
$$\left(\cos\frac{\pi}{3}n-j\sin\frac{\pi}{3}n\right)$$

$$=\cos\frac{\pi n}{3}+0.576\sin\frac{\pi n}{3}$$

$$=1.15\left[0.866\cos\frac{\pi n}{3}+0.5\sin\frac{\pi n}{3}\right]$$

$$=1.15\left[\sin\frac{\pi}{3}\cos\frac{\pi n}{3}+\cos\frac{\pi}{3}\sin\frac{\pi n}{3}\right]$$

$$=1.15\left[\sin\frac{\pi(n+1)}{3}\right]u(n)$$

$$z^2-z+1=0$$
$$z_{1,2}=\frac{1\pm\sqrt{3}}{2}=0.5\pm j0.866$$
$$\frac{A}{z-0.5-j0.866}+\frac{A^*}{z-0.5+j0.866}$$

Solved Problem 10.39 A causal system is represented by the following difference equation

$$y(n)+\frac{1}{4}y(n-1)=x(n)+\frac{1}{2}x(n-1)$$

a) Find the system function $H(z)$ and give the corresponding region of convergence.

b) Find the unit sample response of the system.

c) Find the frequency response $H(e^{j\omega})$ and determine its magnitude and phase.

Solution:

a) Given $y(n)+\frac{1}{4}y(n-1)=x(n)+\frac{1}{2}x(n-1)$

Applying z-transform on both sides and assuming zero initial conditions, we get

$$Y(z)+\frac{1}{4}z^{-1}Y(z)=X(z)+\frac{1}{2}z^{-1}X(z)$$

$$\text{The system function} \Rightarrow Y(z)\left[1+\frac{1}{4}z^{-1}\right]=X(z)\left[1+\frac{1}{2}z^{-1}\right]$$

$$H(z)=\frac{Y(z)}{X(z)}=\frac{1+\frac{1}{2}z^{-1}}{1+\frac{1}{4}z^{-1}}=\frac{z+\frac{1}{2}}{z+\frac{1}{4}}$$

As the system is causal the Region of convergence is given by $|z|>\frac{1}{4}$.

b)

For unit sample sequence $x(n)=\delta(n); X(z)=1$

Therefore

$$H(z)=Y(z)=\frac{z+\frac{1}{2}}{z+\frac{1}{4}}$$

$$=\frac{z}{z+\frac{1}{4}}+\frac{\frac{1}{2}}{z+\frac{1}{4}}$$

$$=\frac{z}{z+\frac{1}{4}}+\frac{1}{2}z^{-1}\frac{z}{z+\frac{1}{4}}$$

Taking inverse z-transform on both sides, we get

$$y(n)=\left(-\frac{1}{4}\right)^n u(n)+\frac{1}{2}\left(-\frac{1}{4}\right)^{n-1}u(n-1)\ (\because \text{ time shifting property})$$

$$\boxed{Z[x(n-1)]=z^{-1}X(z)}$$

c) The pole $z=-\frac{1}{4}$ is inside the unit circle, Therefore the system is stable and the Fourier transform $H(e^{j\omega})$ can be obtained from $H(z)$.

$$H(e^{j\omega})=H(z)|_{z=e^{j\omega}}$$

$$=\left.\frac{z+\frac{1}{2}}{z+\frac{1}{4}}\right|_{z=e^{j\omega}}$$

$$H(e^{j\omega})=\frac{e^{j\omega}+\frac{1}{2}}{e^{j\omega}+\frac{1}{4}}=\frac{\cos\omega+j\sin\omega+\frac{1}{2}}{\cos\omega+j\sin\omega+\frac{1}{4}}$$

$$|H(e^{j\omega})| = \left[\frac{(\cos\omega+\frac{1}{2})^2+\sin^2\omega}{(\cos\omega+\frac{1}{4})^2+\sin^2\omega}\right]^{1/2}$$

$$= \left[\frac{\cos^2\omega+\cos\omega+\frac{1}{4}+\sin^2\omega}{\cos^2\omega+\frac{1}{2}\cos\omega+\frac{1}{16}+\sin^2\omega}\right]^{1/2}$$

$$= \left[\frac{\frac{5}{4}+\cos\omega}{\frac{17}{16}+0.5\cos\omega}\right]^{1/2}$$

$$\angle H(e^{j\omega}) = \tan^{-1}\frac{\sin\omega}{0.5+\cos\omega} - \tan^{-1}\frac{\sin\omega}{0.25+\cos\omega}$$

Practice Problem 10.22

Find the impulse and step response of the systems described by the following difference equations

(i) $y(n)+\frac{1}{3}y(n-1)=x(n)$

(ii) $y(n)+\frac{1}{4}y(n-1)+\frac{1}{6}y(n-2)=x(n)+\frac{1}{3}x(n-1)$

(iii) $y(n)+4y(n-1)+5y(n-2)=3x(n)+2x(n-1)$

Solved Problem 10.40 Find the output of the system whose input and output are related by

$$y(n)=7y(n-1)-12y(n-2)+2x(n)-x(n-2)$$

for the input $x(n)=u(n)$ (Anna University Nov. 2002)

Solution:

Given

$$y(n)=7y(n-1)-12y(n-2)+2x(n)-x(n-2)$$

Taking z-transform on both sides with zero initial conditions, we get

$$Y(z)=7z^{-1}Y(z)-12z^{-2}Y(z)+2X(z)-z^{-2}X(z)$$

$$Y(z)\left[1-7z^{-1}+12z^{-2}\right]=X(z)\left[2-z^{-2}\right]$$

$$H(z)=\frac{Y(z)}{X(z)}=\frac{2-z^{-2}}{1-7z^{-1}+12z^{-2}}$$

$$=\frac{(2z^2-1)}{z^2-7z+12}$$

For input $x(n)=u(n)$

$$X(z)=\frac{z}{z-1}$$

Now $$Y(z)=\frac{z(2z^2-1)}{(z-1)(z-4)(z-3)}$$

$$\frac{Y(z)}{z}=\frac{2z^2-1}{(z-1)(z-4)(z-3)}$$

$$=\frac{A}{z-1}+\frac{B}{z-4}+\frac{C}{z-3}$$

$$A=(z-1)\frac{Y(z)}{z}\Big|_{z=1}$$

$$=(z-1)\frac{2z^2-1}{(z-1)(z-4)(z-3)}\Big|_{z=1}=\frac{2(1)^2-1}{(1-4)(1-3)}=\frac{1}{(-3)(-2)}=\frac{1}{6}$$

$$B=(z-4)\frac{Y(z)}{z}\Big|_{z=4}$$

$$=(z-4)\frac{2z^2-1}{(z-1)(z-4)(z-3)}\Big|_{z=4}=\frac{2(4)^2-1}{(4-1)(4-3)}=\frac{31}{3}$$

$$C=(z-3)\frac{Y(z)}{z}\Big|_{z=3}$$

$$=(z-3)\frac{2z^2-1}{(z-1)(z-4)(z-3)}\Big|_{z=3}=\frac{2(3)^2-1}{(3-1)(3-4))}=\frac{-17}{2}$$

$$\Rightarrow \frac{Y(z)}{z}=\frac{1}{6}\cdot\frac{1}{z-1}+\frac{31}{3}\cdot\frac{1}{z-4}-\frac{17}{2}\cdot\frac{1}{z-3}$$

$$Y(z)=\frac{1}{6}\frac{z}{z-1}+\frac{31}{3}\frac{z}{z-4}-\frac{17}{2}\frac{z}{z-3}$$

Taking inverse z-transform yields

$$y(n)=\frac{1}{6}u(n)+\frac{31}{3}(4)^n u(n)-\frac{17}{2}(3)^n u(n)$$

Solved Problem 10.41 Find the inverse z-transform of (AU Nov. 2002)

$$\frac{z+4}{z^2-4z+3}$$

Solution:

Given

$$X(z) = \frac{z+4}{z^2-4z+3}$$

$$= \frac{z+4}{(z-1)(z-3)}$$

$$= \frac{A}{z-1} + \frac{B}{z-3}$$

$$A = (z-1)\frac{z+4}{(z-1)(z-3)}\bigg|_{z=1} = \frac{-5}{2}$$

$$B = (z-3)\frac{z+4}{(z-1)(z-3)}\bigg|_{z=3} = \frac{7}{2}$$

$$X(z) = \frac{-5}{2(z-1)} + \frac{7}{2(z-3)}$$

$$\Rightarrow x(n) = \frac{-5}{2}u(n-1) + \frac{7}{2}(3)^{n-1}u(n-1)$$

$$Z^{-1}\left[\frac{1}{z-1}\right] = u(n-1)$$

$$Z^{-1}\left[\frac{1}{z-3}\right] = 3^{n-1}u(n-1)$$

Solved Problem 10.42 (i) Find the z-transform of

$$x(n) = u(n) - u(n-10) \qquad \text{(AU April 2003)}$$

(ii) Find the impulse response of the system whose input output relation is given by the equation

$$y(n) - 4y(n-1) + 3y(n-2) = x(n) + 2x(n-1) \qquad \text{(AU April 2003)}$$

Solution:

(i) Given

$$x(n) = u(n) - u(n-10)$$

$$X(z) = \frac{1}{1-z^{-1}} - \frac{z^{-10}}{1-z^{-1}}$$

$$= \frac{1-z^{-10}}{1-z^{-1}}$$

$$Z[x(n-k)] = z^{-k}X(z)$$

(ii) Given

$$y(n) - 4y(n-1) + 3y(n-2) = x(n) + 2x(n-1)$$

Taking z-transform on both sides

$$Y(z) - 4z^{-1}Y(z) + 3z^{-2}Y(z) = X(z) + 2z^{-1}X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1+2z^{-1}}{1-4z^{-1}+3z^{-2}}$$

$$= \frac{z(z+2)}{z^2-4z+3}$$

$$= \frac{z(z+2)}{(z-1)(z-3)}$$

$$\Rightarrow \frac{Y(z)}{X(z)} = \frac{z(z+2)}{(z-1)(z-3)}$$

For an impulse input

$$X(z) = 1$$

Therefore

$$\frac{Y(z)}{z} = \frac{z+2}{(z-1)(z-3)}$$

$$= \frac{-3}{2(z-1)} + \frac{5}{2(z-3)}$$

$$Y(z) = \frac{-3}{2}\cdot\frac{z}{z-1} + \frac{5}{2}\frac{z}{z-3}$$

Taking inverse z-transform on both sides we get

$$y(n) = \frac{-3}{2}u(n) + \frac{5}{2}(3)^n u(n)$$

10.12 Block Diagram Representation

In chapter 7, we have studied the block diagram representation of continuous-time systems. We now discuss the basic block diagram for LTI discrete-time systems. The basic block diagram consists of unit delay elements, multiplier and adders.

Multiplier

The basic diagram of a multiplier is shown in Fig. 10.16.

Fig. 10.16 A multiplier.

Here the signal $x(n)$ is multiplied by a scalar factor a.

Adder

An adder is used to add two discrete-time signals. Such an adder is shown in Fig. 10.17.

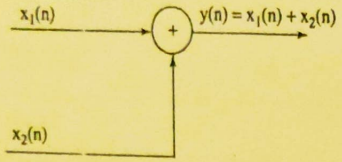


Fig. 10.17 An adder.

Unit-delay element

The block diagram of unit delay element is shown in Fig. 10.18.

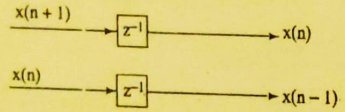


Fig. 10.18 Unit-delay element.

If the input signal is x(n) then the output of the unit delay element is $x(n-1)$.

A discrete-time system can be realized in different ways. In this section, we will study four different types of realizations. They are

1. Direct form-I
2. Direct form-II
3. Cascade form
4. Parallel form

A discrete-time system is realized using delay elements, multiplier and adders.

10.12.1 Direct form I realization

Consider a system with system function

$$H(z)=\frac{Y(z)}{X(z)}=\frac{b_0+b_1z^{-1}+b_2z^{-2}}{1+a_1z^{-1}+a_2z^{-2}} \tag{10.119}$$

$$\Rightarrow \quad Y(z)+a_1z^{-1}Y(z)+a_2z^{-2}Y(z)=b_0X(z)+b_1z^{-1}X(z)+b_2z^{-2}X(z)$$

Let

$$b_0X(z)+b_1z^{-1}X(z)+b_2z^{-2}X(z)=W(z) \tag{10.120}$$

then

$$Y(z)+a_1z^{-1}+a_2z^{-2}Y(z)=W(z) \tag{10.121}$$

or

$$Y(z)=W(z)-a_1z^{-1}Y(z)-a_2z^{-2}Y(z) \tag{10.122}$$

The Eq. (10.120) can be realized as shown in Fig. 10.19.

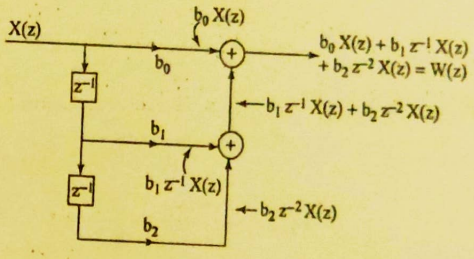


Fig. 10.19 Realization of Eq. (10.120).

Note that b_0, b_1, b_2, the multipliers in Fig. (10.19) are the coefficients of numerator of the system function. The Eq. (10.122) can be realized as shown in Fig. 10.20.

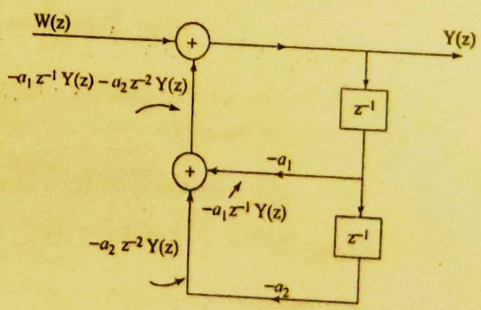


Fig. 10.20 Realization of Eq. (10.122).

Note that the multipliers in Fig. 10.20 are the negative coefficient of the denominator coefficients of system function (Except a_0 which is equal to one). Combining Fig. 10.19 and Fig. 10.20 we get direct form realization of the system function as shown in Fig. 10.21.

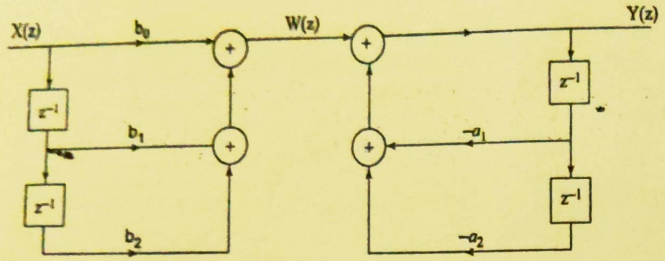


Fig. 10.21 Direct form-I realization of $H(z)$.

Solved Problem 10.43 Obtain the direct form-I realization for the system described by the difference equation

$$y(n)-\frac{5}{6}y(n-1)+\frac{1}{6}y(n-2)=x(n)+2x(n-1)$$

Solution:

Given

$$y(n)-\frac{5}{6}y(n-1)+\frac{1}{6}y(n-2)=x(n)+2x(n-1) \qquad (10.123)$$

Taking z-transform on both sides assuming zero initial conditions, we get

$$Y(z)-\frac{5}{6}z^{-1}Y(z)+\frac{1}{6}z^{-2}Y(z)=X(z)+2z^{-1}X(z)$$

Let

$$X(z)+2z^{-1}X(z)=W(z) \qquad (10.124)$$

then

$$Y(z)-\frac{5}{6}z^{-1}Y(z)+\frac{1}{6}z^{-2}Y(z)=W(z)$$

or

$$Y(z)=W(z)+\frac{5}{6}z^{-1}Y(z)-\frac{1}{6}z^{-2}Y(z) \qquad (10.125)$$

Realizing Eq. (10.124), we get

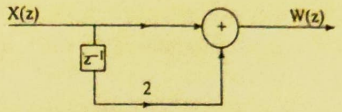


Fig. 10.22 Realization of Eq. (10.124).

Similarly realization of Eq. (10.125) is shown in Fig.10.23

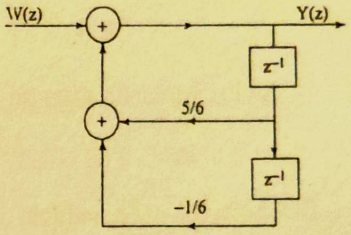


Fig. 10.23 Realization of Eq. (10.125).

Combining both figures yields

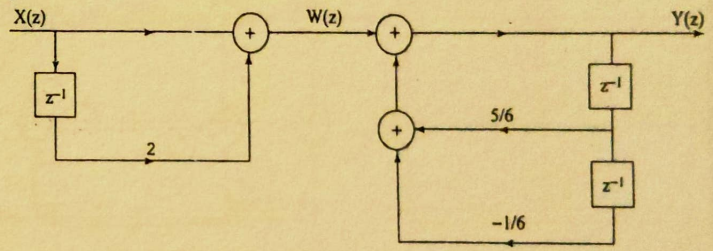


Fig. 10.24 Realization of the system in solved problem 10.43.

The main disadvantage of direct form-I realization is that it uses seperate delays for input and output which increases the number of delay elements required for realization. In direct form-II realization the number of delay elements required for realization is less than that of direct form-I realization.

10.12.2 Direct form II

Consider a system

$$H(z)=\frac{Y(z)}{X(z)}=\frac{b_0+b_1z^{-1}+b_2z^2}{1+a_1z^{-1}+a_2z^{-2}} \qquad (10.126)$$

Let
$$\frac{Y(z)}{X(z)} = \frac{Y(z)}{W(z)} \cdot \frac{W(z)}{X(z)},$$

where $$\frac{W(z)}{X(z)} = \frac{1}{1+a_1z^{-1}+a_2z^{-2}} \quad (10.127)$$

and $$\frac{Y(z)}{W(z)} = b_0+b_1z^{-1}+b_2z^{-2} \quad (10.128)$$

From Eq. (10.127), we have

$$W(z) = X(z) - a_1z^{-1}W(z) - a_2z^{-2}W(z) \quad (10.129)$$

Similarly from Eq. (10.128), we can write

$$Y(z) = b_0W(z) + b_1z^{-1}W(z) + b_2z^{-2}W(z) \quad (10.130)$$

The realization of Eq. (10.129) is shown in Fig. 10.25.

Note that the multipliers in Fig. 10.25 are the negative of denominator coefficients of the system function. The realization of Eq. (10.130) is shown in Fig. 10.26.

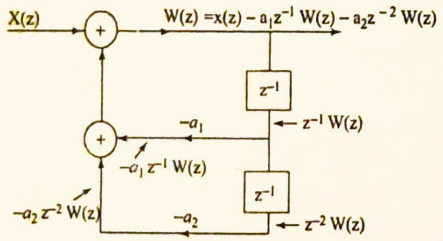


Fig. 10.25 Realization of Eq. (10.129).

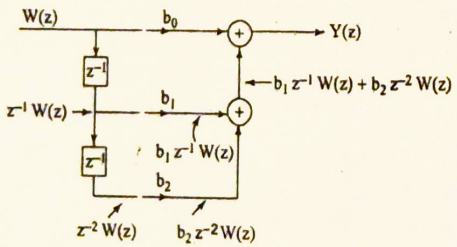


Fig. 10.26 Realization of Eq. (10.130).

Note that the multipliers in Fig. 10.26 is the numerator coefficients of the system function. Combining Fig. 10.25 and Fig. 10.26 we get the direct form II realization.

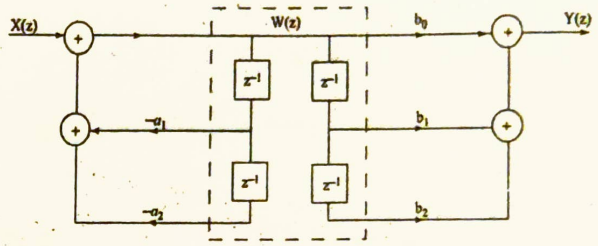


Fig. 10.27

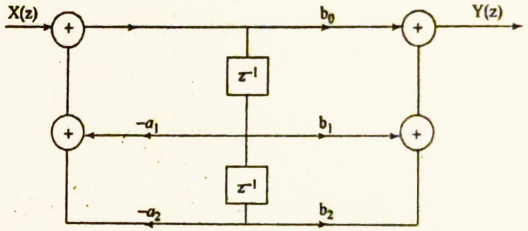


Fig. 10.28 Direct form-II realization of $H(z)$.

Now combine the delay elements to get the direct form-II realization of the system function $H(z)$ as shown in Fig.10.28.

Solved Problem 10.44 Realize the system given in solved problem 10.43 in direct form-II

Solution:

Given
$$y(n) - \frac{5}{6}y(n-1) + \frac{1}{6}y(n-2) = x(n) + 2x(n-1)$$

Taking z-transform on both sides assuming zero initial conditions, we get
$$Y(z) - \frac{5}{6}z^{-1}Y(z) + \frac{1}{6}z^{-2}Y(z) = X(z) + 2z^{-1}X(z) \quad (10.131)$$

$$Y(z)\left[1 - \frac{5}{6}z^{-1} + \frac{1}{6}z^{-2}\right] = X(z)[1+2z^{-1}]$$

$$\Rightarrow \frac{Y(z)}{X(z)} = \frac{1+2z^{-1}}{1-\frac{5}{6}z^{-1}+\frac{1}{6}z^{-2}} \quad (10.132)$$

Let $\frac{Y(z)}{X(z)} = \frac{Y(z)}{W(z)} \cdot \frac{W(z)}{X(z)}$

where $\frac{W(z)}{X(z)} = \frac{1}{1 - \frac{5}{6}z^{-1} + \frac{1}{6}z^{-2}}$

$$\Rightarrow \quad W(z) = X(z) + \frac{5}{6}z^{-1}W(z) - \frac{1}{6}z^{-2}W(z) \qquad (10.133)$$

$$\frac{Y(z)}{W(z)} = 1 + 2z^{-1}$$

$$\Rightarrow Y(z) = W(z) + 2z^{-1}W(z) \qquad (10.134)$$

Realization of Eq. (10.133) is shown in Fig. 10.29.

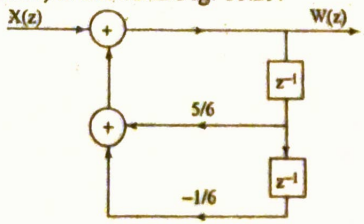


Fig. 10.29 Realization of Eq. (10.133).

Realization of Eq. (10.134) is shown in Fig. 10.30.

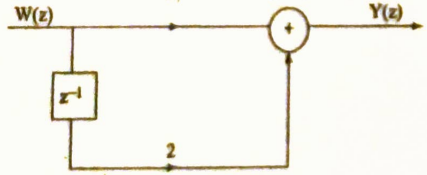


Fig. 10.30 Realization of Eq. (10.134).

Combining Fig. 10.29 and Fig. 10.30 we get the direct form II realization.

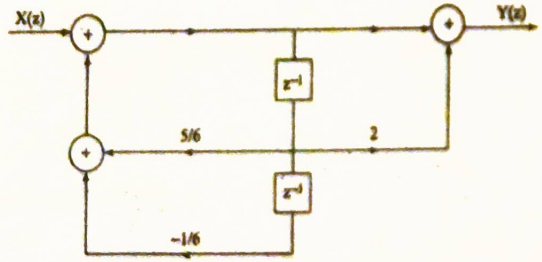


Fig. 10.31 Direct form-II realization of solved problem 10.44.

10.12.3 Cascade form

In cascade form realization, the given system function is expressed as a product of several system functions. Each of the system function is realized in direct form-II and cascaded.

Consider a system with the following system function

$$H(z) = \frac{(b_{k0} + b_{k1}z^{-1} + b_{k2}z^{-2})(b_{m0} + b_{m1}z^{-1} + b_{m2}z^{-2})}{(1 + a_{k1}z^{-1} + a_{k2}z^{-2})(1 + a_{m1}z^{-1} + a_{m2}z^{-2})} \qquad (10.135)$$

$$= H_1(z)H_2(z),$$

where

$$H_1(z) = \frac{b_{k0} + b_{k1}z^{-1} + b_{k2}z^{-2}}{1 + a_{k1}z^{-1} + a_{k2}z^{-2}} \qquad (10.136)$$

and

$$H_2(z) = \frac{b_{m0} + b_{m1}z^{-1} + b_{m2}z^{-2}}{1 + a_{m1}z^{-1} + a_{m2}z^{-2}} \qquad (10.137)$$

Realizing $H_1(z)$ and $H_2(z)$ in direct form II, and cascading we obtain cascade form of the system function $H(z)$.

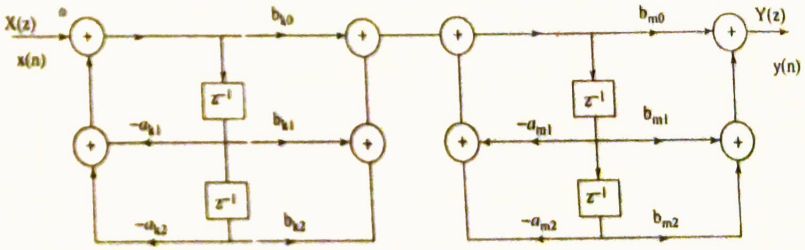


Fig. 10.32 Cascade form realization of Eq. (10.135).

Solved Problem 10.45 Obtain the cascade form realization of the system described by the difference equation

$$y(n) - \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = x(n) + 3x(n-1) + 2x(n-2)$$

Solution:

Given

$$y(n) - \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = x(n) + 3x(n-1) + 2x(n-2) \qquad (10.138)$$

Taking z-transform on both sides we get

$$Y(z)-\frac{1}{4}z^{-1}Y(z)-\frac{1}{8}z^{-2}Y(z)=X(z)+3z^{-1}X(z)+2z^{-2}X(z)$$

$$Y(z)\left[1-\frac{1}{4}z^{-1}-\frac{1}{8}z^{-2}\right]=X(z)[1+3z^{-1}+2z^{-2}]$$

$$\frac{Y(z)}{X(z)}=\frac{1+3z^{-1}+2z^{-2}}{1-\frac{1}{4}z^{-1}-\frac{1}{8}z^{-2}}$$

$$=\frac{(1+z^{-1})(1+2z^{-1})}{(1-\frac{1}{2}z^{-1})(1+\frac{1}{4}z^{-1})} \qquad (10.139)$$

$$=H_1(z)H_2(z)$$

where $$H_1(z)=\frac{1+z^{-1}}{1-\frac{1}{2}z^{-1}} \qquad (10.140)$$

and

$$H_2(z)=\frac{1+2z^{-1}}{1+\frac{1}{4}z^{-1}} \qquad (10.141)$$

The realization of $H_1(z)$ is shown in Fig. 10.33.

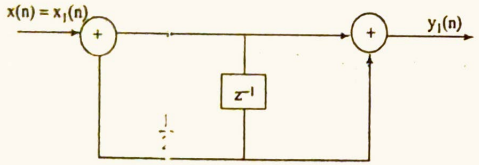


Fig. 10.33 Realization of $H_1(z)$.

The realization of $H_2(z)$ is shown in Fig. 10.34.

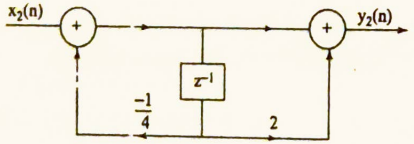


Fig. 10.34 Realization of $H_2(z)$.

Now the cascade realization of $H(z)$ can be obtained by cascading Fig. 10.33 and Fig. 10.34.

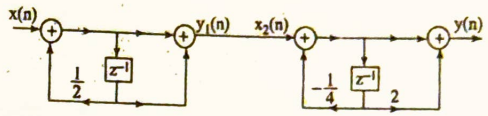


Fig. 10.35 Cascade form realization of solved problem 10.45.

10.12.4 Parallel form realization

The parallel form realization of a system can be obtained by expressing the system function $H(s)$ in partial fraction as

$$H(z)=c+\sum_{k=1}^{N}\frac{c_k}{1-p_k z^{-1}} \qquad (10.142)$$

where $\{p_k\}$ are poles of the system function.

The Eq. (10.142) can be written as

$$H(z)=c+\frac{c_1}{1-p_1z^{-1}}+\frac{c_2}{1-p_2z^{-1}}+\cdots+\frac{c_N}{1-p_nz^{-1}} \qquad (10.143)$$

Now each factor in the system is realized in direct form II and connected in parallel as shown in Fig. 10.36.

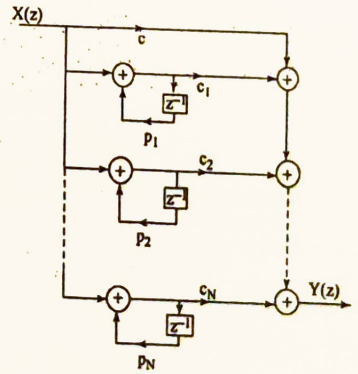


Fig. 10.36 Parallel form realization of Eq. (10.143).

Solved Problem 10.46 Obtain parallel form realization of the discrete-time system given in solved problem 10.45.

Solution:

Given

$$y(n)-\frac{1}{4}y(n-1)-\frac{1}{8}y(n-2)=x(n)+3x(n-1)+2x(n-2)$$

Taking z-transform on both sides, we get

$$Y(z)-\frac{1}{4}z^{-1}Y(z)-\frac{1}{8}z^{-2}Y(z)=X(z)+3z^{-1}X(z)+2z^{-2}X(z)$$

$$\Rightarrow \frac{Y(z)}{X(z)}=\frac{1+3z^{-1}+2z^{-2}}{1-\frac{1}{4}z^{-1}-\frac{1}{8}z^{-2}} \qquad (10.144)$$

Now we express the system function in partial fraction form as follows

$$H(z)=16+\frac{-15+7z^{-1}}{1+3z^{-1}+2z^{-2}}$$

$$=16+\frac{-15+7z^{-1}}{(1+z^{-1})(1+2z^{-1})}$$

$$=16+\frac{22}{1+z^{-1}}-\frac{37}{1+2z^{-1}} \qquad (10.145)$$

$$\begin{array}{r|l} & 16 \\ \hline \frac{1}{8}z^{-2}-\frac{1}{4}z^{-1}+1 & 2z^{-2}+3z^{-1}+1 \\ & 2z^{-2}-4z^{-1}+16 \\ \hline & 7z^{-1}-15 \end{array}$$

The realization of Eq. (10.145) is shown in Fig. 10.37.

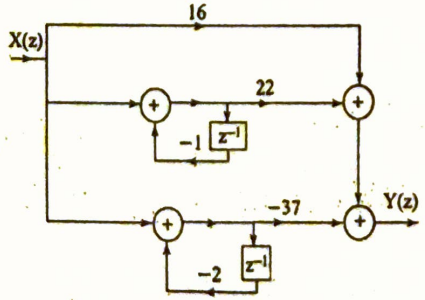


Fig. 10.37 Parallel-form realization of solved problem 10.45.

$$\frac{-15+7z^{-1}}{(1+z^{-1})(1+2z^{-1})}=\frac{A}{1+z^{-1}}+\frac{B}{1+2z^{-1}}$$

$$=\frac{A(1+2z^{-1})+B(1+z^{-1})}{(1+2z^{-1})(1+z^{-1})}$$

$$=\frac{(A+B)+(2A+B)z^{-1}}{(1+2z^{-1})(1+z^{-1})}$$

$$\Rightarrow A+B=-15$$

$$2A+B=7$$

$$\Rightarrow A=22$$

$$B=-37$$

Review Questions

Explain in detail about the following types of realization

1. Direct form - I
2. Direct form - II
3. Cascade form and
4. Parallel form

Practice Problem 10.23

Realize the system given by difference equation

$$y(n)=-0.1y(n-1)+0.72y(n-1)+0.7x(n)-0.252x(n-2)$$

in direct form I, direct form II, cascade and parallel form.

Practice Problem 10.24

Obtain the direct form *I*, direct form II, cascade and parallel form realization for the following system.

$$y(n)=-0.1y(n-1)+0.2y(n-2)+3x(n)+3.6x(n-1)+0.6x(n-2)$$

10.13 State Variable Model for Discrete-Time Systems

In chapter 7, we studied the state variable representation of continuous-time systems. In this section we will discuss the state variable representation for discrete-time systems.

The state of a system at some index n is minimum set of variables $q_1(n)$, $q_2(n),\ldots,q_n(n)$ such that the knowledge of these variables and future values of inputs allows one to calculate all future outputs of the system.

Consider an Nth order difference equation of the form

$$y(n)+\sum_{k=1}^{N}a_k y(n-k)=\sum_{k=0}^{M}b_k x(n-k) \qquad (10.146)$$

$$y(n)+a_1y(n-1)+\cdots\cdots+a_{N-1}y(n-N+1)+a_Ny(n-N)$$
$$=b_0x(n)+b_1x(n-1)+\cdots\cdots+b_{N-1}x(n-N+1)+b_Mx(n-M)$$

The system function of the system is given by

$$H(z)=\frac{Y(z)}{X(z)}=\frac{b_0+b_1z^{-1}+b_2z^{-2}+\cdots+b_{N-1}z^{(N-1)}+b_Mz^{-M}}{1+a_1z^{-1}+a_2z^{-2}+\cdots+a_{N-1}z^{-(N-1)}+a_Nz^{-N}} \qquad (10.147)$$

If $M=N$, the direct-form II realization of the system is as shown in Fig. 10.38.

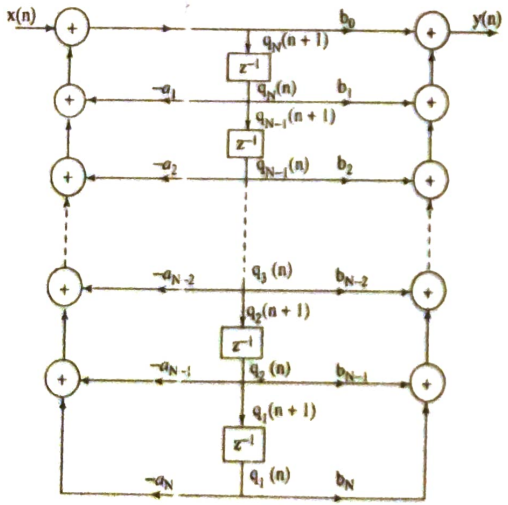


Fig. 10.38 Direct-form II realization of the system in Eq. (10.147).

Let us denote the signals appearing at the outputs of N delay elements as $q_1(n),q_2(n),\ldots,q_N(n)$. The input to the first delay element is $q_N(n+1)$.

We can write the output of each delay element as

$$\begin{aligned}
q_1(n+1)&=q_2(n)\\
q_2(n+1)&=q_3(n)\\
&\vdots\\
q_{N-1}(n+1)&=q_N(n)\\
q_N(n+1)&=-a_1q_N(n)-a_2q_{N-1}(n)\cdots-a_{N-1}\,q_2(n)-a_Nq_1(n)+x(n)
\end{aligned} \qquad (10.148)$$

The above equations are known as state equations and can be represented in matrix form as

$$\underbrace{\begin{bmatrix}q_1(n+1)\\q_2(n+1)\\\vdots\\q_{N-1}(n+1)\\q_N(n+1)\end{bmatrix}}_{Q(n+1)=}=\underbrace{\begin{bmatrix}0&1&0&\ldots&0\\0&0&1&\ldots&0\\\vdots&\vdots&\vdots&\ldots&\vdots\\0&0&0&\ldots&1\\-a_N&-a_{N-1}&-a_{N-2}&\ldots&a_1\end{bmatrix}}_{A}\underbrace{\begin{bmatrix}q_1(n)\\q_2(n)\\\vdots\\q_{N-1}(n)\\q_N(n)\end{bmatrix}}_{Q(n)}+\underbrace{\begin{bmatrix}0\\0\\\vdots\\0\\1\end{bmatrix}}_{B}x(n) \qquad (10.149)$$

$$Q(n+1)=AQ(n)+Bx(n) \qquad (10.150)$$

From Fig. 10.38, the output of the system can be written as

$$y(n)=b_0q_N(n+1)+b_1q_N(n)+b_2q_{N-1}(n)+\cdots+b_{N-1}q_2(n)+b_Nq_1(n) \qquad (10.151)$$

Substituting Eq. (10.148) in Eq. (10.151) we get

$$\begin{aligned}
y(n)&=b_0[-a_1q_N(n)-a_2q_{N-1}(n)-\cdots-a_{N-1}q_2(n)-a_Nq_1(n)+x(n)]\\
&\quad+b_1q_N(n)+b_2q_{N-1}(n)+\cdots+b_{N-1}q_2(n)+b_Nq_1(n)\\
&=(b_1-a_1b_0)q_N(n)+(b_2-a_2b_0)q_{N-1}(n)+\cdots+(b_{N-1}-a_{N-1}b_0)q_2(n)\\
&\quad+(b_N-a_Nb_0)q_1(n)+b_0x(n)
\end{aligned} \qquad (10.152)$$

$$y(n)=\underbrace{\begin{bmatrix}(b_N-a_Nb_0),(b_{N-1}-a_{N-1}b_0),\\\ldots(b_2-a_2b_0),(b_1-a_1b_0)\end{bmatrix}}_{C}\underbrace{\begin{bmatrix}q_1(n)\\q_2(n)\\\vdots\\q_{N-1}(n)\\q_N(n)\end{bmatrix}}_{Q(n)}+b_0x(n) \qquad (10.153)$$

$$y(n)=CQ(n)+Dx(n) \qquad (10.154)$$

Review Question

1. Define state of a system.

Solved Problem 10.47 Find the state variable matrices A,B,C and D for the equation

$$y(n)-3y(n-1)-2y(n-2)=x(n)+5x(n-1)+6x(n-2)$$

(Anna university Nov. 2002)

Solution:

Given the difference equation

$$y(n)-3y(n-1)-2y(n-2)=x(n)+5x(n-1)+6x(n-2) \qquad (10.155)$$

Taking z-transform on both sides with zero initial conditions

$$Y(z)-3z^{-1}Y(z)-2z^{-2}Y(z)=X(z)+5z^{-1}X(z)+6z^{-2}X(z)$$

$$Y(z)\left[1-3z^{-1}-2z^{-2}\right]=X(z)\left[1+5z^{-1}+6z^{-2}\right]$$

$$H(z)=\frac{Y(z)}{X(z)}=\frac{1+5z^{-1}+6z^{-2}}{1-3z^{-1}-2z^{-2}} \qquad (10.156)$$

$$\text{let} \quad \frac{Y(z)}{W(z)}=1+5z^{-1}+6z^{-2} \qquad (10.157)$$

and

$$\frac{W(z)}{X(z)}=\frac{1}{1-3z^{-1}-2z^{-2}} \qquad (10.158)$$

Now the direct form II realization of the system is

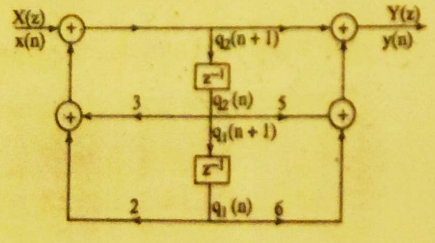


Fig. 10.39

Let the outputs of delay elements are $q_1(n)$ and $q_2(n)$, and the input of first delay element is $q_2(n+1)$.

Then we can write the state equation as

$$q_1(n+1)=q_2(n)$$

$$q_2(n+1)=2q_1(n)+3q_2(n)+x(n) \qquad (10.159)$$

and output equation is

$$\begin{aligned} y(n) &= q_2(n+1)+5q_2(n)+6q_1(n) \\ &= [2q_1(n)+3q_2(n)+x(n)]+5q_2(n)+6q_1(n) \\ &= 8q_1(n)+8q_2(n)+x(n) \end{aligned} \qquad (10.160)$$

The state matrix is

$$\begin{bmatrix} q_1(n+1) \\ q_2(n+1) \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}}_{A} \begin{bmatrix} q_1(n) \\ q_2(n) \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{B} x(n)$$

The output matrix is

$$y(n)=\begin{bmatrix} 8 & 8 \end{bmatrix}\begin{bmatrix} q_1(n) \\ q_2(n) \end{bmatrix}+x(n)$$

$$\Rightarrow A=\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}; B=\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$C=\begin{bmatrix} 8 & 8 \end{bmatrix} \qquad D=[1] \qquad (10.161)$$

Solved Problem 10.48 Find the state variable matrices A,B,C and D for the input-output relation given by the equations

$$y(n)=6y(n-1)+4y(n-2)+x(n)+10x(n-1)+12x(n-2)$$

(Anna University April 2003)

Solution:

Given

$$y(n) = 6y(n-1) + 4y(n-2) + x(n) + 10x(n-1) + 12x(n-2) \quad (10.162)$$

$$Y(z) = 6z^{-1}Y(z) + 4z^{-2}Y(z) + X(z) + 10z^{-1}X(z) + 12z^{-2}X(z)$$

$$\Rightarrow Y(z)\left[1 - 6z^{-1} - 4z^{-2}\right] = X(z)\left[1 + 10z^{-1} + 12z^{-2}\right]$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + 10z^{-1} + 12z^{-2}}{1 - 6z^{-1} - 4z^{-2}} \quad (10.163)$$

$$H(z) = \frac{Y(z)}{W(z)} \cdot \frac{W(z)}{X(z)} \quad (10.164)$$

where

$$\frac{W(z)}{X(z)} = \frac{1}{1 - 6z^{-1} - 4z^{-2}} \quad (10.165)$$

from Eq.(10.165) we can write

$$W(z) = X(z) + 6z^{-1}W(z) + 4z^{-2}W(z) \quad (10.166)$$

and

$$\frac{Y(z)}{W(z)} = 1 + 10z^{-1} + 12z^{-2}$$

$$\Rightarrow \quad Y(z) = W(z) + 10z^{-1}W(z) + 12z^{-2}W(z) \quad (10.167)$$

Realizing Eq. (10.166) and Eq. (10.167) and combining, we get

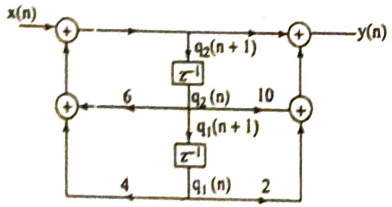


Fig. 10.40

Let us take output of the delay elements q_1 and q_2 as state variables, then from Fig. 10.40 we can write the state equation as

$$q_1(n+1) = q_2(n)$$

$$q_2(n+1) = -4q_1(n) - 6q_2(n) - x(n) \quad (10.168)$$

The output equation is

$$y(n) = q_2(n+1) + 10q_2 + 12q_1(n)$$

$$= [-4q_1(n) - 6q_2(n) - x(n)] + 10q_2(n) + 12q_1(n)$$

$$= 8q_1(n) + 4q_2(n) + x(n) \quad (10.169)$$

The state equation can be expressed in matrix form as

$$\begin{bmatrix} q_1(n+1) \\ q_2(n+1) \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -4 & -6 \end{bmatrix}}_{A} \begin{bmatrix} q_1(n) \\ q_2(n) \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{B} \quad (10.170)$$

The output equation is given by

$$y(n) = \underbrace{\begin{bmatrix} 8 & 4 \end{bmatrix}}_{C} \begin{bmatrix} q_1(n) \\ q_2(n) \end{bmatrix} + x(n)$$

$$A = \begin{bmatrix} 0 & 1 \\ -4 & -6 \end{bmatrix}; B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}; C = \begin{bmatrix} 8 & 4 \end{bmatrix}$$

$$D = [1] \quad (10.171)$$

Practice Problem 10.25

Find the state model for the system with system function

(i) $H(z) = \dfrac{2 + 4z^{-1} + 5z^{-2} + 12z^{-3}}{1 + 6z^{-1} + 9z^{-2} + 10z^{-3} + 12z^{-4}}$

(ii) $H(z) = \dfrac{(1 + 3z^{-1})(1 + 0.6z^{-2})}{(1 + 4z^{-1})(1 + 3z^{-1} + 2z^{-2})}$

Derivation of system function from state model

Consider the state model of discrete-time system

$$Q(n+1) = AQ(n) + Bx(n) \quad (10.172)$$

and

$$y(n) = CQ(n) + Dx(n) \quad (10.173)$$

Taking z-transform on both sides for Eq. (10.172) yields

$$zQ(z) - Q(0) = AQ(z) + BX(z) \tag{10.174}$$

and

$$Y(z) = CQ(z) + DX(z) \tag{10.175}$$

Assuming initial conditions to zero, we get

$$zQ(z) = AQ(z) + BX(z) \tag{10.176}$$

Since A and B are matrices, Eq. (10.176) can be written as follows

$$(zI - A)Q(z) = BX(z) \tag{10.177}$$

where I is unit matrix. Now

$$Q(z) = (zI - A)^{-1}BX(z) \tag{10.178}$$

Substituting Eq. (10.178) in Eq. (10.175) we obtain

$$Y(z) = C(zI - A)^{-1}BX(z) + DX(z) \tag{10.179}$$

$$\Rightarrow H(z) = \frac{Y(z)}{X(z)} = C(zI - A)^{-1}B + D \tag{10.180}$$

$$\boxed{H(z) = C\,\frac{\text{adj}(zI - A)}{\det(zI - A)} + D} \tag{10.181}$$

where $H(z)$ is the system function and

$\text{adj}(zI - A)$ is adjoint matrix of $(zI - A)$

$\det(zI - A)$ is determinant of $(zI - A)$

Solved Problem 10.49 Find the system function for the discrete-time system which is represented in state space representation as follows

$$\begin{bmatrix} q_1(n+1) \\ q_2(n+1) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} q_1(n) \\ q_2(n) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} x(n)$$

$$y(n) = \begin{bmatrix} 8 & 8 \end{bmatrix} \begin{bmatrix} q_1(n) \\ q_2(n) \end{bmatrix} + x(n)$$

Solution:

Given

$$A = \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}; B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}; C = \begin{bmatrix} 8 & 8 \end{bmatrix};\ D = [1]$$

$$H(z) = C(zI - A)^{-1}B + D$$

$$(zI - A) = z\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} z & -1 \\ -2 & z-3 \end{bmatrix}$$

$$(zI - A)^{-1} = \frac{1}{z(z-3) - 2}\begin{bmatrix} z-3 & 1 \\ 2 & z \end{bmatrix}$$

$$= \begin{bmatrix} \dfrac{z-3}{z^2-3z-2} & \dfrac{1}{z^2-3z-2} \\ \dfrac{2}{z^2-3z-2} & \dfrac{z}{z^2-3z-2} \end{bmatrix}$$

$$C[(I - A)^{-1}]B = \begin{bmatrix} 8 & 8 \end{bmatrix} \begin{bmatrix} \dfrac{z-3}{z^2-3z-2} & \dfrac{1}{z^2-3z-2} \\ \dfrac{2}{z^2-3z-2} & \dfrac{z}{z^2-3z-2} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} \dfrac{8(z-3)}{z^2-3z-2} + \dfrac{2}{z^2-3z-2} & \dfrac{8}{z^2-3z-2} + \dfrac{8z}{z^2-3z-2} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \frac{8(z+1)}{z^2-3z-2}$$

$$H(z) = C(zI - A)^{-1}B + D$$

$$= \frac{8(z+1)}{z^2-3z-2} + 1$$

$$= \frac{z^2+5z+6}{z^2-3z-2}$$

or

$$\boxed{H(z) = \frac{1 + 5z^{-1} + 6z^{-2}}{1 - 3z^{-1} - 2z^{-2}}}$$

10.14 Deconvolution using z-Transform

In section 3.16 we have studied about deconvolution, which is used to find the input $x(n)$ applied to the system once the impulse response $h(n)$ and the output $y(n)$ of the system are known. The z-transform also can be used for deconvolution operation.

We have the relation

$$Y(z) = X(z)H(z) \tag{10.182}$$

where $Y(z), X(z)$ and $H(z)$ are the z-transforms of system's output, input and impulse response respectively. Using Eq. (10.182) we can write

$$X(z) = \frac{Y(z)}{H(z)} \tag{10.183}$$

For a given impulse response $h(n)$ and output $y(n)$ of the system, we can compute their z-transforms $H(z)$ and $Y(z)$. Using Eq. (10.183) we can obtain $X(z)$. Now one can easily find the input sequence $x(n)$ by finding inverse z-transform of $X(z)$. Thus, the deconvolution is reduced to the procedure of evaluating an inverse z-transform.

Solved Problem 10.50 Find the input $x(n)$ of the system, if the impulse response $h(n)$ and the output $y(n)$ as shown below.

$$h(n) = \{\underset{\uparrow}{1}, 2, 3, 2\}$$

$$y(n) = \{\underset{\uparrow}{1}, 3, 7, 10, 10, 7, 2\}$$

Solution:

Given $h(n) = \{\underset{\uparrow}{1}, 2, 3, 2\}$

$\Rightarrow h(0) = 1; \quad h(1) = 2; \quad h(2) = 3; \quad h(3) = 2$

$$H(z) = \sum_{n=-\infty}^{\infty} h(n)z^{-n}$$

$$= \sum_{n=0}^{3} h(n)z^{-n}$$

$$= 1 + 2z^{-1} + 3z^{-2} + 2z^{-3}$$

$$Y(z) = \sum_{n=-\infty}^{\infty} y(n)z^{-n}$$

$$= \sum_{n=0}^{6} y(n)z^{-n}$$

$$= 1 + 3z^{-1} + 7z^{-2} + 10z^{-3} + 10z^{-4} + 7z^{-5} + 2z^{-6}$$

$$X(z) = \frac{Y(z)}{H(z)}$$

$$= \frac{1 + 3z^{-1} + 7z^{-2} + 10z^{-3} + 10z^{-4} + 7z^{-5} + 2z^{-6}}{1 + 2z^{-1} + 3z^{-2} + 2z^{-3}}$$

$$\begin{array}{r|l}
 & 1 + z^{-1} + 2z^{-2} + z^{-3} \\
\hline
1 + 2z^{-1} + 3z^{-2} + 2z^{-3} & 1 + 3z^{-1} + 7z^{-2} + 10z^{-3} + 10^{-4} + 7z^{-5} + 2z^{-6} \\
 & 1 + 2z^{-1} + 3z^{-2} + 2z^{-3} \\
\hline
 & z^{-1} + 4z^{-2} + 8z^{-3} + 10^{-4} \\
 & z^{-1} + 2z^{-2} + 3z^{-3} + 2z^{4} \\
\hline
 & 2z^{-2} + 5z^{-3} + 8z^{-4} + 7z^{-5} \\
 & 2z^{-2} + 4z^{-3} + 6z^{-4} + 4z^{-5} \\
\hline
 & z^{-3} + 2z^{-4} + 3z^{-5} + 2z^{-6} \\
 & z^{-3} + 2z^{-4} + 3z^{-5} + 2z^{-6} \\
\hline
 & 0
\end{array}$$

$\Rightarrow X(z) = 1 + z^{-1} + 2z^{-2} + z^{-3}$

$x(n) = \{\underset{\uparrow}{1}, 1, 2, 1\}$

Solved Problem 10.51 Given the sequence

$$x(n) = (-1)^n u(n) + a^n u(-u - k)$$

Determine the constraints on the complex number a and the integer k, given that the ROC of $x(z)$ is $1 < |z| < 3$

Solution:

The z-transform of the sequence $(-1)^n u(n)$ is

$$Z[(-1)^n u(n)] = \frac{1}{1 + z^{-1}} \quad \text{ROC: } |z| > 1$$

The z-transform of the sequence $a^n u(-n-1)$ is

$$Z[a^n u(-n-1)] = -\frac{1}{1 - az^{-1}} \quad \text{ROC: } |z| < a$$

$$Z[a^n u(-n-k)] = \frac{-z^{-k}}{1 - az^{-1}} \quad \text{ROC: } |z| < a$$

That is for any k, the ROC: $|z| < a$

Therefore for the given ROC $1 < |z| < 3$ the value of $|a| = 3$ and there is no constraint on 'k'.

Solved Problem 10.52 Find the inverse z- transform of

$$X(z)=\frac{1}{1024}\left[\frac{1024-z^{-10}}{1-\frac{1}{2}z^{-1}}\right],\quad |z|>0$$

Solution:

The ROC is entire z-plane. Therefore the sequence is of finite duration and the summation is also finite. The $X(z)$ can be written as

$$X(z)=\frac{1}{2^{10}}\left[\frac{2^{10}-z^{-10}}{1-\frac{1}{2}z^{-1}}\right]$$

$$=\frac{1-\left(\frac{1}{2}z^{-1}\right)^{10}}{1-\frac{1}{2}z^{-1}}$$

Comparing the above with finite sum formula $\frac{a(1-r^n)}{1-r}$

we find $a=1$; $r=\frac{1}{2}z^{-1}$ and $n=10$ terms

$$\sum_{n=0}^{n-1}r^n=\frac{1-r^n}{1-r}$$

$$\sum_{n=0}^{9}\left(\frac{1}{2}z^{-1}\right)^n=\frac{1-\left(\frac{1}{2}z^{-1}\right)^{10}}{1-\frac{1}{2}z^{-1}}$$

Therefore the $x(n)=\left(\frac{1}{2}\right)^n$ for $0\le n\le 9$

$=0$ otherwise

Solved Problem 10.53 Determine the z-transform of the sequence $x(n)=n\left(\frac{1}{2}\right)^{|n|}$. Find whether the Fourier transform of the sequence exists.

Solution:

Given $x(n)=n\left(\frac{1}{2}\right)^{|n|}$

Let us consider the sequence $x_1(n)=\left(\frac{1}{2}\right)^{|n|}$

For $n\ge 0$ $x_1(n)=\left(\frac{1}{2}\right)^n u(n)$

For $n<0$ $x_1(n)=2^n u(-n-1)$

The z-transform of $x_1(n)$ for $n\ge 0$ is $\frac{1}{1-\frac{1}{2}z^{-1}}$ with ROC: $|z|>\frac{1}{2}$

The z-transform of $x_1(n)$ for $n<0$ is $\frac{-1}{1-2z^{-1}}$ with ROC: $|z|<2$

Therefore $X_1(z)=\frac{1}{1-\frac{1}{2}z^{-1}}-\frac{1}{1-2z^{-1}}$ ROC: $\frac{1}{2}<|z|<2$

We know $Z[nx(n)]=-z\frac{d}{dz}X(z)$. Therefore

$$Z\left[n\left(\frac{1}{2}\right)^{|n|}\right]=-z\frac{d}{dz}\left\{\frac{1}{1-\frac{1}{2}z^{-1}}\right\}+z\frac{d}{dz}\left\{\frac{1}{1-2z^{-1}}\right\}$$

$$-\frac{\left(\frac{1}{2}z^{-1}\right)}{(1-0.5z^{-1})^2}+\frac{2z^{-1}}{(1-2z^{-1})}\quad \text{ROC: } \frac{1}{2}<|z|<2$$

since ROC is $\frac{1}{2}<|z|<2$ Fourier transform exists.

Solved Problem 10.54 Given the following five facts about a discrete-time signal $x(n)$ with z-transform $X(z)$

(i) $x(n)$ is real and right-sided.

(ii) $X(z)$ has exactly two poles.

(iii) $X(z)$ has two zeros at the origin.

(iv) $X(z)$ has a pole at $z=\frac{1}{2}e^{j\pi/3}$.

(v) $x(1)=\frac{8}{3}$.

Determine $X(z)$ and find ROC.

Solution:

The signal $x(n)$ is real. Its z-transform has two poles. One pole is at $z=\frac{1}{2}e^{-j\pi/3}$. The other pole is at $z=\frac{1}{2}e^{-j\pi/3}$. $X(z)$ has two zeros at origin. Therefore the numerator is of the form Az^2

$$X(z)=\frac{Az^2}{\left(z-\frac{1}{2}e^{j\pi/3}\right)\left(z-\frac{1}{2}e^{-j\pi/3}\right)}$$

Given $x(1)=\frac{8}{3}$

$$x(1)=\frac{A}{\left(1-\frac{1}{2}e^{j\pi/3}\right)\left(1-\frac{1}{2}e^{-j\pi/3}\right)}=\frac{A}{1-\frac{1}{2}+\frac{1}{4}}=\frac{4A}{3}$$

$$\frac{4A}{3}=\frac{8}{3}\Rightarrow A=2$$

$$X(z)=\frac{2z^2}{\left(z-\frac{1}{2}e^{j\pi/3}\right)\left(z-\frac{1}{2}e^{-j\pi/3}\right)}$$

ROC: $|z|>\left|\frac{1}{2}e^{j\pi/3}\right| \Rightarrow |z|>\frac{1}{2}$

Solved Problem 10.55 A causal LTI system is described by the difference equation

$$y(n)=y(n-1)+y(n-2)+x(n-1)$$

(a) Find the system function (b) Find the unit sample response

Solution:

Given $y(n)=y(n-1)+y(n-2)+x(n-1)$

Applying z-transform

$$Y(z)=z^{-1}Y(z)+z^{-2}Y(z)+z^{-1}X(z)$$

The system function

$$H(z)=\frac{z^{-1}}{1-z^{-1}-z^{-2}}$$

$$H(z)=\frac{z^2}{z^2-z-1}$$

$$\frac{H(z)}{z}=\frac{z}{z^2-z-1}$$ The pole are $\frac{1\pm\sqrt{5}}{2}$

$$=\frac{A}{\left[z-\left(\frac{1+\sqrt{5}}{2}\right)\right]}+\frac{B}{\left[z-\left(\frac{1-\sqrt{5}}{2}\right)\right]}$$

$$H(z)=\frac{(-1/\sqrt{5})z}{\left[z-\left(\frac{1+\sqrt{5}}{2}\right)\right]}+\frac{\frac{1}{\sqrt{5}}z}{\left[z-\left(\frac{1-\sqrt{5}}{2}\right)\right]}$$

$$h(n)=\left[-\frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n+\frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n\right]u(n)$$

Solved Problem 10.56 Prove that for an even sequence $x(n)$ with rational z-transform $X(z)$.

(i) $X(z)=X(z^{-1})$

(ii) Show that if a pole of $X(z)$ occurs at $z=a$ then a pole must also occur at $z=\frac{1}{a}$. Verify the result for $x(n)=\delta(n+1)+2\delta(n)+\delta(n-1)$

Solution:

$$Z[x(-n)]=\sum_{n=-\infty}^{\infty}x(-n)z^{-n}$$

since $x(n)=x(-n)$

$$Z[x(-n)]=\sum_{n=-\infty}^{\infty}x(n)z^{n}$$

$$=\sum_{n=-\infty}^{\infty}x(n)(z^{-1})^{-1}$$

$$=X(z^{-1})$$

Since $x(n)=x(-n)$

$$X(z)=X(z^{-1})$$

If $z=a$ is a pole then $X(a)=\infty$ or $\frac{1}{X(a)}=0$. Since $X(a)=X(1/a)$ we can write

$$\frac{1}{X(1/a)}=0$$

Therefore there is a pole at $z=1/a$.

(c)

$$x(n)=\delta(n+1)+2\delta(n)+\delta(n-1)$$

$$x(z)=z+2+z^{-1}=\frac{z^2+2z+1}{z}\qquad \text{ROC:}|z|>0$$

$X(z)$ has two zeros at $z=-1$; pole at $z=0$ and $z=\infty$

$X(z)$ has zeros at $z_1=-1$ and $z_2=\frac{1}{-1}=-1$

Short Questions and Answers

1. **Defind z-transform.**

The z-tranform of a discrete time sequence $x(n)$, is defined as

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

where z is a complex variable. In polar form z can be expressed as

$$z = re^{j\omega}$$

where r is radius of the circle.

2. **Define inverse z-transform.**

The inverse z-transform of $X(z)$ is defined as

$$x(n) = \frac{1}{2\pi j}\oint_c X(z)z^{n-1}dz$$

3. **What is meant by Region of Convergence?** **AU Nov 2008**

The region of convergence (ROC) of $X(z)$ is the set of all values of z for which $X(z)$ attains a finite value.

4. **What is ROC of a finite duration causal sequence?**

The ROC of a finite duration causal sequence is entire z- plane except at $z = 0$.

5. **What is ROC of a finite duration anticausal sequence?**

The ROC of a finite duration anticausal sequence is entire z-plane except at $z = \infty$.

6. **What is ROC of infinite duration causal sequence?**

The ROC of an infinite duration causal sequence is exterior to the circle of radius r.

7. **What is ROC of an infinite duration anticausal sequence?**

The ROC of an infinite duration anticausal sequence is interior to the circle of radius r.

8. **State time shifting property of z-transform.** **AU Nov 2008**

If $X(z) = z\{x(n)\}$ with the initial conditions for $x(n)$ are zero, then

$$Z\{x(n-m)\} = z^{-m}X(z)$$

where m is a positive or a negative integer.

9. **State linearity property of z-transform.**

If $Z\{x_1(n)\} = X_1(z)$ and
$Z\{x_2(n)\} = X_2(z)$
then
$$Z\{a_1x_1(n) + a_2x_2(n)\} = a_1X_1(z) + a_2X_2(z)$$

10. **State the scaling property of the z-transform. AU Nov 2008**

If $Z\{x(n)\} = X(z)$, ROC: $r_1 < |z| < r_2$
then
$$Z[a^n x(n)] = X(a^{-1}z) \quad \text{ROC: } |a|r_1 < |z| < |a|r_2$$

11. **State the time reversal property of the z-transform.**

If $Z\{x(n)\} = X(z)$, ROC: $r_1 < |z| < r_2$
then
$$Z\{x(-n)\} = X(z^{-1}) \quad \text{ROC: } \frac{1}{r_2} < |z| < \frac{1}{r_1}$$

12. **State the convolution property of the z-transform.**

If $Z\{x_1(n)\} = X(z)$ and
$Z\{x_2(n)\} = X_2(z)$
then
$$Z\{x_1(n)\} * x_2(n)\} = X_1(z)X_2(z)$$

13. **State the conjugation property of z-transform.**

If $Z\{x(n)\} = X(z)$
then
$$Z\{x^*(n)\} = X^*(z^*)$$

14. **What is the z-transform of the sequence $x(n) = a^n u(n)$?**

$$X(z) = \frac{z}{z-a} \quad \text{ROC: } |z| > |a|$$

15. **Find the z-transform of (a) impulse (b) unit step.**

(a) For an impulse sequence $x(n) = \delta(n)$; the z-transform $X(z) = 1$

(b)

For a unit step $u(n) = 1$ for $n \geq 0$

$= 0$ for $n < 0$

$$X(z) = \sum_{n=0}^{\infty} 1 \cdot z^{-n} = \frac{1}{1 - z^{-1}}$$

16. What is the z-transform of the sequence $x(n) = nu(n)$?

$$Z[u(n)] = \frac{z}{z-1}$$

$$Z[nu(n)] = -z\frac{d}{dz}\left(\frac{z}{z-1}\right) \qquad \boxed{Z[nx(n)] = -z\frac{d}{dz}X(z)}$$

$$= \frac{-z\{(z-1)-z\}}{(z-1)^2} = \frac{z}{(z-1)^2}$$

17. What is the relationship between z-transform and DTFT?

The z-transform of the sequence $x(n)$ is given by

$$X(z) = Z\{x(n)\} = \sum_{n=-\infty}^{\infty} x(n)z^{-n} \tag{1}$$

where $z = re^{j\omega}$ (2)

Substituting Eq. (2) in Eq. (1) we get

$$X(re^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)r^{-n}e^{-j\omega n} \tag{3}$$

The Fourier transform of $x(n)$ is given by

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} \tag{4}$$

Eq. (3) and Eq. (4) are identical when $r = 1$.

In this z-plane this corresponds to the locus of points on the unit circle $|z| = 1$. Hence $H(e^{j\omega})$ is equal to $H(z)$ evaluated along the unit circle.

$$X(e^{j\omega}) = X(z)\big|_{z=e^{j\omega}}$$

18. State the multiplication property of the z-transform.

If $Z\{x_1(n)\} = X_1(z)$

and

$Z\{x_2(n)\} = X_2(z)$

then

$$Z\{x_1(n)x_2(n)\} = \frac{1}{2\pi j}\oint_c X_1(v)X_2\left(\frac{z}{v}\right)v^{-1}dv$$

19. State the initial value theorem in z-transform.

If $x(n)$ is causal, then

$$x(0) = \lim_{z\to\infty} X(z)$$

20. State the final value theorem in z-transform.

If $x(n)$ is causal, $z\{x(n)\} = X(z)$, where the ROC for $X(z)$ includes, but is not necessarily confined to $|z| > 1$ and $(z-1)X(z)$ has no poles on or outside the unit circle then

$$x(\infty) = \lim_{z\to 1}(z-1)X(z)$$

21. State Parseval's relation in z-transform.

If $x_1(n)$ and $x_2(n)$ are complex-valued sequences, then

$$\sum_{n=-\infty}^{\infty} x_1(n)x_1^*(n) = \frac{1}{2\pi j}\oint_c X_1(v)X_2^*\left(\frac{1}{v^*}\right)v^{-1}dv$$

22. What are the properties of Region of convergence?

1. The ROC is a ring or disc in the z-plane centered at the origin.
2. The ROC cannot contain any poles.
3. The ROC of an LTI stable system contain the unit circle.
4. The ROC must be a connected region.

23. What are the different methods of evaluating inverse z-transform?

1. Long division method.
2. Partial fraction expansion method.
3. Residue method.
4. Convolution method.

24. Define system function.

Let $x(n)$ and $y(n)$ are the input and output sequences of an LTI system with impulse $h(n)$. Then the system function of the LTI system is defined as the ratio of $Y(z)$ and $X(z)$. That is

$$H(z) = \frac{Y(z)}{X(z)}$$

where $Y(z)$ is the z-transform of the output signal $y(n)$ and $X(z)$ is the z-transform of the input signal $x(n)$.

25. Determine the system function of the discrete time system described by the difference equation.

$$y(n) - \frac{1}{2}y(n-1) + \frac{1}{4}y(n-2) = x(n) - x(n-1)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 - z^{-1}}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}$$

Multiple Choice Questions

1. The z-transform of the following real exponential sequence

$$x(n) = a^n \quad n \geq 0$$
$$= 0 \quad \text{for } n < 0 \quad \text{is given by}$$

(a) $1 - az^{-1}; |z| > a$ (b) $\frac{1}{1 - az^{-1}}; |z| > a$

(c) $-\frac{1}{1 - az}; |z| > a$ (d) $1 + az^{-1}; |z| < a$

2. A linear discrete-time system has the characteristic equation $z^3 - 0.81z = 0$. The system

(a) is stable (b) is marginally stable (c) is unstable

3. The z-transform $f(z)$ of the function $f(nT) = a^{nT}$ is

(a) $\frac{z}{z - a^T}$ (b) $\frac{z}{z + a^T}$ (c) $\frac{z}{z - a^T}$ (d) $\frac{z}{z + a^{-T}}$

4. The z-transform of a signal is given by $X(z) = \frac{z^{-1}(1 - z^{-4})}{4(1 - z^{-1})^2}$. Its final value is

(a) $\frac{1}{4}$ (b) 0 (c) 1.0 (d) ∞

5. The region of convergence of the z-transform of a unit step function is

(a) $|z| > 1$ (b) $|z| < 1$
(c) (Real part of z) > 0 (d) (Real part of z) < 0

6. If the impulse response of a discrete time system is $h(n) = -5^n u(-n-1)$ then the system function $H(z)$ is equal to

(a) $\frac{-z}{z-5}$ and the system is stable (b) $\frac{z}{z-5}$ and the system is stable
(c) $\frac{-z}{z-5}$ and the system is unstable (d) $\frac{z}{z-5}$ and the system is unstable

7. A sequence $x(n)$ with z-transform $X(z) = z^4 + z^2 - 2z + 2 - 3z^{-4}$ is applied as an input to a linear, time invariant system with impulse response $h(n) = 2\delta(n-3)$ where

$$\delta(n) = \begin{cases} 1 & n = 0 \\ 0 & \text{otherwise} \end{cases}$$

The output at $n = 4$ is

(a) -6 (b) zero (c) 2 (d) -4

8. The z-transform of a system is $H(z) = \frac{z}{z - 0.2}$. If the ROC is $|z| < 0.2$, then the impulse response of the system is

(a) $(0.2)^n u(n)$ (b) $(0.2)^n u(-n-1)$
(c) $-(0.2)^n u(n)$ (d) $-(0.2)^n u(-n-1)$

9. The region of convergence of z-transform of the sequence $\left(\frac{5}{6}\right)^n u(n) - \left(\frac{6}{5}\right)^n u(-n-1)$ must be

(a) $|z| < \frac{5}{6}$ (b) $|z| > \frac{6}{5}$

(c) $\frac{5}{6} < |z| < \frac{6}{5}$ (d) $\frac{6}{5} < |z| < \infty$

10. Given that $F(z)$ and $G(z)$ are the one sided z-transforms of discrete time functions $f(nT)$ and $g(nT)$, the z-transform of $\sum f(kT)g(nT - kT)$ is given by

(a) $\sum f(nT)g(nT)z^{-n}$ (b) $\sum f(nT)z^{-n}\sum g(nT)z^{-n}$

(c) $\sum f(kT)g(nT - kT)z^{-n}$ (d) $\sum f(nT - kT)g(nT)z^{-n}$

11. Which one of the following represents the impulse response of a system defined by

$$H(z) = z^{-m}$$

(a) $u(n-m)$ (b) $\delta(n-m)$ (c) $\delta(m)$ (d) $\delta(m-n)$

12. If the function $H_1(z) = 1 + 1.5z^{-1} - z^{-2}$ and $H_2(z) = z^2 + 1.5z - 1$ then

(a) The poles and zeros of the functions will be the same.
(b) The poles of the functions will be identical but not zero.
(c) The zeros of the functions will be identical but not the poles.
(d) Neither the poles nor the zeros of the two functions will be identical

13. Consider the following statements regarding a linear discrete time system

$$H(z) = \frac{z^2+1}{(z+0.5)(z-0.5)}$$

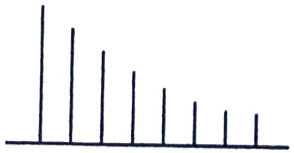
1. The system is stable.

2. The initial value $h(0)$ of the impulse response is -4.

3. The steady state output is zero for a sinusoidal discrete time input of frequency equal to one-fourth the sampling frequency. Which of these statements are correct?

(a) 1, 2 and 3 (b) 1 and 2 (c) 1 and 3 (d) 2 and 3

14. The impulse response of a discrete time system with a pole is shown in figure. The pole of the system must be located on the



(a) real axis at $z = -1$.

(b) real axis between $z = 0$ and $z = 1$.

(c) imaginary axis at $z = j$.

(d) imaginary axis between $z = 0$ and $z = j$.

15. Consider the compound system shown in the figure. Its output is equal to input with a delay of two units. If the transfer function of the first system is given by $H_1(z) = \frac{z-0.5}{z-0.8}$, then the transfer function of the second system would be

(a) $H_2(z) = \frac{z^{-2}-0.2z^{-3}}{1-0.4z^{-1}}$ (b) $H_2(z) = \frac{z^{-2}-0.8z^{-3}}{1-0.5z^{-1}}$

(c) $H_2(z) = \frac{z^{-1}-0.2z^{-3}}{1-0.4z^{-1}}$ (b) $H_2(z) = \frac{z^{-2}+0.8z^{-3}}{1+0.5z^{-1}}$

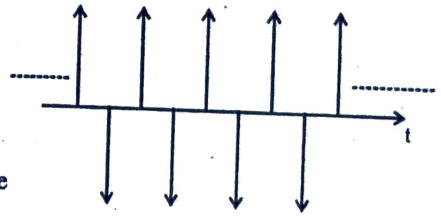
16. The minimum number of delay elements required in realizing a digital filter with the transfer function

$$H(z) = \frac{1+az^{-1}+bz^{-2}}{1+cz^{-1}+dz^{-2}+ez^{-3}}$$

is

(a) 2 (b) 3 (c) 4 (d) 5

17. The impulse response of a discrete system with a simple pole is shown in the given figure



The pole must be located at

(a) on the real axis at $z = 1$

(b) on the real axis at $z = -1$

(c) at the origin of the z-plane

(d) at $z = \infty$

18. If the output of discrete-time LTI system is always identical to the input signal, then the unit-impulse response $h(n)$ is

(a) unit step (b) unit impulse (c) all ones (d) ramp

19. If the unit impulse response of an LTI system and the input signal are both rectangular pulses, then the response will be a

(a) ramp function (b) parabola (c) triangular pulse (d) rectangular pulse

20. If an LTI system has a unit impulse response with a finite number of non-zero values and the input signal has a finite number of non-zero values, then the output signal is

(a) all ones (b) all zeros (c) constant (d) has a finite number of non-zero values

Answers to Multiple Choice Questions

1. (b)	6. (b)	11. (b)	16. (b)
2. (a)	7. (b)	12. (c)	17. (b)
3. (a)	8. (d)	13. (b)	18. (b)
4. (c)	9. (c)	14. (b)	19. (c)
5. (a)	10. (b)	15. (b)	20. (d)

11

Signal Transmission Through Linear Systems

11.1 Introduction

A system is one which do some operation on an input signal $x(t)$ to give an output signal $y(t)$. For an input signal $x(t)$, the output $y(t)$ is the convolution of input signal $x(t)$ and impulse response $h(t)$ of the system. That is

$$y(t) = x(t) * h(t) \tag{11.1}$$

Fig. 11.1 A continuous - time system

In frequency domain Eq. (11.1) can be represented as

$$Y(j\Omega) = X(j\Omega)H(j\Omega) \tag{11.2}$$

Eq. (11.2) shows that the output spectrum can be obtained by multiplying the input spectrum and the spectral response (transfer function) of the system.

Expressing Eq. (11.2) in polar form, we obtain

$$|Y(j\Omega)|e^{j\underline{/Y(j\Omega)}} = |X(j\Omega)||H(j\Omega)|e^{j(\underline{/X(j\Omega)}+\underline{/H(j\Omega)})} \tag{11.3}$$

from which

$$|Y(j\Omega)| = |X(j\Omega)||H(j\Omega)| \tag{11.4a}$$

and

$$\underline{/Y(j\Omega)} = \underline{/X(j\Omega)} + \underline{/H(j\Omega)} \tag{11.4b}$$

From Eq. (11.4) we find that the input amplitude spectrum $|X(j\Omega)|$ is changed to $|H(j\Omega)|X(j\Omega)|$ and the input phase spectrum $\underline{/X(j\Omega)}$ is changed to $\underline{/X(j\Omega)} +$

$\angle H(j\Omega)$. That is the amplitude and phase spectra of $H(j\Omega)$ dictates how the system modifies the amplitude and phase of various sinusoidal inputs. For this reason $H(j\Omega)$ is called the frequency response of the system. When the input signal is passed through the system, the system acts as a filter which may boost certain frequency components and attenuate others depending upon the nature of $H(j\Omega)$. It also changes the relative phases of various frequency components. As a result, the output may not resemble the input. Then the signal is said to be distorted. In many applications the systems are designed to modify the input signal. For example, a lowpass filter is an example of a system that distorts the input signal by allowing low frequency components and attenuating high frequency components.

11.2 Distortionless Transmission through a System

In applications such as amplifier, it is desirable that the output resembles the input. That is, the output must be free from all types of distortions. Distortions in linear systems are two types.

1. The amplitudes of various components may undergo different amplification. This type of distortion is known as amplitude distortion. This arises if $|H(j\Omega)|$ is not constant for all values of Ω.
2. The second one is the relative phases of various components may be disturbed (phase distortion); this occurs if $\angle H(j\Omega)$ is not linear through the origin.

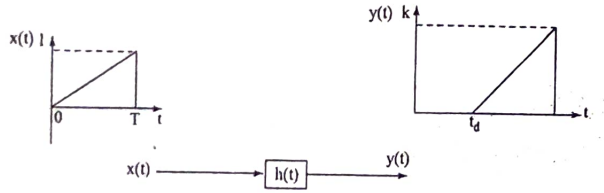


Fig. 11.2 A distortionless system

A transmission is said to be ideal or distortionless if the output wave form is the same as input wave form, except that the magnitude is scaled by a constant k and the waveform is delayed by t_d sec. Thus if $x(t)$ is the input signal, a distortionless system has an output

$$y(t) = kx(t - t_d) \tag{11.5}$$

This concept is illustrated in Fig. 11.2.

11.3 Linear Phase Systems

In previous section, we studied that if the phase shift of the filter is linear, then the system has no phase distortion. Therefore, the transfer function of a linear phase system is

$$H(j\Omega) = |H(j\Omega)|e^{-j\Omega t_d} \tag{11.6}$$

The impulse response of such system can be obtained by finding inverse Fourier transform of Eq. (11.6).

That is,

$$\begin{aligned}
h(t) &= F^{-1}[H(j\Omega)] \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty} |H(j\Omega)|e^{-j\Omega t_d}e^{j\Omega t}\,d\Omega \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty} |H(j\Omega)|e^{j\Omega(t-t_d)}\,d\Omega \\
&= \frac{1}{2\pi}\left[\int_{-\infty}^{0} |H(j\Omega)|e^{j\Omega(t-t_d)}\,d\Omega + \int_{0}^{\infty} |H(j\Omega)|e^{j\Omega(t-t_d)}\,d\Omega\right] \\
&= \frac{1}{2\pi}\left[\int_{0}^{\infty} |H(j\Omega)|e^{-j\Omega(t-t_d)}\,d\Omega + \int_{0}^{\infty} |H(j\Omega)|e^{j\Omega(t-t_d)}\,d\Omega\right] \\
&= \frac{1}{\pi}\left[\int_{0}^{\infty} |H(j\Omega)|\cos\,\Omega(t-t_d)\,d\Omega\right]
\end{aligned} \tag{11.7}$$

The impulse response is symmetrical about t_d, because

$$h(t+t_d) = h(t_d - t) \tag{11.8}$$

and the maximum value of $h(t)$ is at $t = t_d$.

$$h_{max} = h(t_d) = \frac{1}{\pi}\int_{0}^{\infty} |H(j\Omega)|\,d\Omega \tag{11.9}$$

From Eq. (11.7)

$$h(t+t_d) = \frac{1}{\pi}\int_{0}^{\infty} |H(j\Omega)|\cos\,\Omega t\,d\Omega$$

and

$$h(t_d - t) = \frac{1}{\pi}\int_{0}^{\infty} |H(j\Omega)|\cos\,\Omega t\,d\Omega$$

Therefore $h(t+t_d) = h(t_d - t)$.

That is, for a linear phase system the impulse response $h(t)$ is symmetrical about $t = t_d$ and it is noncausal (nonzero for $t < 0$).

The Fourier transform of Eq. (11.5) gives

$$Y(j\Omega) = kF[x(t-t_d)] \tag{11.10}$$

$$= ke^{-j\Omega t_d}X(j\Omega) \quad (\because \text{ by time shifting property of Fourier transform}) \tag{11.11}$$

From Eq. (11.2) we have

$$Y(j\Omega) = H(j\Omega)X(j\Omega) \tag{11.12}$$

Comparing Eq. (11.11) and Eq. (11.12) we obtain

$$H(j\Omega) = ke^{-j\Omega t_d} \tag{11.13}$$

This is the transfer function required for distortionless transmission.

From Eq. (11.13) we find that

$$|H(j\Omega)| = k$$
$$\underline{/H(j\Omega)} = -\Omega t_d \tag{11.14}$$

The magnitude of the transfer function is k, which is independent of Ω and the phase shift of the transfer function is $-\Omega t_d$; a linear function of Ω. If $k > 1$, the system is said to have gain and if $k < 1$, the system is said to have attenuation.

A plot of $|H(j\Omega)|$ and $\underline{/H(j\Omega)}$ is shown in Fig. 11.3. A system is said to have amplitude distortion if $|H(j\Omega)|$ is not constant, and phase distortion if $\underline{/H(j\Omega)}$ is not linear through the origin.

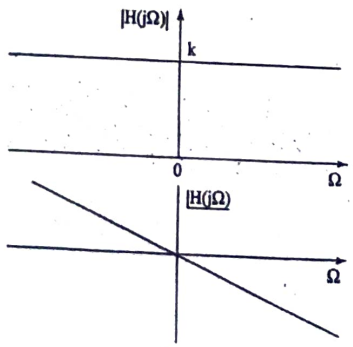


Fig. 11.3 Frequency response of distortionless system.

Review Questions

1. What is a distortionless system? Explain.
2. What are the different types of distortions? Explain.
3. Obtain the conditions for the distortionless transmission through a system.

11.4 Ideal Filters

An ideal filter is a system that allows distortionless transmission for all frequencies in a certain band. The amplitude spectrum over the band is constant and the phase spectrum over the band is linear. In Fig. 11.4 the magnitude characteristics of lowpass, highpass, bandpass, and bandreject filters are shown.

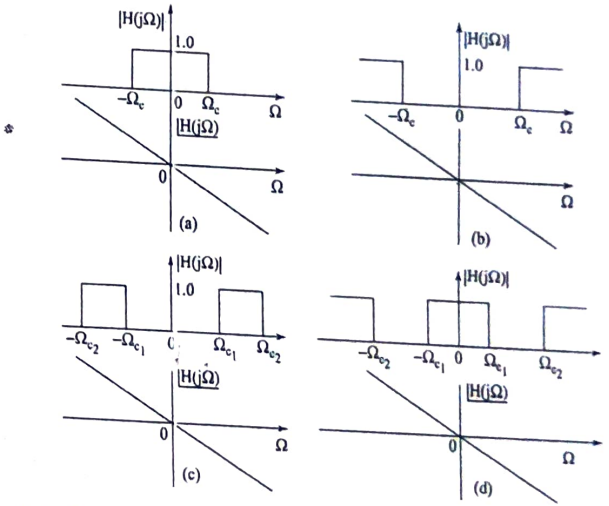


Fig. 11.4 Frequency response of ideal filters (a) Lowpass (b) Highpass (c) Bandpass (d) Bandreject.

The ideal lowpass filter characteristics is defined by the transfer function

$$H(j\Omega) = e^{-j\Omega t_d} \quad \Omega \le |\Omega_c|$$
$$= 0 \qquad \Omega > |\Omega_c| \tag{11.15}$$

where Ω_c is called as cutoff frequency of the filter. We can find the impulse response of the lowpass filter by finding the inverse Fourier transform of Eq. (11.15). That is

$H(j\Omega) = |H(j\Omega)|e^{-j\Omega t_d}$
For an ideal filter, in passband
$|H(j\Omega)| = 1$
Therefore
$H(j\Omega) = e^{-j\Omega t_d}$

$$h(t) = F^{-1}[H(j\Omega)]$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} H(j\Omega)e^{j\Omega t}d\Omega = \frac{1}{2\pi}\int_{-\Omega_c}^{\Omega_c} e^{-j\Omega t_d}e^{j\Omega t}d\Omega$$

$$= \frac{1}{2\pi}\int_{-\Omega_c}^{\Omega_c} e^{j\Omega(t-t_d)}d\Omega = \frac{1}{2\pi j(t-t_d)}e^{j\Omega(t-t_d)}\Big|_{-\Omega_c}^{\Omega_c}$$

$$= \frac{1}{2\pi j(t-t_d)}\left[e^{j\Omega_c(t-t_d)} - e^{-j\Omega_c(t-t_d)}\right]$$

$$= \frac{1}{\pi(t-t_d)}\left[\frac{e^{j\Omega_c(t-t_d)} - e^{-j\Omega_c(t-t_d)}}{2j}\right]$$

$$= \frac{1}{\pi(t-t_d)}\sin(\Omega_c(t-t_d))$$

$$= \frac{\Omega_c}{\pi}\frac{\sin(\Omega_c(t-t_d))}{\Omega_c(t-t_d)}$$

$$= \frac{\Omega_c}{\pi}\,\text{sinc}[\Omega_c(t-t_d)] \qquad (11.16)$$

The impulse response $h(t)$ is sketched in Fig. 11.5. From Fig.11.5 we can find that

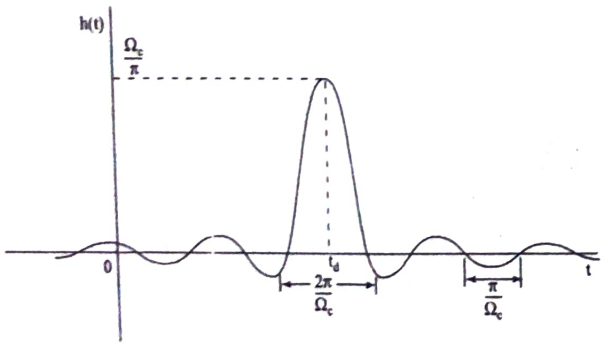


Fig. 11.5 The impulse response of an ideal lowpass filter.

the impulse response has peak value when $t = t_d$. This value, $\frac{\Omega_c}{\pi}$, is proportional to the cutoff frequency. The width of the main lobe is $\frac{2\pi}{\Omega_c}$. Notice that as $\Omega_c \to \infty$, the filter becomes an allpass filter, as $t_d \to 0$, the output response peak $\to \infty$. That is, the output response approaches input. Further we can observe that the impulse response $h(t)$ is non-zero for $t < 0$, even though the input $\delta(t)$ is applied at $t = 0$. That is the impulse response begins even before the input is applied. In real life, no system exhibits such type of characteristics. Hence we can conclude that the ideal lowpass filter of Fig. 11.4a is physically unrealizable. Similarly we can show that other ideal filters like highpass and bandpass filters are also physically unrealizable.

From the above discussion we find that for a physically realizable system, $h(t)$ must be causal, that is

$$h(t) = 0 \quad \text{for} \quad t < 0 \qquad (11.17)$$

In the frequency domain, the condition for physical realizability is given by

$$\int_{-\infty}^{\infty} \frac{|\ln|H(j\Omega)||}{1+\Omega^2}d\Omega < \infty \qquad (11.18)$$

where $|H(j\Omega)|$ is the magnitude function, such that the area under $|H(j\Omega)|^2$ from $\Omega = -\infty$ to ∞ is finite. This is known as **Paley-Wiener Criterion**. If $|H(j\Omega)| = 0$, over a finite bandwidth, then the integral in Eq. (11.18) becomes infinite because $|\ln|H(j\Omega)|| = \infty$. Hence all the ideal filters are physically unrealizable.

Review Questions

1. Draw the ideal characteristics of all types of filters.
2. Ideal filters are physically unrealizable. Justify.
3. State Paley - Wiener criterion.

11.5 Signal Bandwidth

The spectral components of a signal extends from $\Omega = -\infty$ to ∞. Any practical signal have finite energy. As a result, the spectral components approaches to zero as $\Omega \to \infty$. Therefore, we can neglect the spectral components that has negligible energy and select a band of frequency components that contain most of the signal energy. This band of frequencies that contains most of the signal energy is known as bandwidth of the signal and is denoted by f_m. Normally the band

is selected such that it contains around 95% (lower or higher than 95%) of total energy depending on the precision.

11.6 System Bandwidth

In section (11.2) we studied about the requirement of a distortionless system. But in practice, it is impossible to design an ideal distortionless system, because the response of all practical systems diminishes at higher frequencies. But it is possible to design a system that is distortionless over a certain range of frequencies. This range of frequencies is known as bandwidth of the system. For example if $H(j\Omega)$ is the transfer function of a system, the bandwidth of the system can be defined as the interval of frequencies for which $|H(j\Omega)|$ remain with in $\frac{1}{\sqrt{2}}$ of its maximum value ($-3dB$ down). Fig. 11.6 shows the frequency response of a practical bandpass filter, from which one can find the bandwidth as $(\Omega_2 - \Omega_1)$ rad/sec.

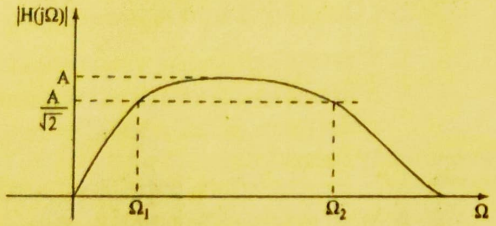


Fig. 11.6 Bandwidth of a system.

Definition:

Bandwidth of a system: The range of frequencies for which the magnitude $|H(j\Omega)|$ of the system remains with in $\frac{1}{\sqrt{2}}$ of its maximum value.

11.7 Relationship between Bandwidth and Rise Time

In this section, we are interested in the response of a lowpass filter when a step input is applied. If the impulse response is known, then the step response can be obtained by convolution; that is

$$y(t) = h(t) * u(t) \tag{11.19}$$

$$= \int_{-\infty}^{t} h(\tau)d\tau \tag{11.20}$$

From Eq.(11.6) we have

$$y(t) = \int_{-\infty}^{t} \frac{\Omega_c}{\pi} \frac{\sin \Omega_c(\tau - t_d)}{\Omega_c(\tau - t_d)} d\tau \tag{11.21}$$

let $x = \Omega_c(\tau - t_d)$, then we have

$$dx = \Omega_c d\tau \tag{11.22}$$

$$\text{or} \quad d\tau = \frac{dx}{\Omega_c}$$

$$\Rightarrow \quad y(t) = \frac{1}{\pi} \int_{-\infty}^{\Omega_c(t-t_d)} \frac{\sin x}{x} dx$$

$$= \frac{1}{\pi} si(x) \Big|_{-\infty}^{\Omega_c(t-t_d)} \tag{11.23}$$

where *si* is the sine integral function.

The interesting properties of the *si* function are

(i) $si(x)$ is an odd function, that is $si(-x) = -si(x)$.

(ii) $si(0) = 0$.

(iii) $si(\infty) = \frac{\pi}{2}$; $si(-\infty) = -\frac{\pi}{2}$.

A sketch of $si(x)$ is shown in Fig. 11.7.

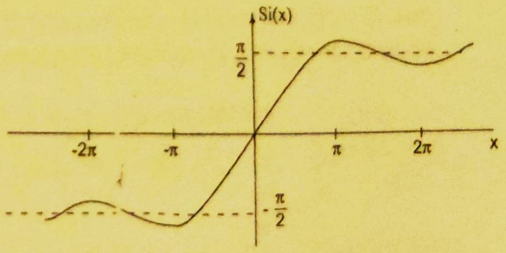


Fig. 11.7 Sine integral function.

The step response can be expressed as

$$y(t) = \frac{1}{\pi}[si[\Omega_c(t-t_d)] - si(-\infty)]$$
$$= \frac{1}{\pi}\{si[\Omega_c(t-t_d)] + \frac{\pi}{2}\}$$
$$= \frac{1}{2} + \frac{1}{\pi} si[\Omega_c(t-t_d)]$$

if $\Omega_c \to \infty$, the response $y(t)$ can be obtained as

$$y(t) = \frac{1}{2} + \frac{1}{\pi} si(\infty) = \frac{1}{2} + \frac{1}{\pi}\left(\frac{\pi}{2}\right) = 1$$

if $\Omega_c \to -\infty$, the response $y(t)$ can be obtained as

$$y(t) = \frac{1}{2} + \frac{1}{\pi} si(-\infty) = \frac{1}{2} - \frac{1}{\pi}\left(\frac{\pi}{2}\right) = 0$$

The step response $y(t)$ is shown in Fig. 11.8.

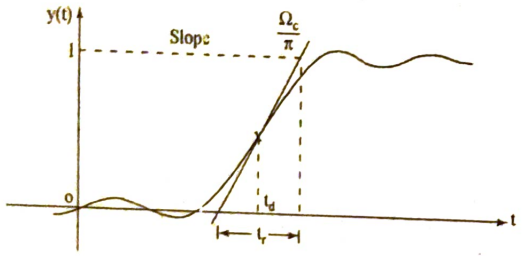


Fig. 11.8 Step response of an ideal lowpass filter.

From Fig. 11.8 we can find that $y(t)$ approaches a delayed unit step $u(t-t_d)$. But the abrupt rise of input $u(t)$, corresponds to more gradual rise of the output.

Rise time: The rise time t_r, is defined as the time required for the response to rise from 10 to 90% of the final value. To find it draw a tangent at $t = t_d$, with the line $y(t) = 0$ and $y(t) = 1$. From Fig. 11.8 we have

$$\left.\frac{dy(t)}{dt}\right|_{t=t_d} = \frac{1}{t_r} = \left.\frac{\Omega_c}{\pi}\frac{\sin \Omega_c(t-t_d)}{\Omega_c(t-t_d)}\right|_{t=t_d}$$

$\therefore$ For a lowpass filter the cutoff frequency = bandwidth

$$= \frac{\Omega_c}{\pi}$$

$$\Rightarrow \quad t_r = \frac{\pi}{\Omega_c}$$

The rise time is inversely proportional to the bandwidth.

$$\Rightarrow \quad \text{Bandwidth} \times \text{rise time} = \text{constant}$$

Review Questions

1. What do you understand by the term signal bandwidth?
2. Derive the relationship between bandwidth and rise time.
3. Define bandwidth and rise time.
4. What is the condition on impulse response for a physically realizable system?

Short Questions and Answers

1. What is distortionless transmission?

A transmission is said to be distortionless if the output is same as input, except the magnitude is scaled by a constant k and the waveform is delayed by t_d sec.

2. What are the different types of distortions in linear system?

(i) Amplitude distortion
(ii) Phase distortion

3. What is amplitude distortion? Explain.

In amplitude distortion the amplitudes of various frequency components under go different amplication. This type of distortion arises when the magnitude response $|H(j\Omega)|$ is not constant for all frequencies.

4. What is phase distortion? Explain.

In phase distortion the relative phases of various frequency components may be disturbed. This type of distortion arises when the phase response $\angle H(j\Omega)$ is not linear through the origin.

5. What is the output of a distortionless system?

If $x(t)$ is the input signal, then the output $y(t)$ of a distortionless system is

$$y(t) = kx(t-t_d)$$

6. **What is the condition on impulse response for a linear phase system?**
   For a linear phase system, the impulse response is symmetrical about $t = t_d$.

7. **What is the transfer function required for distortionless transmission?**
   The transfer function required for distortionless transmission is

$$H(j\Omega) = ke^{-j\Omega t_d}$$

8. **What is the condition on impulse response for a physically realizable system?**
   A system is said to be physically realizable if impulse response $h(t) = 0$ for $t < 0$.

9. **What is Paley-Wiener Criterion?**
   According to Paley-Wiener Criterion, the condition for physically realizability is given by

$$\int_{-\infty}^{\infty} \frac{|\ln|H(j\Omega)||}{1+\Omega^2} d\Omega < \infty$$

   where $|H(j\Omega)|$ is the magnitude function, such that the area under $|H(j\Omega)|^2$ from $\Omega = -\infty$ to ∞ is finite.

10. **Define (a) Signal bandwidth (b) System bandwidth.**

    a) Signal bandwidth: The band of frequencies that contains most of the signal energy is known as bandwidth of the signal.

    b) System bandwidth: The range of frequencies for which the magnitude $|H(j\Omega)|$ of the system remains with in $\frac{1}{\sqrt{2}}$ of its maximum value is known as bandwidth of a system.

11. **What is the relationship between bandwidth and risetime?**

    The risetime is inversely proportional to the bandwidth

$$t_r = \frac{\pi}{\omega_c}$$

12. **Define Rise time.**

    The rise time t_r, is the time required for the response to rise from 10 to 90% of the final value.

Multiple Choice Questions

1. A system has a phase response given by $\phi(\Omega)$, where Ω is the angular frequency. The phase delay and group delay at $\Omega = \Omega_0$ are respectively given by

   (a) $\frac{-\phi(\Omega_0)}{\Omega_0}, \left.\frac{-d\phi(\Omega)}{d\Omega}\right|_{\Omega=\Omega_0}$ (b) $\phi(\Omega_0), \left.\frac{-d^2\phi(\Omega_0)}{d\Omega^2}\right|_{\Omega=\Omega_0}$

   (c) $\frac{\Omega_0}{\phi(\Omega_0)}, \left.\frac{-d\phi(\Omega)}{d\Omega}\right|_{\Omega=\Omega_0}$ (d) $\Omega_0\phi(\Omega_0), \int_{-\infty}^{\Omega_0} \phi(\lambda)d\lambda$

2. A linear phase channel with phase delay T_p and group delay T_g must have

   (a) $T_p = T_g =$ constant (b) $T_p \propto f$ and $T_g \propto f$
   (c) $T_p =$ constant and $T_g \propto f$ (d) $T_p \propto f$ and $T_g =$ constant

3. Let $x(t)$ be the input to a linear, time invariant system. The required output is $4x(t-2)$. The transform function of the system should be

   (a) $4e^{j4\pi f}$ (b) $2e^{-j8\pi f}$ (c) $4e^{-j4\pi f}$ (d) $2e^{j8\pi f}$

4. The group delay function $\tau(\Omega)$ is related to phase function $\phi(\Omega)$ as

   (a) $\tau(\Omega) = \frac{d\phi(\Omega)}{d\Omega}$ (b) $\tau(\Omega) = \frac{d\phi(\Omega)}{d\Omega^2}$

   (c) $\tau(\Omega) = \frac{d^2\phi(\Omega)}{d\Omega^2}$ (d) $\tau(\Omega) = \frac{d^2\phi(\Omega)}{d\Omega}$

5. For a distortion less transmission the input $x(t)$ and the output $y(t)$ satisfy the condition

   (a) $y(t) = kx(t + t_d)$ (b) $y(t) = kx(t - t_d)$
   (c) $y(t) = kx(t/t_d)$ (d) $y(t) = k[x(t - t_d)]^2$

6. For a distortion less transmission the transfer function is of the form

   (a) $H(j\Omega) = ke^{j\Omega t_d}$ (b) $H(j\Omega) = ke^{-j\Omega t_d}$
   (c) $H(j\Omega) = ke^{j\Omega(t+t_d)}$ (d) $H(j\Omega) = ke^{j(\Omega-\Omega_d)}$

7. According to Paley-wiener criterion

   (a) $\int_{-\pi}^{\pi} \frac{|\ln|H(j\Omega)||}{1+\Omega^2} d\Omega < \infty$ (b) $\int_{-\pi}^{\pi} \frac{|\ln|H(j\Omega)||}{1-\Omega^2} d\Omega < \infty$

   (c) $\int_{-\infty}^{\infty} \frac{|\ln|H(j\Omega)||}{1+\Omega^2} d\Omega < \infty$ (d) $\int_{-\infty}^{\infty} \frac{|\ln|H(j\Omega)||}{1-\Omega^2} d\Omega < \infty$

8. The relationship between bandwidth and rise time is given by

(a) $t_r = \Omega_c \pi$ (b) $\Omega_c = t_r \pi$ (c) $t_r = \dfrac{\pi}{\Omega_c}$ (d) $t_r = \pi - \Omega_c$

Answers to Multiple Choice Questions

1. (a)
2. (a)
3. (c)
4. (a)
5. (b) $y(t) = kx(t - t_d)$
6. (b) $H(j\Omega) = ke^{-j\Omega t_d}$
7. (c) $\int_{-\infty}^{\infty} \frac{|\ln|H(j\Omega)||}{1+\Omega^2} d\Omega < \infty$
8. (c) $t_r = \dfrac{\pi}{\Omega_c}$

Appendix A

Simulation Examples using MATLAB

MATLAB (**MAT**rix **LAB**oratoty) is a high-performance language for high-efficiency Engineering and scientific numerical calculations. It was originally developed to provide easy access to matrix software developed by the LINPACK and EISPACK matrix computation software. The MATLAB environment allows us to integrate user-friendly tools with superior computational capabilities. As a result, MATLAB is one of the most useful tools for scientific and Engineering calculations and computing.

Some useful MATLAB commands:

abs(x) computes the absolute value of the elements of x. When x is complex, abs(x) is the complex magnitude of the elements of x.

angle(x) computes the phase angles, in radians, of the vector x.

log10(x) computes the logarithm to the base 10 of the elements of x

log(x) computes the natural logarithm of each of the elements in x

rem(n,N) determines the remainder after dividing n by N

ones(N) is an N-by-N matrix of ones.

ones(M,N) is an M-by-N matrix of ones.

zeros(N) is an N-by-N matrix of zeros.

zeros(M,N) is an M-by-N matrix of zeros.

plot(x,y) plots vector y versus vector x.

sin(x) is the sine of the elements of x.

cos(x) is the cosine of the elements of x.

square(T) generates a square wave with period 2π for the elements of given time vector T.

sinc(x) returns a matrix whose elements are the sinc of the elements of x, i.e.

$y = \sin(\pi x)/(\pi x)$ if $x \neq 0$

$= 1$ if $x \neq 0$

where x is an element of the input matrix and y is the resultant output element.

sawtooth(T) generates a sawtooth waveform with period 2π for the elements of given time vector T.

rectpuls(T) generates samples of a continuous, aperiodic, unity-height rectangle at the points specified in array T, centered about T=0. By default, the rectangle has width 1.

rectpuls(T,W) generates a rectangle of width W.

tripuls(T) generates samples of a continuous, aperiodic, unity-height triangle at the points specified in array T, centered about T=0. By default, the triangle is symmetric and has width 1.

tripuls(T,W) generates a triangle of width W.

sign(x) returns 1 if the element is greater than zero, 0 if it equals zero and -1 if it is less than zero. For the nonzero elements of complex x, sign(x) = x ./ abs(x).

subplot (a,b,c), breaks the figure window into an a-by-b matrix of small axes and selects the c-th axes for the current plot.

[h,w] = freqz(b,a,n) returns the N-point frequency response vector h and N-point corresponding frequency vector w in radians/sample of the filter, where n is an integer and b, a are numerator and the denominator coefficient vectors.

h = freqz(b,a,w) returns the frequency response at frequencies designated in vector w, in radians/sample (normally between 0 and π)

y = filter(b,a,x) filters the data in vector x with the filter described by vectors a and b to create the filtered data y.

y = conv(x,h) convolves two sequences x and h.

hold on holds the current plot and all axis properties so that subsequent graphing commands add to the existing graph.

axis([xmin xmax ymin ymax]) sets scaling for the x- and y-axes on the current plot.

axis off turns off all axis labeling, tick marks and background.

[q,r] = deconv(b,a) deconvolves vector a out of vector b. The result is returned in vector q and the remainder in vector r such that b = conv(a,q) + r.

ceil(x) rounds the elements of x to the nearest integers towards infinity.

floor(x) rounds the elements of x to the nearest integers towards minus infinity.

Round(x) rounds the elements of x to the nearest integers.

mod(x,y) is x - y.*floor(x /y) if y= 0. By convention, mod(x,0) is x.
The input x and y must be real arrays of the same size, or real scalars.

Creation of user defined functions

New functions may be added to MATLAB's vocabulary if they are expressed in terms of other existing functions. The commands and functions that comprise the new function must be put in a file whose name defines the name of the new-function, with a filename extension of '.m'. At the top of the file must be a line that contains the syntax definition for the new function as shown below where new-function is the user-defined function.

FUNCTION new-function

Example A.1:

Generate the following signals using MATLAB.
i)Sine ii)square iii) sawtooth iv) exponential waves

```
% To plot continuos time signals
%(i) to plot a sinusoidalsignal
  clear all
  T=.200  % time period in sec
  t=-1:0.01:1;
  x1=sin(2*pi*t/T);
  subplot(2,2,1),plot(t,x1);
  xlabel('\itt'),ylabel('x_1({\itt})')
```

```
%(ii) to plot an exponential signal
  x2=exp(-2*t);
  subplot(2,2,2),plot(t,x2);
  xlabel("\itt"),ylabel("x_2({\itt})")
%(iii) to plot a sawtooth signal
  t1=-20:0.01:20;
  x3=sawtooth(t1);
  subplot(2,2,3),plot(t1,x3);
  xlabel("\itt"),ylabel("x_3({\itt})")
%to plot a square signal
  t1=-20:0.01:20;
  x4=square(t1);
  subplot(2,2,4),plot(t1,x4);
  xlabel("\itt"),ylabel("x_4({\itt})")
  axis([-20 20 -1.5 1.5])
```

Output:

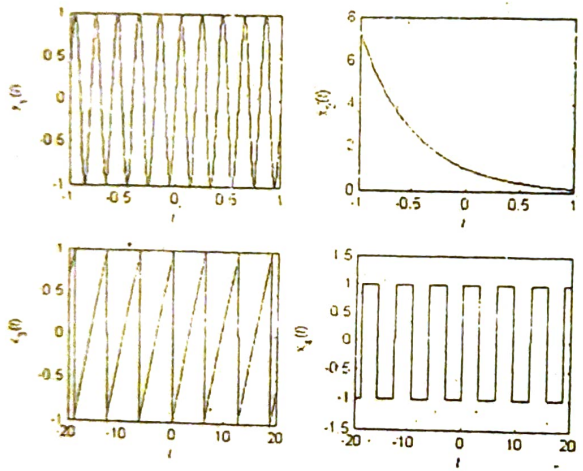


Fig. A.1. Examples for continuous-time signals

Example A.2

Generate the following signals

i) sinc ii) triangular pulse iii) rectangular pulse iv) signum function

```
% To plot continuos time signals
%(i) to plot a sinc function
  clear all
  t=-20:0.01:20;
  x1=sinc(t/2);
  subplot(2,2,1),plot(t,x1);
  xlabel('\itt'),ylabel('x_1({\itt})')
  axis([-20 20 -1.5 1.5] )
%(ii) to plot a rectangular pulse
  x2=rectpuls(t/10);
  subplot(2,2,2),plot(t,x2);
  xlabel('\itt'),ylabel('x_2({\itt})')
  axis([-20 20 -1.5 1.5] )
%(iii) to plot a triangular pulse
  x3=tripuls(t/10);
  subplot(2,2,3),plot(t,x3);
  xlabel("\itt"),ylabel("x_3({\itt})")
  axis([-20 20 -1.5 1.5] )
%to plot a signum function
  x4=sign(t/3);
  subplot(2,2,4),plot(t,x4);
  xlabel('\itt'),ylabel('x_4({\itt})')
  axis([-20 20 -1.5 1.5] )
```

Output

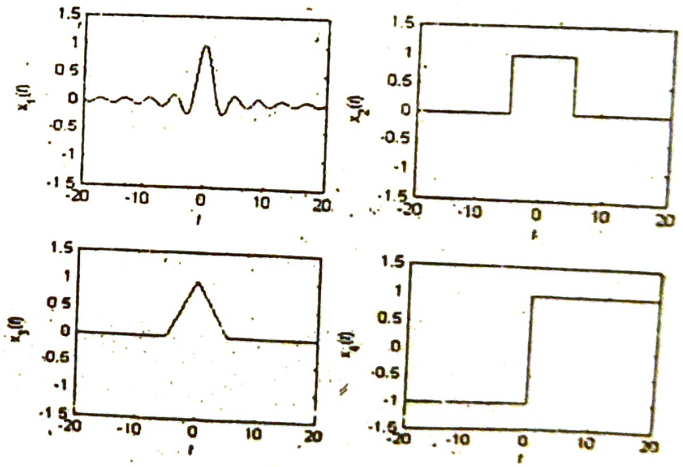


Fig. A.2. Examples of continuous-time signals

Example A.3

Generate the following signals

i) x1=sin(2*pi*t/T).*exp(-2*t);

ii) x2=2*cos(2*pi*t/T2).*sin(2*pi*t/T3);

iii) x3=sin(2*pi*t/T).*exp(-2*t)+sin(2*pi*t/T1).*exp(-4*t);

iv) x4=sinc(t).*sin(2*pi*t/T)

```
% To generate product of two functions
  clear all
%example A.2  i)
  t=-2:0.01:2
  T=.2;T1=.4
  x1=sin(2*pi*t/T).*exp(-2*t);
  subplot(2,2,1),plot(t,x1);
  xlabel("\itt"),ylabel("x_1({\itt})")
%example A.2  ii)
  T2=.3; T3=2
  x2=2*cos(2*pi*t/T2).*sin(2*pi*t/T3);
  subplot(2,2,2),plot(t,x2);
  xlabel("\itt"),ylabel("x_2({\itt})")
%example A.2  iii)
  x3=sin(2*pi*t/T).*exp(-2*t)+sin(2*pi*t/T1).*exp(-4*t);
  subplot(2,2,3),plot(t,x3);
  xlabel("\itt"),ylabel("x_3({\itt})")
%example A.2 iv)
  x4=sinc(t).*sin(2*pi*t/T)
  subplot(2,2,4),plot(t,x4);
  xlabel('\itt'),ylabel('x_4({\itt})')
```

Output

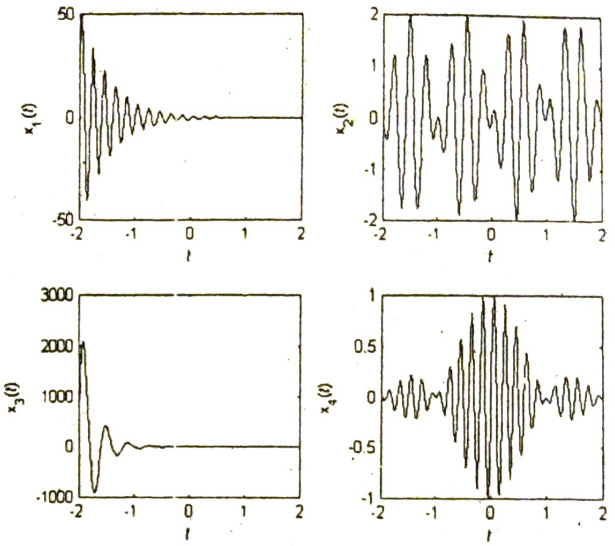


Fig.A.3. Examples for product of two continuous-time signals

Example A.4

i) Create the function x = y(t) such that

y(t) = t + 5 when –5 < t £ -2
= 11 + 4t when –2 < t £ 1
= 24 - 19 when 1 < t £ 3
= t - 6 when 3 < t £ 6

ii) Hence plot y(t), y(t–4), 2y(t-3), y(2t), y(2t-3), y(t/2)

Program i)

Include the following function y(t) in your working directory before executing the Program A.4.ii.

```
function x=y(t)
% (i) to generate the given signal
x1=t+5;  x2=11+4*t;  x3=24-9*t;  x4=t-6;
x=x1.*(-5<t & t<=-2)+x2.*(-2<t & t<=1)+x3.*(1<t & t<=3)+x4.*(3<t & t<=6);
```

Program ii)

```
% program to plot the given function y(t) and then to plot y(t+4), 2y(t-3)
%y(2t), y(2t-3), y(t/2)

clear all
tmin=-15; tmax=20;
t=tmin:0.1:tmax;
y0=y(t);
y1=y(t+4);
y2=2*y(t-3);
y3=y(2*t);
y4=y(2*t-3);
y5=y(t/2);
ymax=max([max(y0),max(y1),max(y2),max(y3),max(y4),max(y5)]);
ymin=min([min(y0),min(y1),min(y2),min(y3),min(y4),min(y5)]);
subplot(3,2,1),plot(t,y0);
xlabel("\itt"),ylabel("y_0({\itt})")
axis([tmin tmax  ymin ymax] )
subplot(3,2,2),plot(t,y1);
xlabel("\itt"),ylabel("y_1({\itt})")
axis([tmin tmax  ymin ymax] )
subplot(3,2,3),plot(t,y2);
xlabel("\itt"),ylabel("y_2({\itt})")
axis([tmin tmax  ymin ymax] )
subplot(3,2,4),plot(t,y3);
xlabel("\itt"),ylabel("y_3({\itt})")
axis([tmin tmax  ymin ymax] )
subplot(3,2,5),plot(t,y4);
xlabel("\itt"),ylabel("y_4({\itt})")
axis([tmin tmax  ymin ymax] )
subplot(3,2,6),plot(t,y5);
xlabel('\itt'),ylabel('y_5({\itt})')
axis([tmin tmax  ymin ymax] )
```

Output

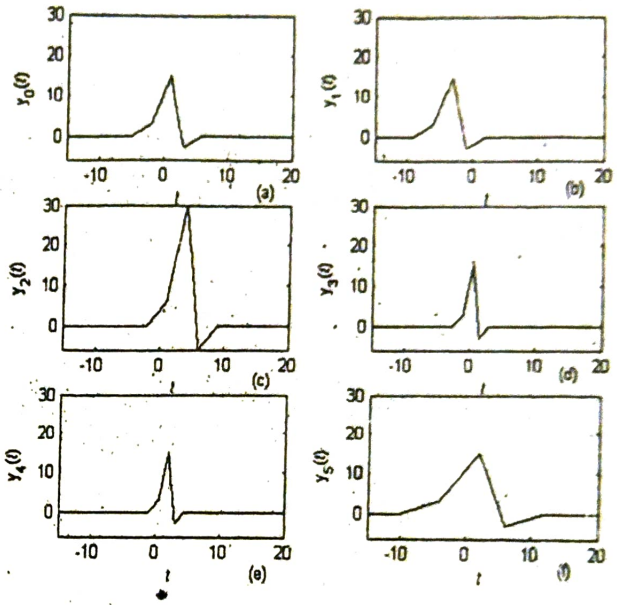


Fig.A.4. a) The signal y(t) b) y(t + 4) c) 2y(t - 3)
d) y(2t) e) y(2t - 3) f) y(t/2)

Example A.5:

Consider the function generated Example A.4, plot the functions y(t), y(-t), even and odd parts of y(t).

Include the following function y(t) in your working directory before executing the Program.

```
function x=y(t)
% (i) to generate the given signal
  x1=t+5;  x2=11+4*t;  x3=24-9*t;  x4=t-6;
  x=x1.*(-5<t & t<=-2)+x2.*(-2<t & t<=1)+x3.*(1<t & t<=3)+x4.*(3<t & t<=6);
```

Program:

```
% program to plot the function y(t), y(-t),odd and even parts of y(t)
  clear all
  tmin=-10; tmax=10;
  t=tmin:1:tmax;
```

```
% The given signal
  y0=y(t):
% time reversal of the signal
  y1=y0(end:-1:1)
  ymax=max([max(y0),max(y1)])
  ymin=min([min(y0),min(y1)])
  subplot(2,2,1),plot(t,y0)
  xlabel("\itt"),ylabel("y({\itt})")
  subplot(2,2,2),plot(t,y1)
  xlabel("\itt"),ylabel("y({-\itt})")
%Even part of the signal
  ye=(y0+y1)/2;
%Odd part of the signal
  yo=(y0-y1)/2;
  subplot(2,2,3),plot(t,ye)
  xlabel('\itt'),ylabel('y_e({\itt})')
  subplot(2,2,4),plot(t,yo)
  xlabel('\itt'),ylabel('y_o({\itt})')
```

Output

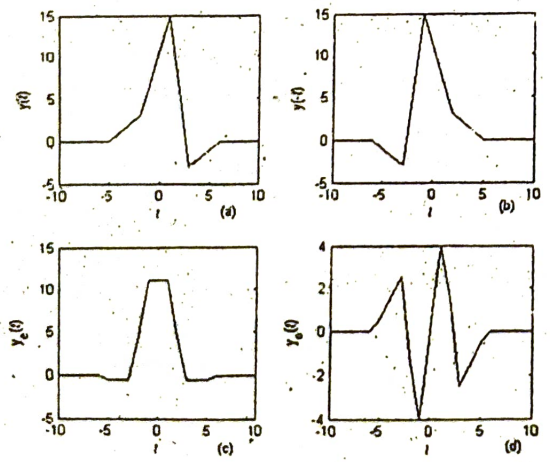


Fig.A.5. a) The signal y(t) b) y(-t) c) $y_e(t-3)$ d) $y_o(t)$

Example A.6

Plot unit step sequence, sinusoidal sequence, Exponential sequence.

```
%To generate a unit step sequence
  N=31;
  x1=ones(1,N);
  n=0:1:N-1;
  subplot(2,2,1),stem(n,x1);
  xlabel('n'),ylabel('x_1({n})')
  title('Unit Step Sequence')
%Sinusoidal Sequence
  x2=2*cos(.1*pi*n);
  subplot(2,2,2),stem(n,x2);
  xlabel('n'),ylabel('x_2({n})')
  title('Sinusoidal Sequence')
%Exponential Sequence
  x3=.6.^(n);
  subplot(2,2,3),stem(n,x3);
  xlabel('n'),ylabel('x_3{n})')
  title('Exponential Sequence')
%Addition of two sinusoidal sequences
  x4=sin(.5*pi*n)+sin(.25*pi*n);
  subplot(2,2,4),stem(n,x4);
  xlabel('n'),ylabel('x_4({n})')
  title('Addition of two sinusoidal sequences')
```

Output

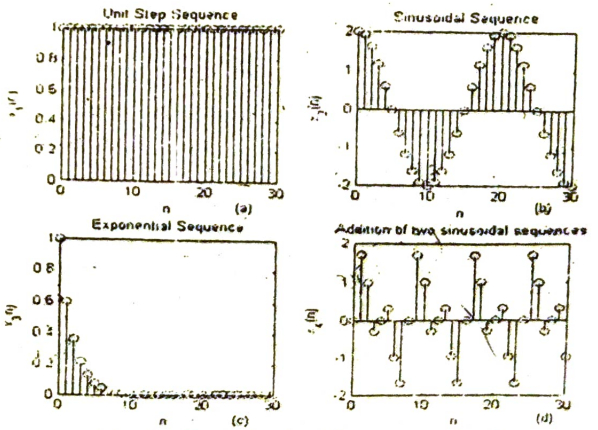


Fig.A.6. Examples for discrete-time signals

Example A.7

Plot a typical exponential sequence a^n when (i) 0<a<1, ii) -1<a<0, (iii) a >1, (i) a<-1

```
% to generate exponential sequence for various values of 'a'
  clear all
  a=.8;
  n=-10:1:10;
  x1=(a).^n;
  subplot(2,2,1),stem(n,x1);
  xlabel('\itn'),ylabel('x_1(\itn)');
  a=-0.8;
  x2=(a).^n;
  subplot(2,2,2),stem(n,x2);
  xlabel('\itn'),ylabel('x_2(\itn)');
  a=1.25;
  x3=a.^(n);
  subplot(2,2,3),stem(n,x3);
  xlabel('\itn'),ylabel('x_3(\itn)');
  a=-1.25;
  x4=(a).^(n);
  subplot(2,2,4),stem(n,x4);
  xlabel('\itn'),ylabel('x_4(\itn)');
```

Output

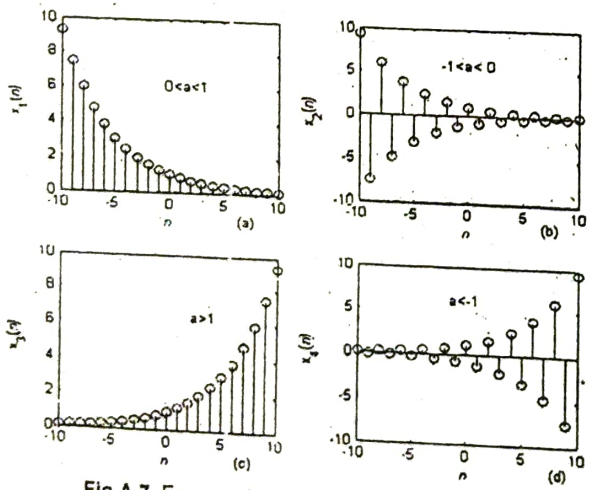


Fig.A.7. Exponential signal for different values of 'a'

Example A.8

Plot impulse, unit step and ramp sequences.

```
%To generate impulse, step and ramp sequences
  clear all
  n=-4:1:4;
% impulse sequence
  x1=[zeros(1,4),1,zeros(1,4)];
  subplot(3,1,1),stem(n,x1);
  xlabel('n'),ylabel('x_1(n)');
  title('Unit impulse sequence');
%unit step sequence
  x2=[zeros(1,4),ones(1,5)];
  subplot(3,1,2),stem(n,x2);
  xlabel('n'),ylabel('x_2(n)');
  title('Unit step sequence');
%unit ramp sequence
  n1=0:1:10
  x3=n1;
  subplot(3,1,3),stem(n1,x3);
  title('Unit ramp sequence');
  xlabel('n'),ylabel('x_3(n)');
```

Output

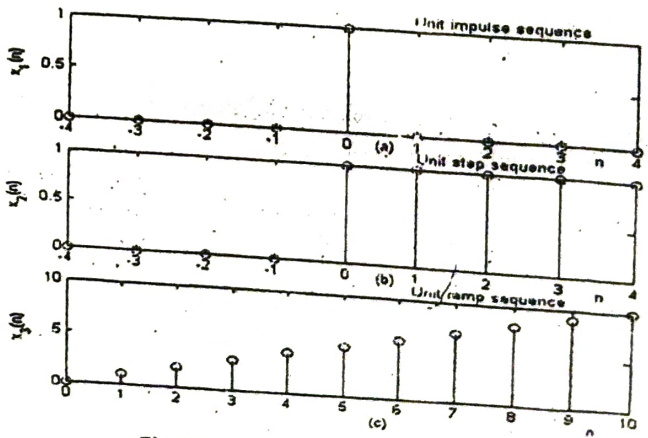


Fig.A.8. Elementary discrete-time signals

Example A.9:

Write a function call x = [1 –1 2 –5 1 1 3 –2 –1 0 1 –1 2] and write a program for amplitude scaling, time scaling and time shifting of given signal.

Program i)

Include the following function y(t) in your working directory before executing the Program.

```
function x=yd(n)
% (i) to generate the given signal
x=[1 -1 2 -.5 1 1 3 -2 -1 0 1 -1 -2]
```

Program ii)

```
% program to plot the function y(n)
% To generate amplitude scaling,time scaling and time shifting of a given signal
  clear all
  nmin=-6; nmax=6;
  n=nmin:1:nmax;
%To generate the signal y(n)
  y0=yd(n);
%To generate the signal y(-n)
  yr=y0(end:-1:1);
%To generate the signal ye(n)
  ye=(y0+yr)/2;
%To generate the signal yo(n)
  yo=(y0-yr)/2;
  ymax=max([max(y0),max(yr),max(ye),max(yo)]);
  ymin=min([min(y0),min(yr),min(ye),min(yo)]);
  subplot(2,2,1),stem(n,y0);
  xlabel("n"),ylabel("y({n})")
  axis([nmin nmax  ymin ymax] )
  subplot(2,2,2),stem(n,yr);
  xlabel('n'),ylabel('y_r({n})')
  axis([nmin nmax  ymin ymax] )
  subplot(2,2,3),stem(n,ye);
  xlabel('n'),ylabel('y_e({n})')
  axis([nmin nmax  ymin ymax] )
  subplot(2,2,4),stem(n,yo);
  xlabel('n'),ylabel('y_o({n})')
  axis([nmin nmax  ymin ymax] )
```

Output

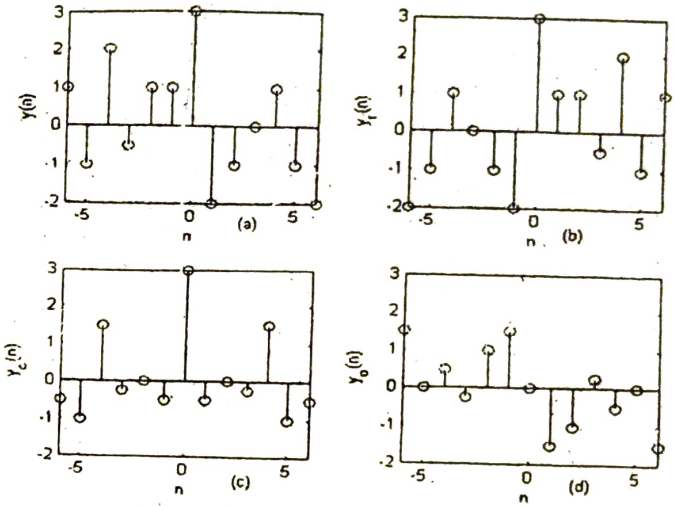


Fig.A.9. a) The signal y(n) b) y(-n) c) $y_e(n)$ d)$y_o(n)$

Example A.10

Obtain the convolution of two given sequences x and h.

```
clear all
x=[1,2,1,2,1,3,2];
N1=length(x);
n=0:1:N1-1;
subplot(2,2,1),stem(n,x);
xlabel('n'),ylabel('x(n)');
title('input sequence x(n)')
h=[1,-1,2,-2,1,1];
N2=length(h);
n1=0:1:N2-1;
subplot(2,2,2),stem(n1,h);
xlabel('n'),ylabel('h(n)');
title('impulse sequence h(n)')
y=conv(x,h)
n2=0:1:N1+N2-2;
```

```
subplot(2,1,2),stem(n2,y);
xlabel('n'),ylabel('y(n)');
title('Convolution of two sequences x(n) and h(n)')
```

Output

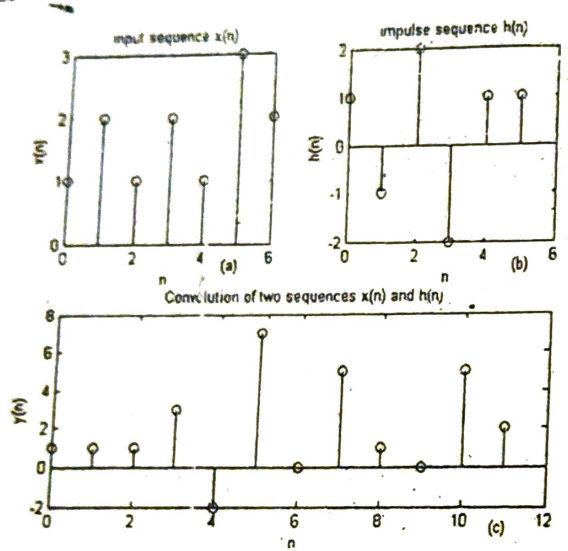


Fig.A.10. Convolution of two sequences

Example A.11

Plot the square wave and hence verify Gibb's phenomena using first 10 terms of Fourier series.

```
clear all
T=input('enter the time period of the square wave\n')
n1=input('enter the number of cycles to be plotted\n')
n=input('enter the number of harmonics to be considered apart from dc\n')
k=n1*T/2;
i=0;
for t=-k:k/100:k
  x=0;
   for j=1:2:(2*n-1)
```

```
      xnew=x+(4*(cos((t*2*pi*j/T)-(pi*floor(j/2)))))/(j*pi));
        x=xnew;
      end
      i=i+1;
      p(:,i)=x;
        end
t=-k:k/100:k;
plot(t,p);
enter the time period of the square wave
1
T =
     1
enter the number of cycles to be plotted
5
n1 =
     5
enter the number of harmonics to be considered apart from dc
10
n =
    10
```

Output

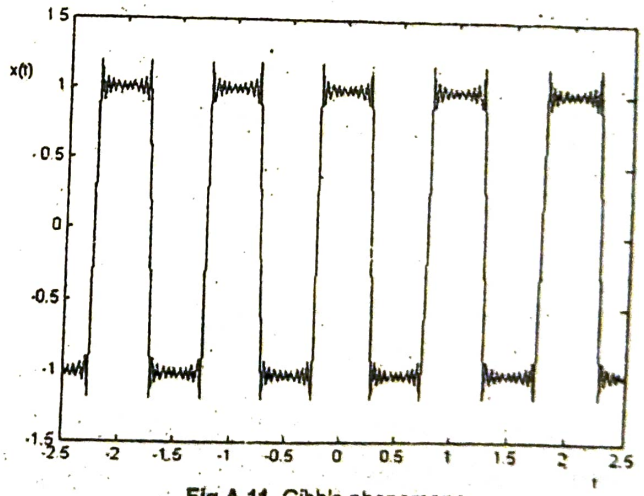


Fig.A.11. Gibb's phenomena

Example A.12

Write a program to verify sampling theorem.

```
% To verify sampling theorem
  clear all
  t=-100:.01:100;
  fm=.02
  x=cos(2*pi*t*fm);
  subplot(2,2,1),plot(t,x);
  xlabel('Time in sec'),ylabel('x(t)');
  title('Continuous time signal');
  fs1=.02
  n=-2:2;
  x1=cos(2*pi*fm*n/fs1);
  subplot(2,2,2),stem(n,x1);
  hold on
  subplot(2,2,2),plot(n,x1,':')
  title('Discrete time signal x(n) with fs<2fm');
  xlabel('n'),ylabel('x(n)');
  fs2=0.04
  n1=-4:4
  x2=cos(2*pi*fm*n1/fs2);
  subplot(2,2,3),stem(n1,x2);
  hold on
  subplot(2,2,3),plot(n1,x2,':')
  title('Discrete time signal x(n) with fs=2fm');
  xlabel('n'),ylabel('x(n)');
  n2=-50:50
  fs3=.5
  x3=cos(2*pi*fm*n2/fs3);
  subplot(2,2,4),stem(n2,x3);
  hold on
  subplot(2,2,4),plot(n2,x3,':')
  xlabel('n'),ylabel('x(n)');
  title('Discrete time signal x(n) with fs>2fm');
```

Output

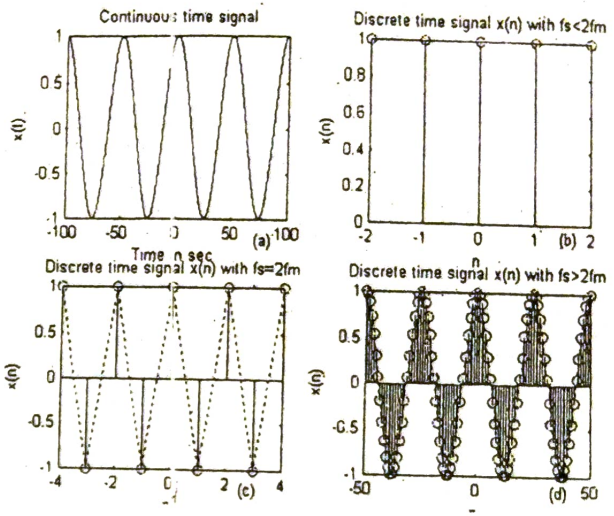


Fig.A.12 a) Continuous-time signal
b) discrete-time signal when $f_s < 2f_m$
c) discrete-time signal when $f_s = 2f_m$
d) discrete-time signal when $f_s > 2f_m$

Example A.13

Write a program to illustrate the effect of zeropadding.

```
% To illustrate the effect of zeropadding
  clear all
  x=input('enter a five point sequence');
  w=0:.01:2*pi;
  h=freqz(x,1,w);
  subplot(3,2,1),plot(w/pi,abs(h))
  xlabel('\omega/\pi')
  ylabel(' magnitude')
  subplot(3,2,2),plot(w/pi,angle(h))
  xlabel('\omega/\pi')
  ylabel(' Phase')
  N=5;
  n=0:1:N-1;
  y=fft(x,N);
  subplot(3,2,3);stem(n,abs(y));
```

Output

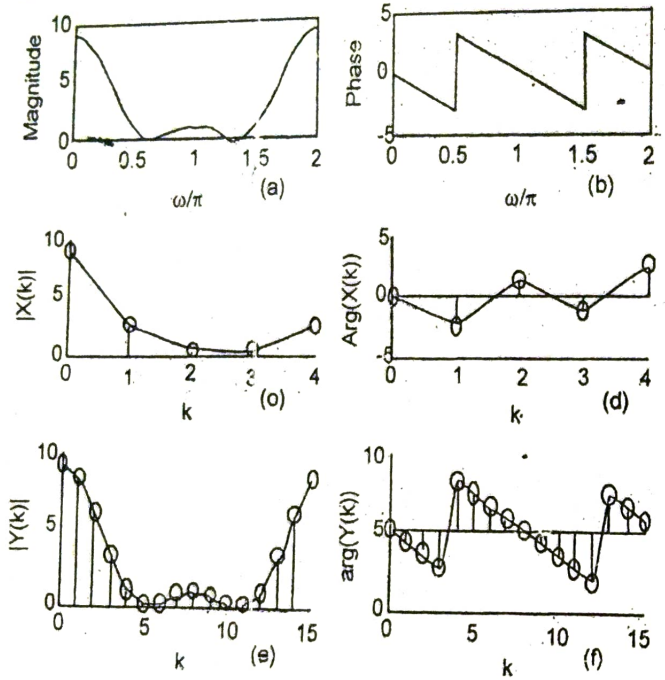


Fig.A.15 (a) Magnitude plot
(b) Phase plot
(c) Magnitude plot when N = 5
(d) Phase plot when N = 5
(e) Magnitude plot when N = 16
(f) Phase plot when N = 16

```
hold on
subplot(3,2,3);plot(n,abs(y),'-.');
xlabel('k')
ylabel('|X(k)|');
subplot(3,2,4);stem(n,angle(y));
hold on
subplot(3,2,4);plot(n,angle(y),'-.');
xlabel('k')
ylabel('arg(X(k))')
x1=[x,zeros(1,11)];
N1=16;
n1=0:1:N1-1;
y1=fft(x1,N1);
subplot(3,2,5);stem(n1,abs(y1));
hold on
subplot(3,2,5);plot(n1,abs(y1),'-.');
xlabel('k')
ylabel('|Y(k)|')
subplot(3,2,6);stem(n1,angle(y1));
hold on
subplot(3,2,6);plot(n1,angle(y1),'-.');
xlabel('k')
ylabel('arg(Y(k))')
```

enter a five point sequence[1 2 3 2 1]

Example A.14

Write a program to find n-point DFT of a given sequence.

```
% To find the n-point DFT of the given sequence
% and plot magnitude and phase plot
  clear all
  xn=input('enter a sequence');
  L=length(xn); % Length of the sequence
  N=input('enter the length of the DFT');
  xk=dft(xn,N)
  subplot(2,1,1),stem(abs(xk))
  xlabel('\itk')
  ylabel( ' |X(k)|')
  subplot(2,1,2),stem(angle(xk))
  xlabel('\itk')
  ylabel( 'arg(X(k))')
```

Include the following function y(t) in your working directory before executing the Program.

```
  function X=dft(xn,N)
  %To compute the DFT of the sequence x(n)
  L=length(xn);%length of the sequence
  %Checking for the length of the DFT
  if(N<L)
    error('N must be >=L')
  end
```

```
x1=[xn zeros(1,N-L)];%Appending zeros
 % Computation of twiddle factors
 for k=0:1:N-1;
  for n=0;1:N-1;
   p=exp(-i*2*pi*n*k/N);
   x2(k+1,n+1)=p;
  end
 end
 X=x1*x2.';
```

Output

```
enter a sequence[1 1 1 1 0 0 0 0]
enter the length of the DFT. 8

xk =

 Columns 1 through 6

  4.0000   1.0000 - 2.4142i  -0.0000 - 0.0000i   1.0000 - 0.4142i   0 - 0.0000i
1.0000 + 0.4142i

 Columns 7 through 8

 0.0000 - 0.0000i   1.0000 + 2.4142i
```

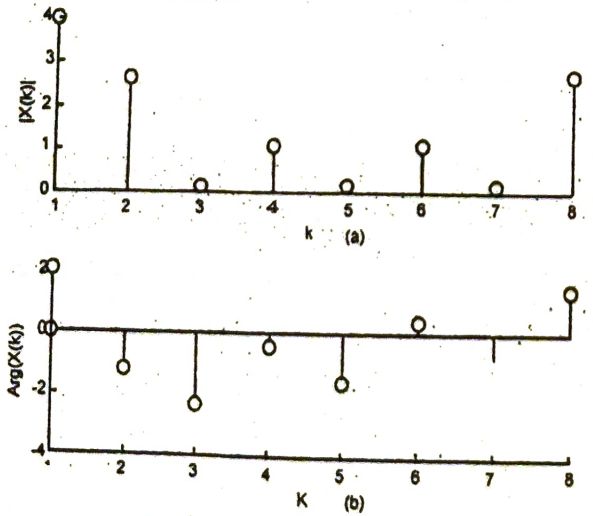


Fig.A.14. (a) Magnitude plot
(b) Phase plot

Example A.15

Write a program find circular convolution of two sequences.

```
% To find the circonvolution of two sequences
  clear all
  x1=input('enter the sequence1');
  x2=input('enter the sequence2');
  N=input('Number of points in the output sequence')
  yc=circonv(x1,x2,N);
  subplot(3,1,1),stem(x1)
  xlabel('\itn')
  ylabel( ' x_1(n)')
  subplot(3,1,2),stem(x2)
  xlabel('\itn')
  ylabel( 'x_2(n)')
  subplot(3,1,3),stem(yc)
  xlabel('\itn')
  ylabel( 'y_c(n)')
```

Include the following function y(t) in your working directory before executing the Program.

```
  function[y]= circonv(x,h,N);
  % To find the circular convolution of two sequences
  %x=input sequence1
  %h=Impulse sequence2
  %N=Number of points in the output sequence
  N2=length(x);
  N3=length(h);
  x=[x zeros(1,N-N2)]% Append N-N2 zeros to the input sequence 1
  h=[h zeros(1,N-N3)] %Append N-N1 zeros t the sequence 2
  %circular shift of the sequence 2
  m=[0:1:N-1];
  M=mod(-m,N);
  h=h(M+1);
  for n=1:1:N
  m=n-1;
  p=0:1:N-1;
  q=mod(p-m,N);
  hm=h(q+1);
```

```
H(n,:)=hm;
end
%Matrix convolution
y=x*H';
```

Output

```
enter the sequence1 [ 1 2 3 4]
enter the sequence2 [4 3 2 1]Number of points in the output sequence  4
yc =   24   22   24   30
```

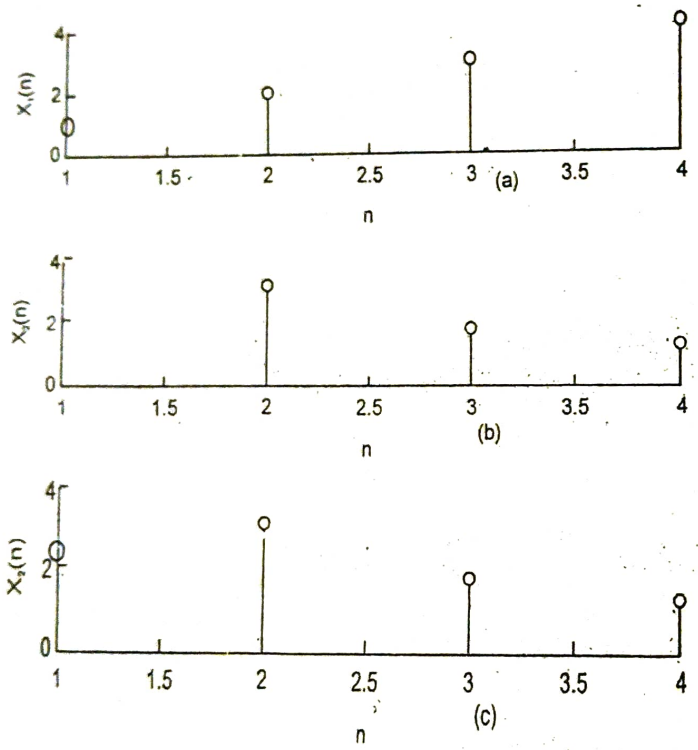


Fig.A.15. (a) Sequence 1
(b) Sequence 2
(c) Circular convolution of two sequences

Example A.16

Write a program to find IDFT of a given sequence.

```
% To find the IDFT of the given sequence
% and plot magnitude and phase plot
  clear all
  xk=input('enter the sequence');
  N=length(xk);
  if mod(N,2)==0
  for i=1:N/2-1
    if xk(i+1)~=conj(xk(N-i+1))
      error('sequence is not satisfying conjugate property')
    end
  end
  end
  if mod(N,2)==1
    for i=1:(N-1)/2
      if xk(i+1)~=conj(xk(N-i+1))
      error('sequence is not satisfying conjugate property')
    end
  end
  end
  xn=idft(xk,N)
```

Include the following function y(t) in your working directory before executing the Program.

```
function xn=idft(X,N)
%To compute the inverse DFT of the sequence X(k)
%L=length(xn);%length of the sequence
  % Computation of twiddle factors
  for k=0:1:N-1;
    for n=0:1:N-1;
      p=exp(i*2*pi*n*k/N);
      x2(k+1,n+1)=p;
    end
  end
  xn=(X*x2.')/N;
```

Output

```
enter the sequence[10 -0.4142 - 7.2426i  -2 + 2i   2.4142 - 1.2426i  -2   2.4142 +
1.2426i  -2 - 2i  -0.4142 + 7.2426i]
```

```
xn =
 Columns 1 through 6
  1.0000 - 0.0000i  2.0000 - 0.0000i  3.0000 - 0.0000i  4.0000 - 0.0000i  -0.0000
+ 0.0000i  0.0000 + 0.0000i
 Columns 7 through 8
-0.0000 - 0.0000i  0.0000 + 0.0000i
```

Example A.17

Write a program to verify Parseval's theorem.

%Verification of Parseval's theorem

```
clear all
n=0:1:100;
x=(0.5).^n;
L=length(x);
N=256;
xk=fft(x,N);
sum1=sum(x.^2)
sum2=sum(abs(xk).^2)/N
eps=0.0001;
if(abs(sum1-sum2)<eps)
  display('Parseval,s Theorem verified')
end
```

Output

```
   sum1 =   1.3333
   sum2 =   1.3333
Parseval,s Theorem verified
```

Example A.18

Write a program to find partial fraction expansion for the given Laplace transfer function.

%Partial fraction expansion in Laplace transform

```
clear all
b=input('enter the numerator coefficients');
a=input('enter the denominator coefficients')
[r,p,k]=residue(b,a) ;
```

Output

```
enter the numerator coefficients   [ 1  2]
enter the denominator coefficients  [1  3  4]
  r =
    0.5000 - 0.1890i
    0.5000 + 0.1890i
  p =
   -1.5000 + 1.3229i
   -1.5000 - 1.3229i
  k =

     []
```

Example A. 19

Write a program to find partial fraction expansion for the given Z transfer function.

%Partial fraction expansion in Z-transform

```
clear all
b=input('enter the numerator coefficients');
a=input('enter the denominator coefficients')
[r,p,k]=residuez(b,a)
```

Output

```
enter the numerator coefficients  [0 0 5]
enter the denominator coefficients[1 2 2]
  r =
   -1.2500 + 1.2500i
   -1.2500 - 1.2500
  p =
   -1.0000 + 1.0000i
   -1.0000 - 1.0000i
  k =
    2.5000
```

Example A.20

Write a program to find impulse response of a digital filter.

%To compute the impulse response of the filter

```
clear all
```

```
b=input('enter the numerator coefficients');
a=input('enter the denominator coefficients');
[H,T] = impz(b,a)
stem(T,H)
xlabel('n')
ylabel('h(n)')
```

Output

enter the numerator coefficients[1]
enter the denominator coefficients[1 -0.7 0.12]

```
H = 1.0000  0.7000  0.3700  0.1750  0.0781  0.0337  0.0142  0.0059  0.0024  0.0010
T =  0       1       2       3       4       5       6       7       8       9
```

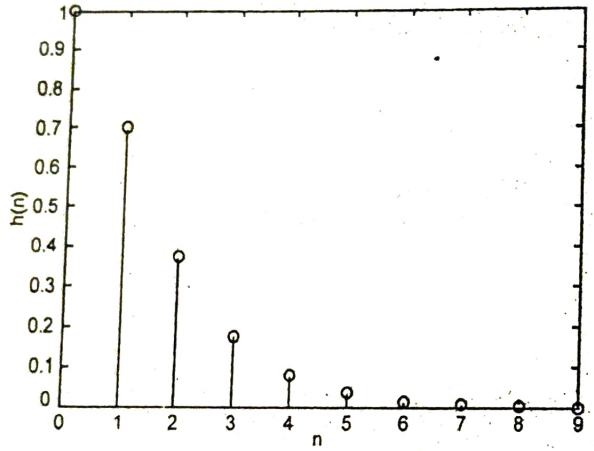


Fig.A.20. Impulse response of the digital filter

Example A.21

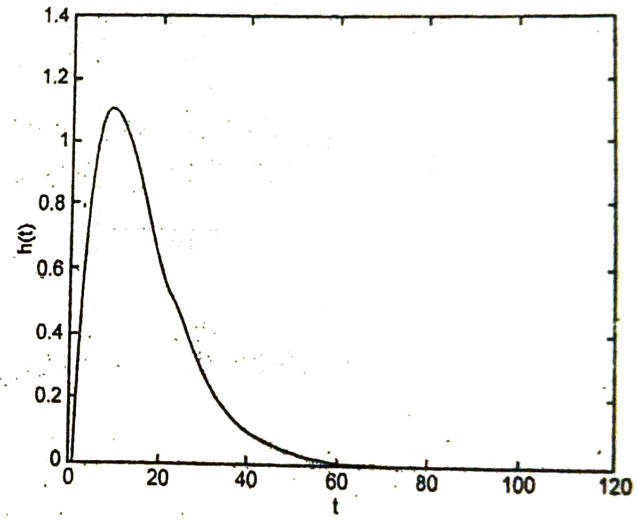
Write a program to find impulse response of an analog filter.

%To compute the impulse response of the analog filter.

```
clear all
b=input('enter the numerator coefficients');
a=input('enter the denominator coefficients');
[H,T] = impulse(b,a)
plot(H)
xlabel('t')
ylabel('h(t)')
```

Output

enter the numerator coefficients [1]
enter the denominator coefficients[1 -0.7 0.12]



Example A.22

Write a program to find step response of an analog filter.

%To compute the step response of the filter

```
clear all
b=input('enter the numerator coefficients');
a=input('enter the denominator coefficients');
[H] = step(b,a)
plot(H)
xlabel('t')
ylabel('s(t)')
```

Output

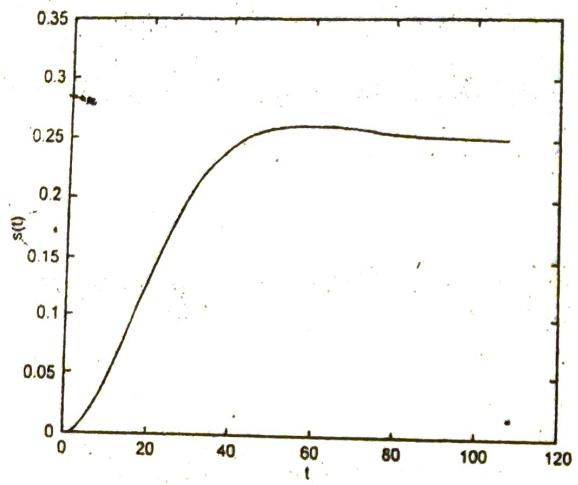


Fig.A.22. Step response of the filter

Example A.23

Write a program to find sinusoidal response of a system.

```
% Response of a system for sinusoidal input
 clear all
  s = tf('s');
 g = (s+1)/(s^2+4*s+4);
  t = 0:0.1:15;
  u = sin(t);
  lsim(g,u,t);
```

Output

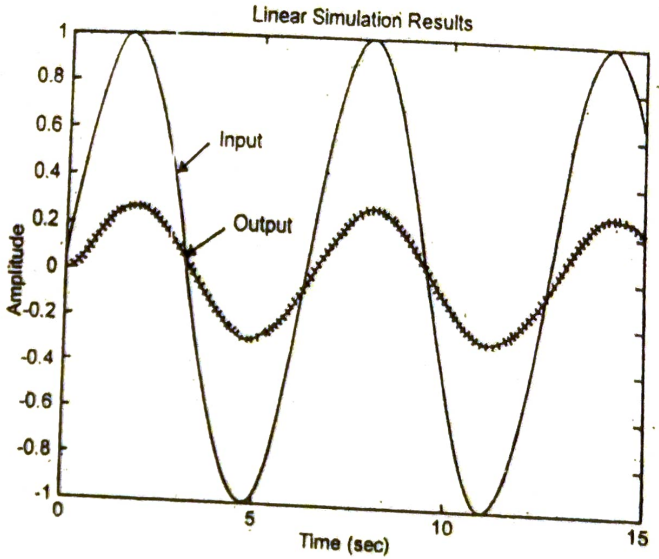


Fig.A.23. Sinusoidal response of a system

Example A.24

Write a program to convert from state space to transfer function.

```
%statespace to transfer function conversion
  A=[2 -1;-1 2];
  B=[0;1];
  C=[0 1];
  D=[0];
  [num,den]=ss2tf(A,B,C,D)
```

Output

```
num =
    0    1   -2
den =
    1   -4    3
```

Example A.25

Write a program to convert from transfer function to state space.

```
%transfer function to statespace conversion
num=[1 6 5];
den=[1 3 5 6];
[A,B,C,D]=tf2ss(num,den)
```

Output

```
A =
    -3    -5    -6
     1     0     0
     0     1     0
B =
     1
     0
     0
C =
     1     6     5
D =
     0
```

Example A.26

Write a program to plot exponential response of a system.

```
clear all
s=tf('s');
g=(s+5)/(s^2+5*s+6);
 t = 0:0.1:15;
 u = exp(-3*t);
 lsim(g,u,t)
```

Output

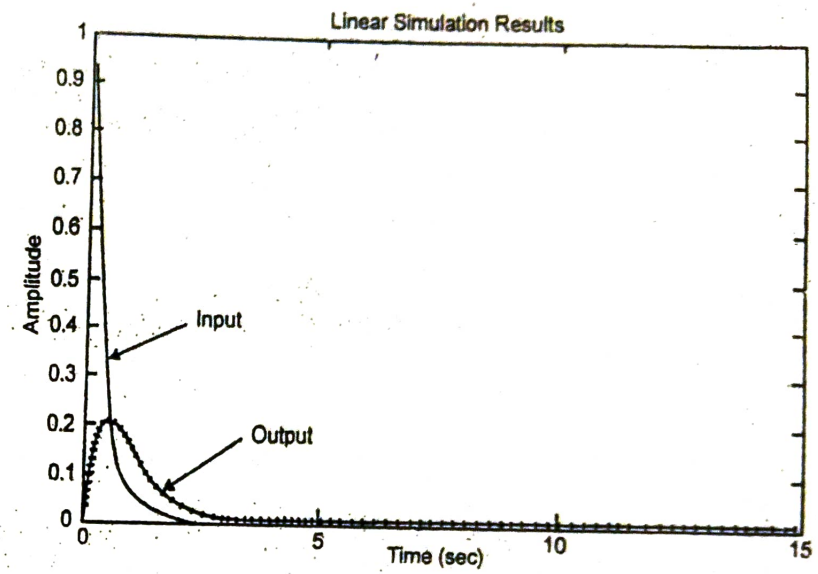
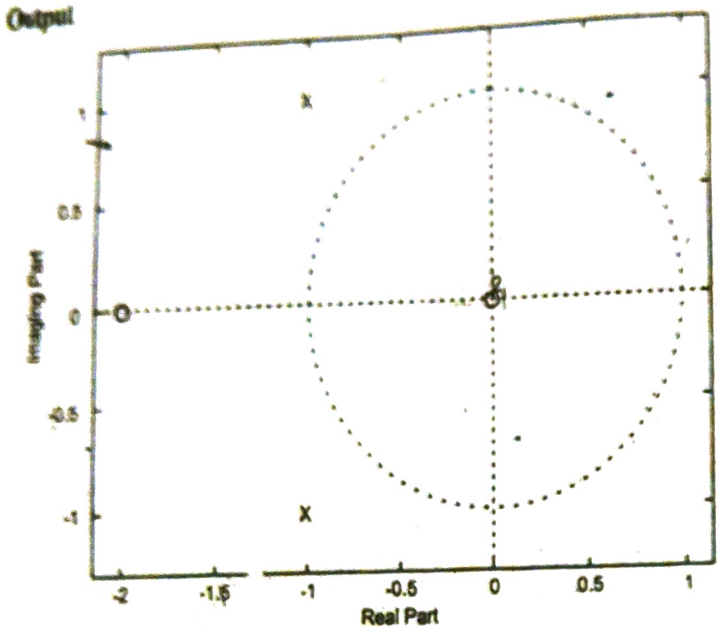


Fig.A.26. Exponential response of a system

Example A.27

Write a program to plot pole zero plot.

```
%To plot pole-zero plot
clear all
b=input('enter the numerator coefficients');
a=input('enter the denominator coefficients');
zplane(b,a)
```



Appendix-B

Infinite Summation Formulas

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}, \quad |a| < 1$$

$$\sum_{n=0}^{\infty} n a^n = \frac{a}{(1-a)^2}, \quad |a| < 1$$

$$\sum_{n=0}^{\infty} n^2 a^n = \frac{a^2+a}{(1-a)^3}, \quad |a| < 1$$

Finite Summation Formulas

$$\sum_{n=0}^{N} a^n = \frac{1-a^{N+1}}{1-a}, \quad a \neq 1$$

$$\sum_{n=0}^{N} 1 = N+1$$

$$\sum_{n=N_1}^{N_2} 1 = N_2 - N_1 + 1$$

$$\sum_{n=0}^{N} n = N(N+1)/2$$

$$\sum_{n=0}^{N} n^2 = N(N+1)(2N+1)/6$$

$$\sum_{n=0}^{N} n^3 = N^2(N+1)^2/4$$

Series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\ldots$$

$$a^x = 1 + x\ln a + \frac{(x\ln a)^3}{3!} + \ldots$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots$$

$$\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \ldots$$

$$\sin x = \frac{e^{jx} - e^{-jx}}{2j}; \quad \cos x = \frac{e^{jx} + e^{-jx}}{2}$$

$$e^{jx} = \cos x + j\sin x; \quad e^{-jx} = \cos x - j\sin x$$

Complex Variables

$Z = x + jy$ x = real part

y = imaginary part

$j = \sqrt{-1}$

Magnitude $|Z| = \sqrt{x^2 + y^2}$

Phase angle $\angle Z = \tan^{-1}\frac{y}{x}$

$$|Z|^2 = x^2 + y^2 = Z \cdot Z^*$$

Trigonometric Identities

$$\sin^2 A + \cos^2 A = 1$$

$$\sin\left(A \pm \frac{\pi}{2}\right) = \pm\cos A$$

$$\cos\left(A \pm \frac{\pi}{2}\right) = \mp\sin A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$\sin 2A = 2\sin A\cos A$$

$$\cos(A+B) = \cos A\cos B - \sin A\sin B$$

$$\cos(A-B) = \cos A\cos B + \sin A\sin B$$

$$\sin(A+B) = \sin A\cos B + \cos A\sin B$$

$$\sin(A-B) = \sin A\cos B - \cos A\sin A$$

$$\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$$

$$\cos A + \cos B = 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$$

$$\cos^2 A = \frac{1}{2}[1 + \cos 2A]$$

$$\sin 2A = \frac{2\tan A}{1 + \tan^2 A}$$

$$\cos 2A = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1$$

$$= \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$\cos 3A = 4\cos^3 A - 3\cos A$$

$$\sin 3A = 3\sin A - 4\sin^3 A$$

$$\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$$

$$\cos^4 A = \frac{1}{8}[3 + 4\cos 2A + \cos 4A]$$

$$\sin^4 A = \frac{1}{8}[3 - 4\cos 2A + \cos 4A]$$

$$\tan 3A = \frac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$$

$$\tan^{-1} A \pm \tan^{-1} B = \tan^{-1}\frac{(A \mp B)}{1 \mp AB}$$

$$A\sin C + B\cos C = (A^2 + B^2)^{1/2}\cos(C - \tan^{-1} A/B)$$

$$= (A^2 + B^2)^{1/2}\sin(C + \tan^{-1} B/A)$$

Definite Integrals

$$\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}$$

$$\int_0^\infty \left(\frac{\sin x}{x}\right)^2 dx = \frac{\pi}{2}$$

$$\int_0^\infty e^{-ax}\sin(bx)\,dx = \frac{b}{a^2+b^2}$$

$$\int_0^\infty e^{-ax}\cos(bx)\,dx = \frac{a}{a^2+b^2}$$

$$\int_0^\infty e^{-a^2x^2}\cos(bx)\,dx = \frac{\sqrt{\pi}e^{-b^2/4a^2}}{2a}, \quad a>0$$

$$\int_0^\infty \frac{\cos(ax)}{b^2+x^2}\,dx = \frac{\pi}{2b}e^{-ab}, \quad a>0, \quad b>0$$

$$\int_0^\infty \frac{x\sin ax}{b^2+x^2}\,dx = \frac{\pi}{2a}e^{-ab} \quad a>0 \quad b>0$$

$$\int_0^\infty x^{2n}e^{-ax^2}\,dx = \frac{1\cdot 3\cdot 5\cdots(2n-1)}{2^{n+1}a^n}\sqrt{\frac{\pi}{a}}$$

$$\int_0^\infty x^2 e^{-x^2}\,dx = \frac{\sqrt{\pi}}{4}$$

Indefinite integrals

$$\int \frac{dx}{a+bx} = \frac{1}{b}\ln[a+bx]$$

$$\int (a+bx)^n = \frac{(a+bx)^{n-1}}{b(n+1)}, \quad 0<n$$

$$\int \frac{xdx}{a^2+x^2} = \frac{1}{2}\ln(a^2+x^2)$$

$$\int \frac{x^2dx}{a^2+x^2} = x - a\tan^{-1}\left(\frac{x}{a}\right)$$

$$\int \frac{xdx}{(a^2+x^2)^2} = \frac{-1}{2(a^2+x^2)}$$

$$\int x\cos x\,dx = \cos x + x\sin x$$

$$\int e^{ax}\sin x\,dx = \frac{e^{ax}}{a^2+1}(a\sin x - \cos x)$$

$$\int e^{ax}\,dx = \frac{e^{ax}}{a}$$

$$\int xe^{ax}\,dx = e^{ax}\left[\frac{x}{a} - \frac{1}{a^2}\right]$$

$$\int x^2e^{ax}\,dx = e^{ax}\left[\frac{x^2}{a} - \frac{2x}{a^2} + \frac{2}{a^3}\right]$$

$$\int \sin x\,dx = -\cos x$$

$$\int x\sin x\,dx = \sin x - x\cos x$$

Derivatives

$$\int udv = uv - \int vdu$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(\sin x) = -\cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(e^{ax}) = ae^{ax}$$

$$\frac{d}{dx}(a^x) = a^x\ln a$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$\frac{d}{dx}(\log_a x) = \frac{1}{x}\log a$$

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